

Scaling Robotic Manipulation via Structured World Models and Tactile Sensing

Yunzhu Li

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Robotics & Simulation @ World Labs

IIIS, Tsinghua University
October 27, 2026

World Labs

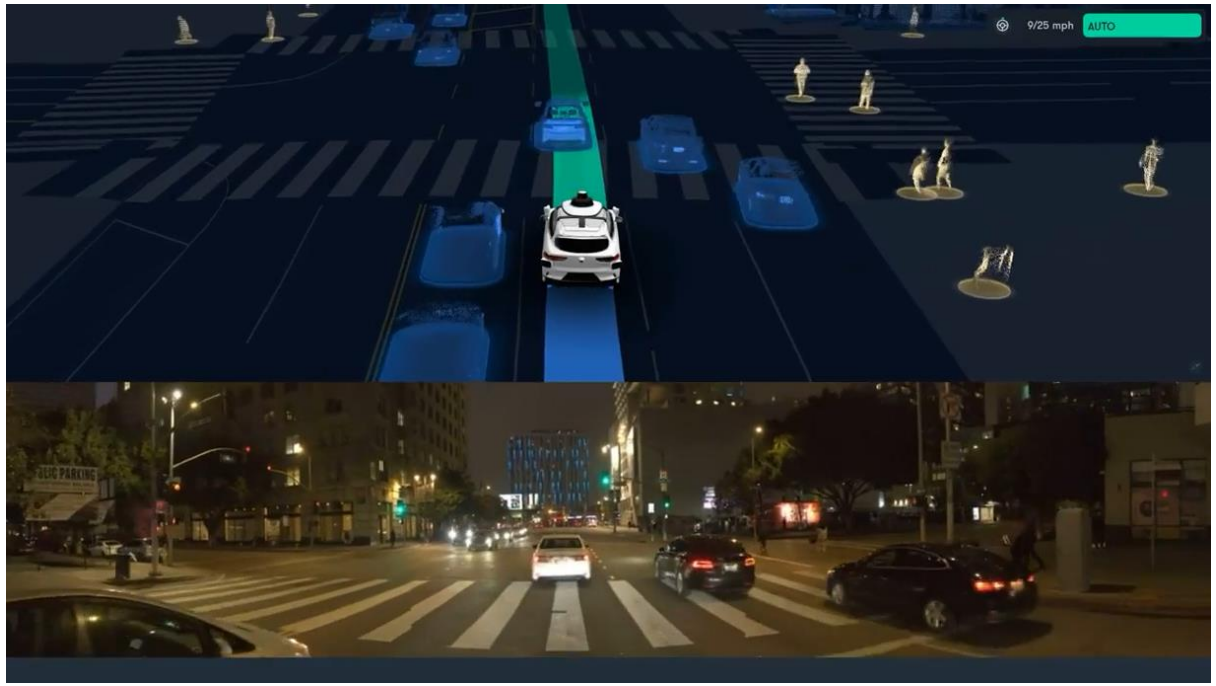
Columbia | Engineering

Progress in robotics

- In certain (restricted) areas, robots have been on par or surpass humans.



Humanoid



Autonomous Driving

Progress in robotics

- Complicated physical interactions with the environment?
- How far can we achieve human-level robotic manipulation?



Boston Dynamics Stretch

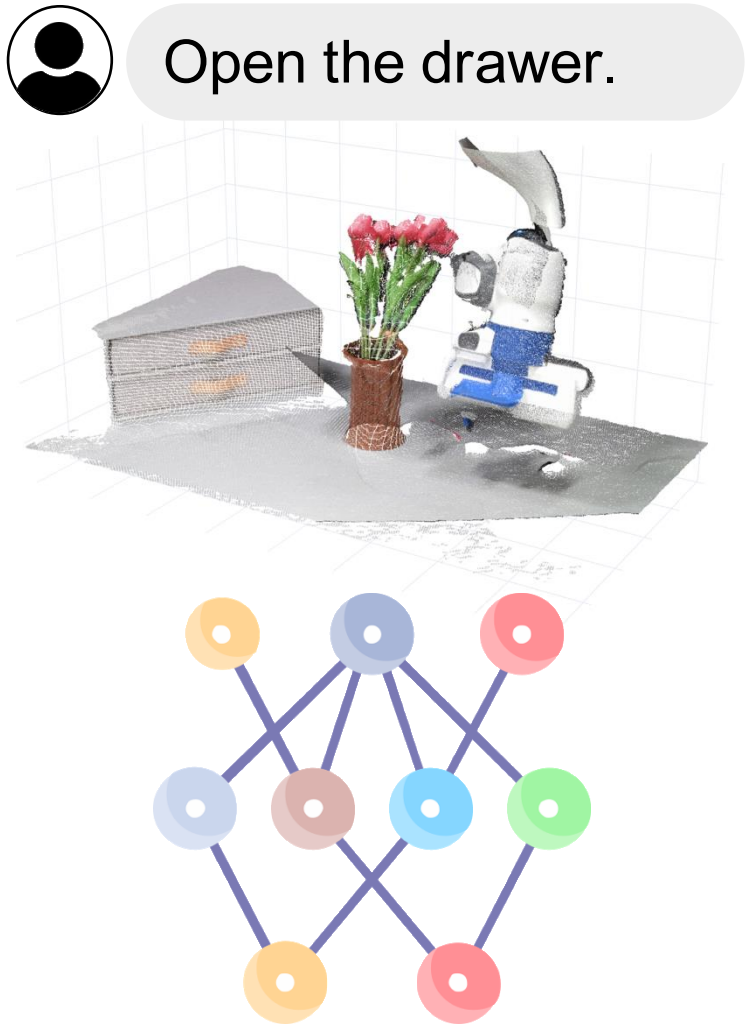


What we want the robots to do







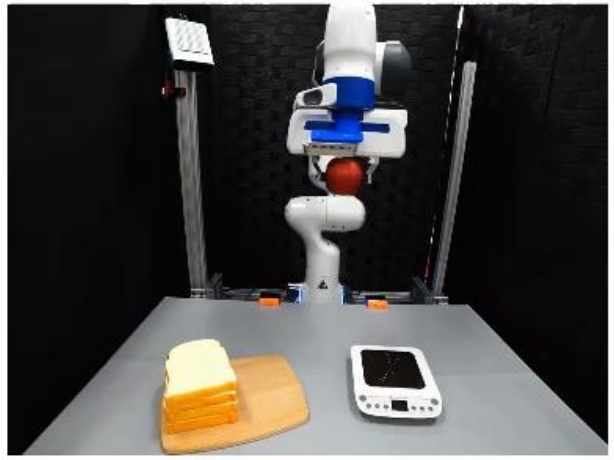



- Foundation Models for Robotics
  - GPT, SAM, DINO, FoundationPose/Stereo…
    - Code-as-Policies, VoxPoser, ReKep…

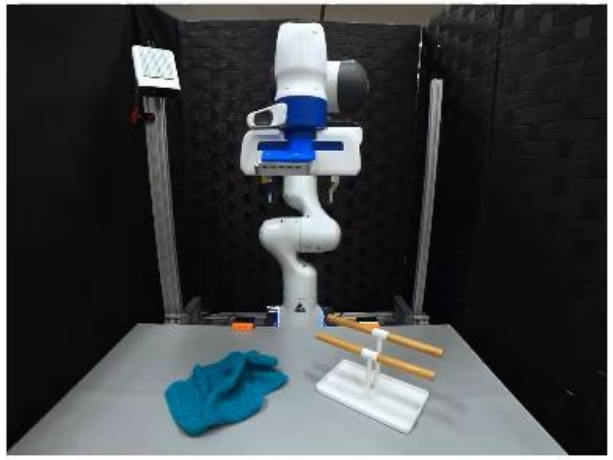
Our prior work on open-vocabulary open-set object manipulation



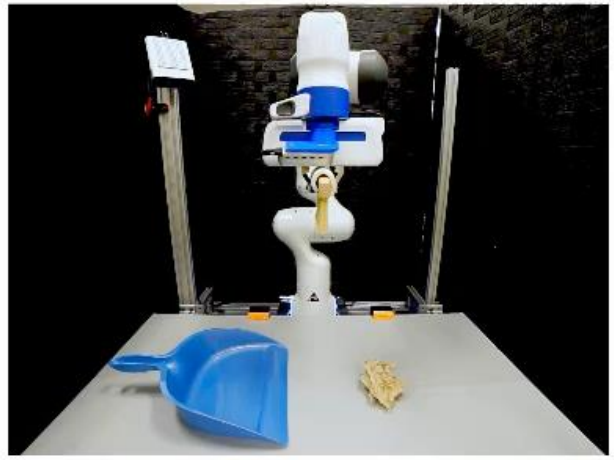
"Turn open vitamin bottle"



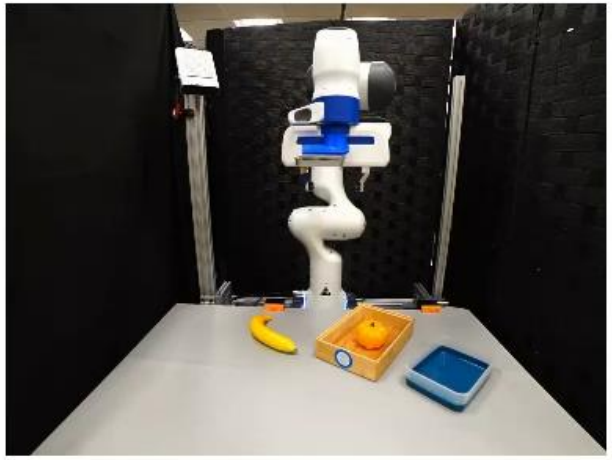
"Measure weight of apple"



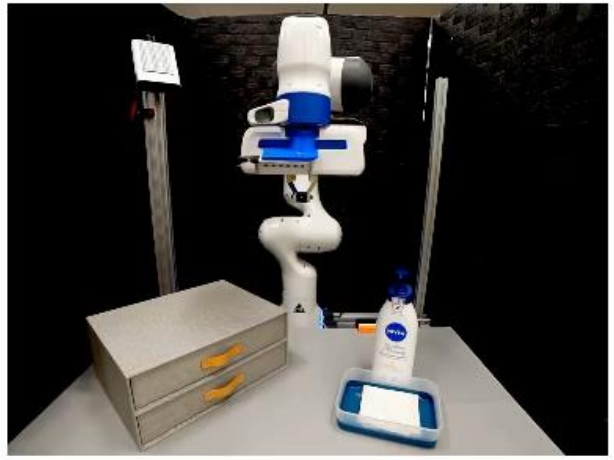
"Hang towel on rack"



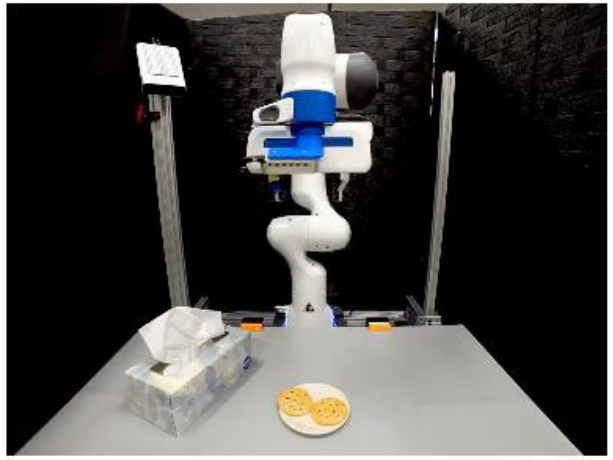
"Sweep trash into dustpan"



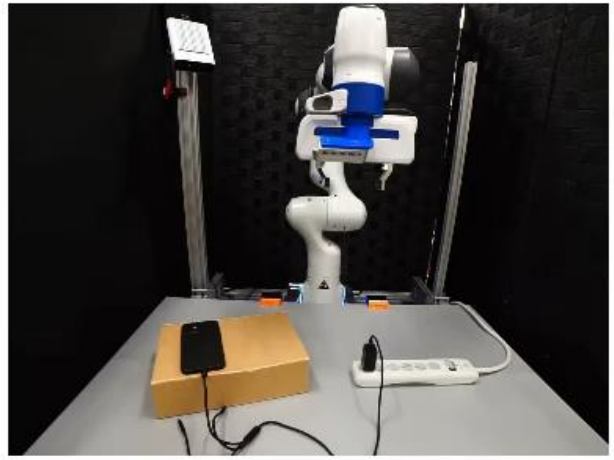
"Sort trash to blue tray"



"Press down moisturizer pump"

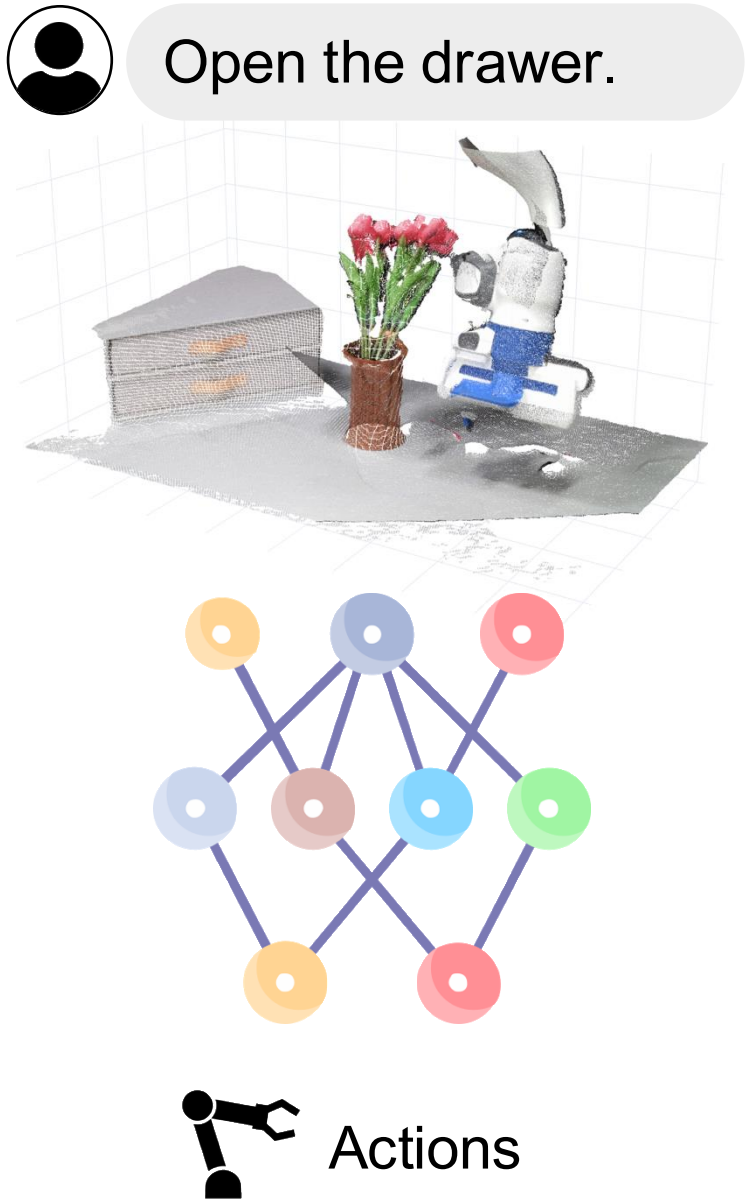


"Take out a napkin"



"Unplug charger for phone"

Wenlong Huang et al., VoxPoser: Composable 3D Value Maps for Robotic Manipulation with Language Models, CoRL-23, **Oral Presentation**

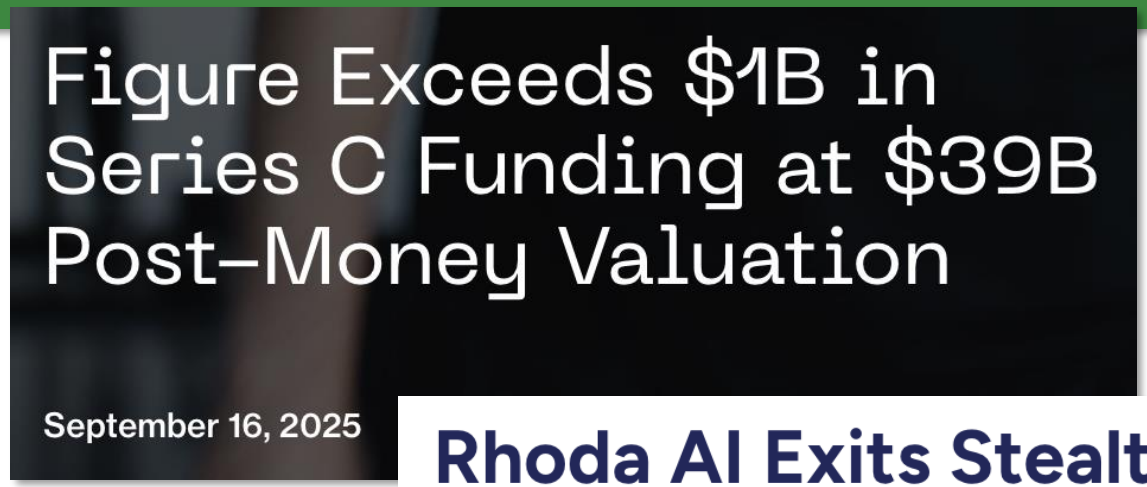


- Foundation Models for Robotics
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- Robot Foundation Models
  - Pretrained on Vision-Language Models
    - RT-X, OpenVLA, Helix, Pi-0.5, Pi*0.6, DYNA-1…
  - Pretrained on World / Video Models
    - WAM, Direct Video-Action Models (DVA)…

Robotics in the news

Physical Intelligence eyes $1B raise at $11B valuation, Founders Fund and Lightspeed in talks

Rivian spin-out Mind Robotics raises $500M for industrial AI-powered robots

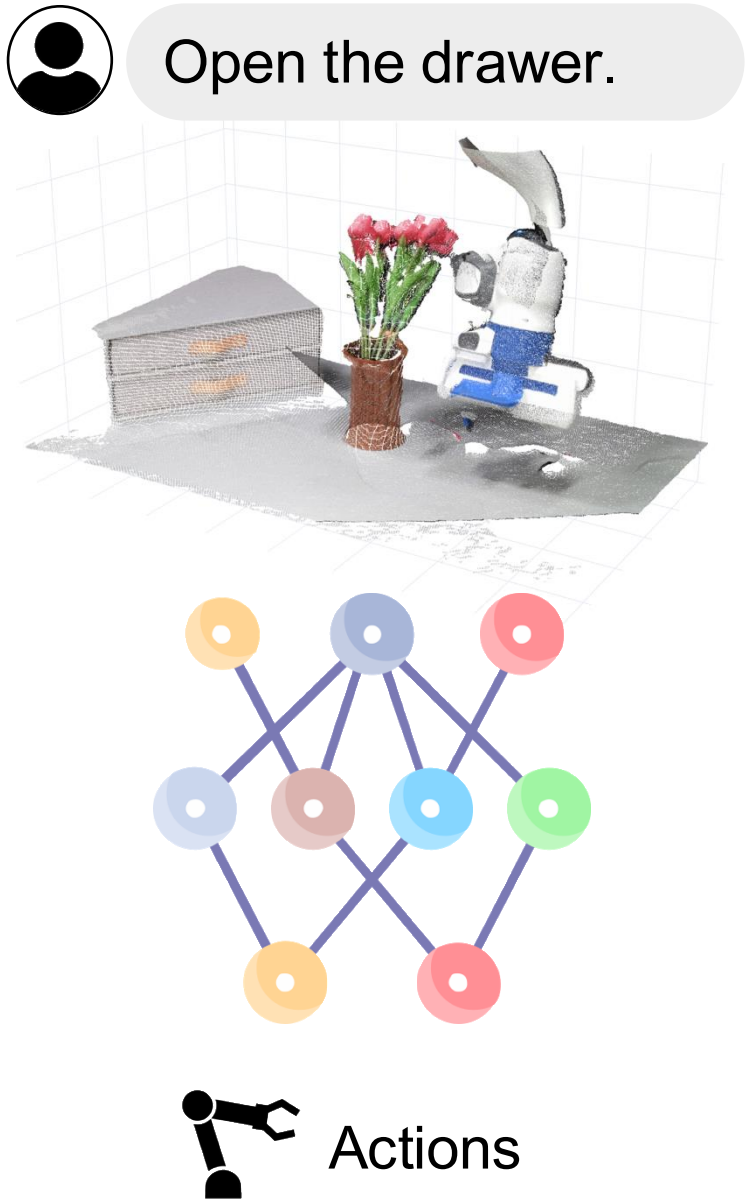




Rhoda AI Exits Stealth with $450 Million Series A to Bring Robots Out of the Lab and Into the Real World

Generalist AI Raises $400M Series B at $2B Valuation

CHRIS WALKER • JUNE 7, 2026 • FUNDING NEWS



- Foundation Models for Robotics
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    - WAM, Direct Video-Action Models (DVA)…

- **Steerability:** What does it take to do a new task?
- **Physical grounding:** Respect real-world physics?
- **Debuggability:** Things went wrong. Now what?
- **Scalability:** Larger-scale? Longer-horizon?

Towards **foundational robotic intelligence**



Tactile sensing plays a critical role in our daily activities

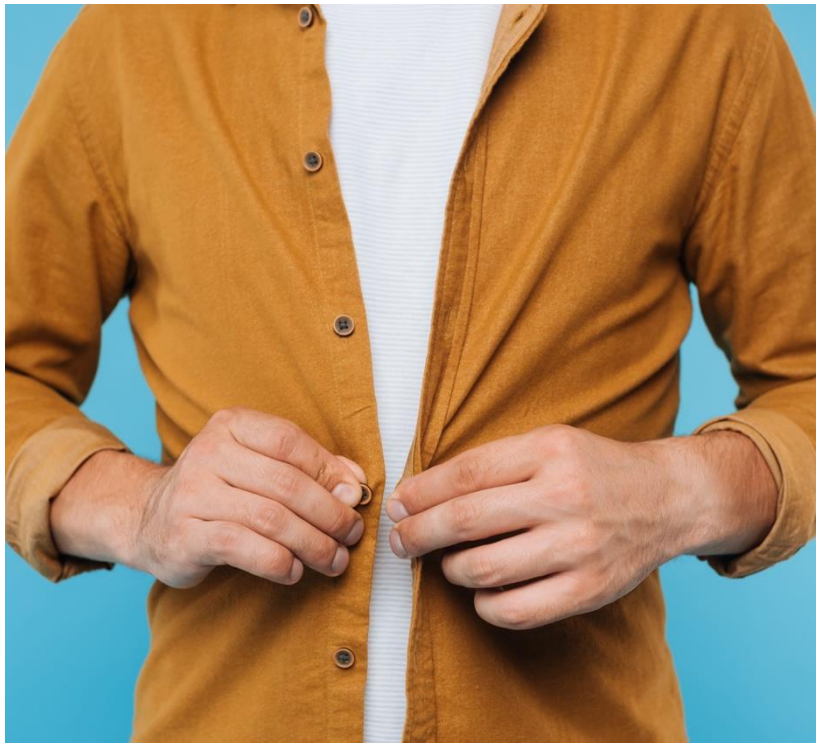
Pick up water bottle



Pull keys from pocket

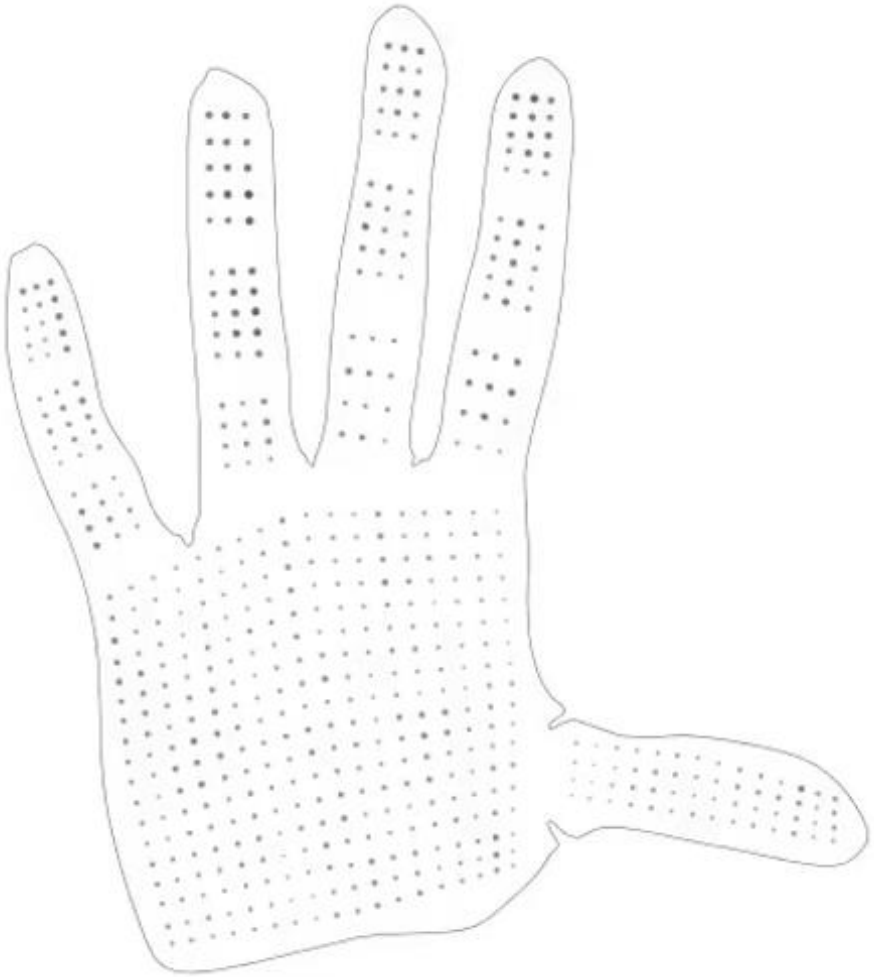


Buttoning a shirt w/o looking



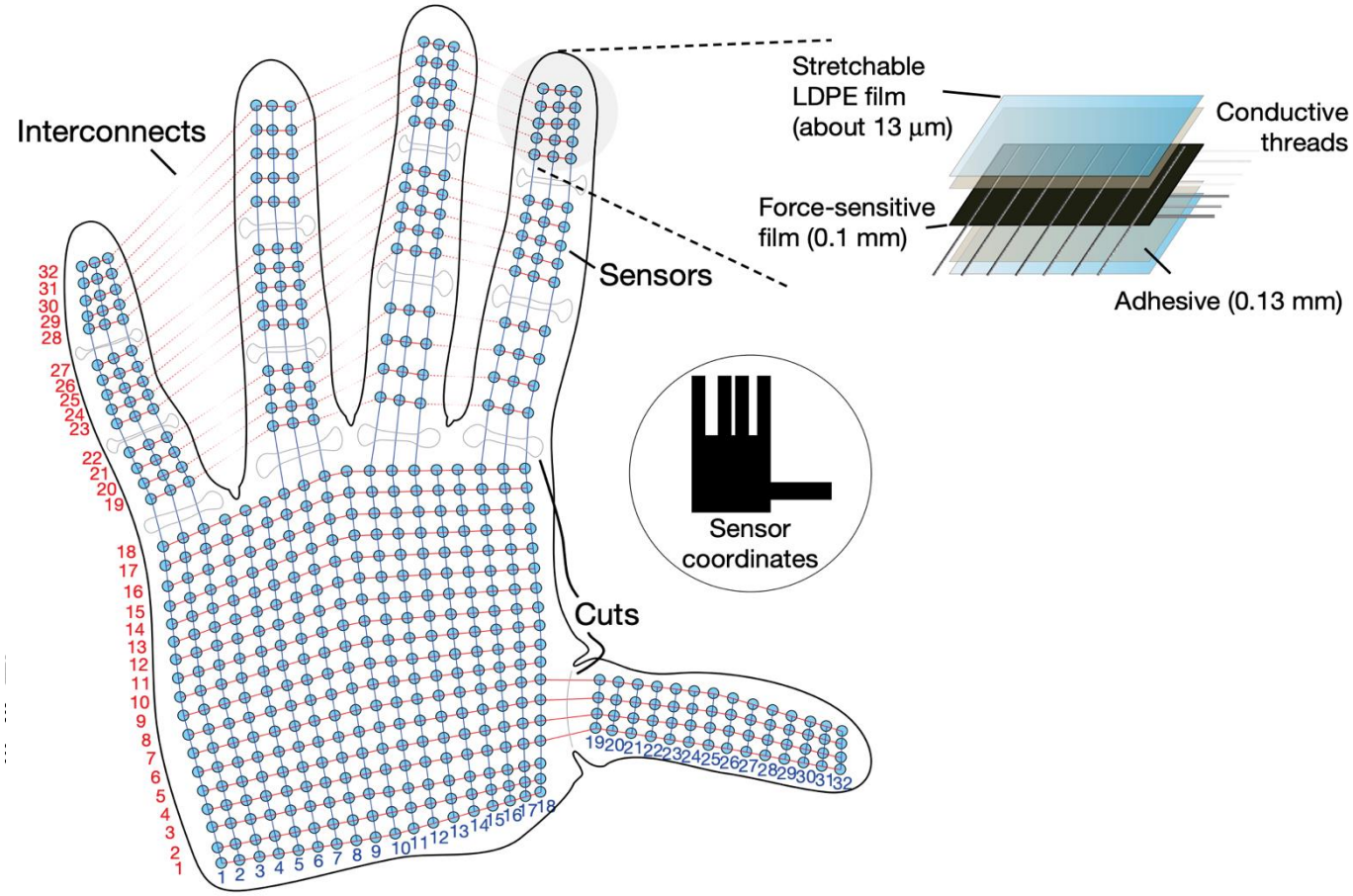
Scalable dense tactile glove

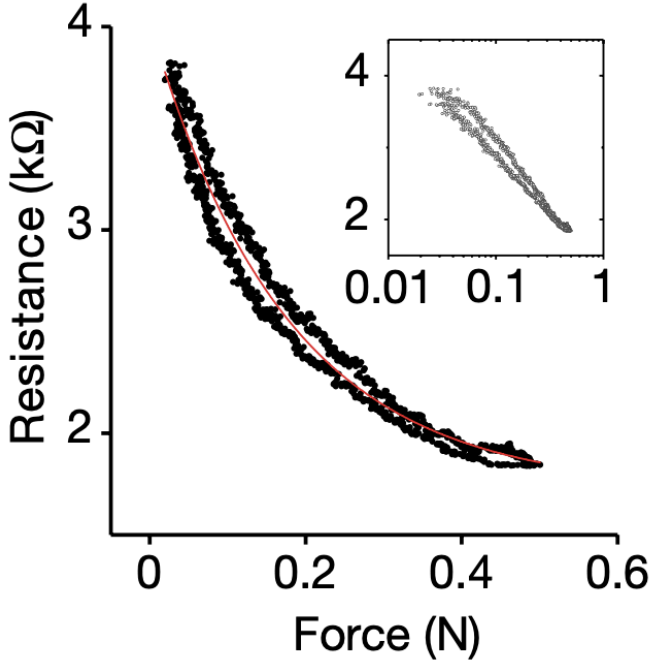




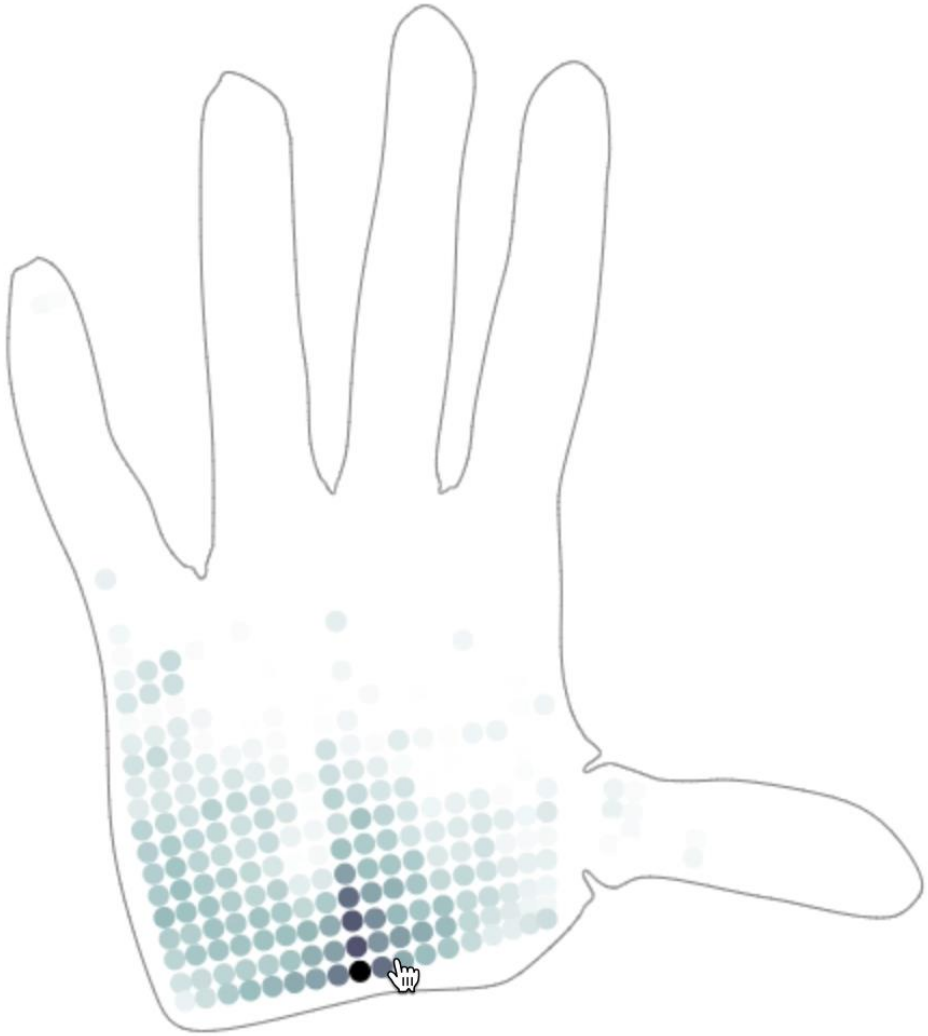
Subramanian Sundaram, Petr Kellnhofer, **Yunzhu Li**, Jun-Yan Zhu, Antonio Torralba, and Wojciech Matusik
Learning the Signatures of the Human Grasp Using a Scalable Tactile Glove, **Nature 2019**

Scalable & flexible dense tactile glove

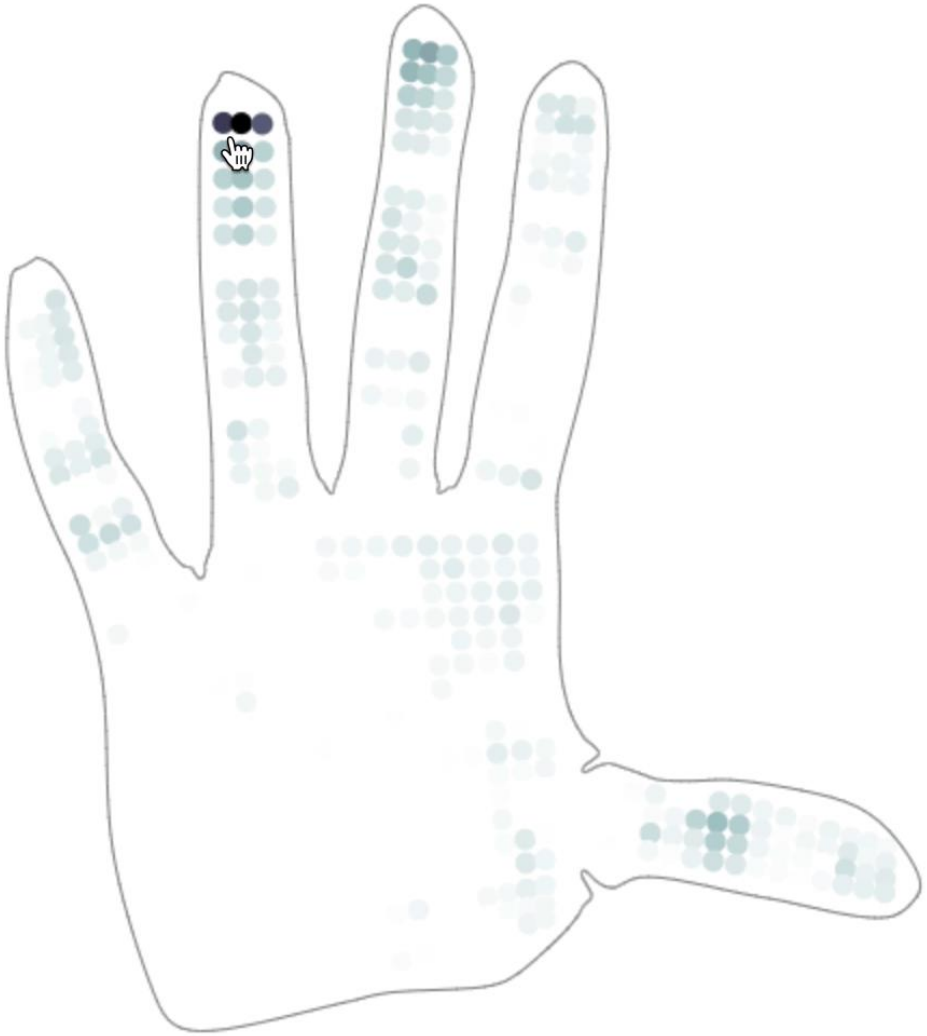




Sensor correlation between different regions of the hand



Palm



Fingertip

Collected by the MIT Museum

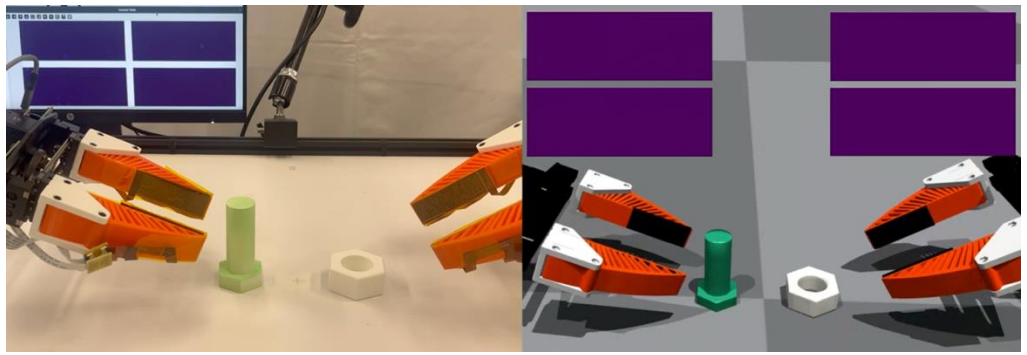


Scalable Tactile Sensing for Robotic Manipulation





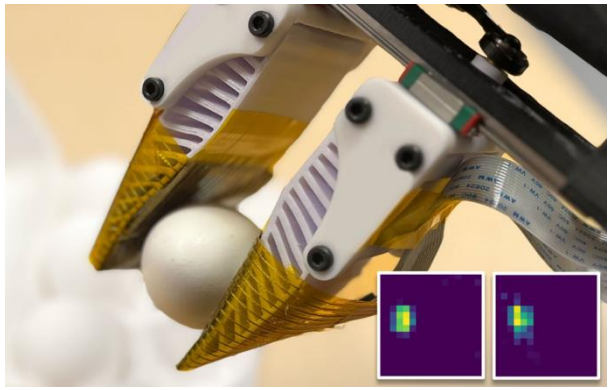
Sundaram et al., Learning the Signatures of the Human Grasp Using a Scalable Tactile Glove, Nature 2019



Huang et al., VT-Refine: Learning Bimanual Assembly with Visuo-Tactile Feedback via Simulation Fine-Tuning, CoRL 2025

Scale up **sim data**

Scale up **real data**

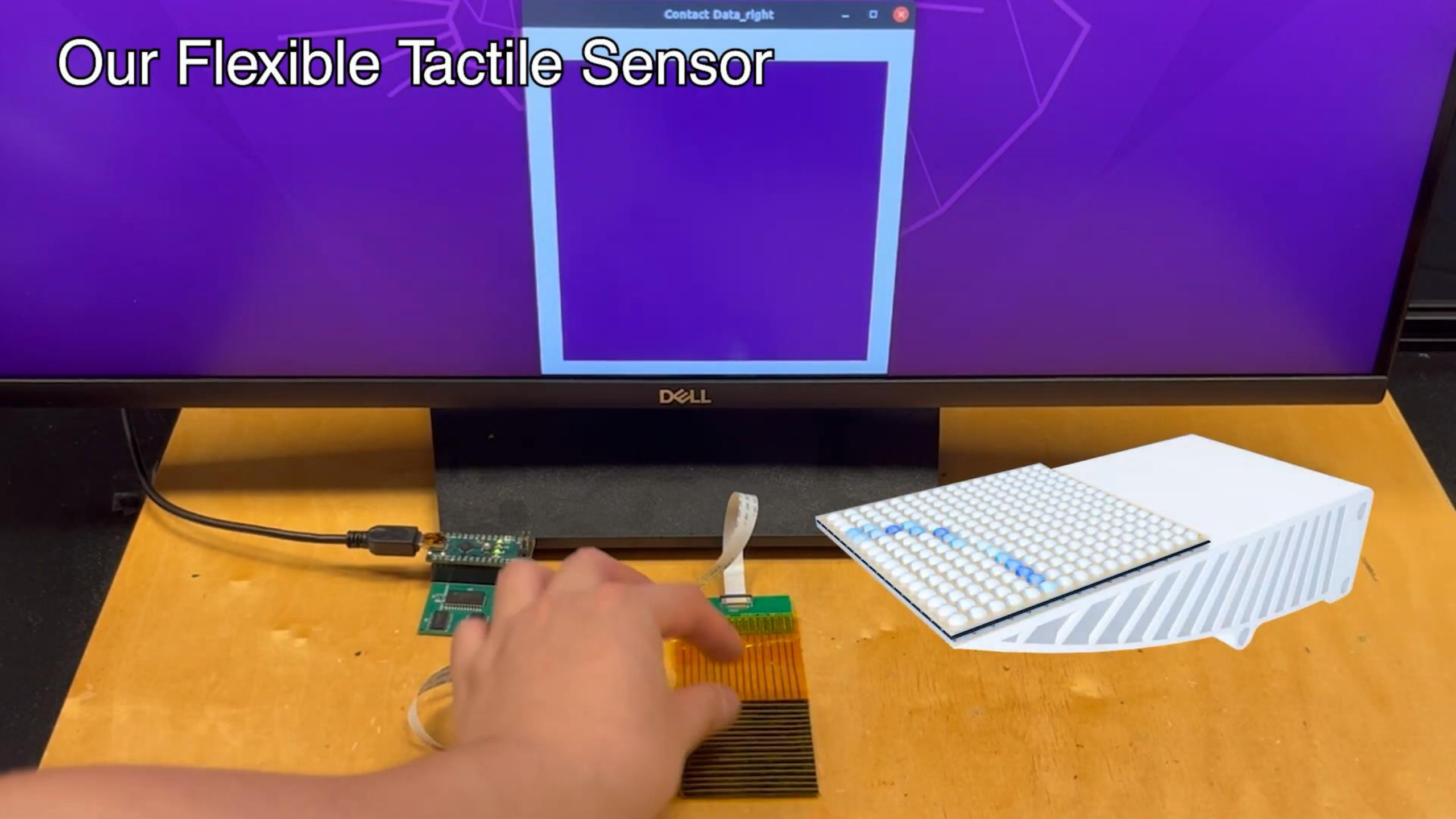


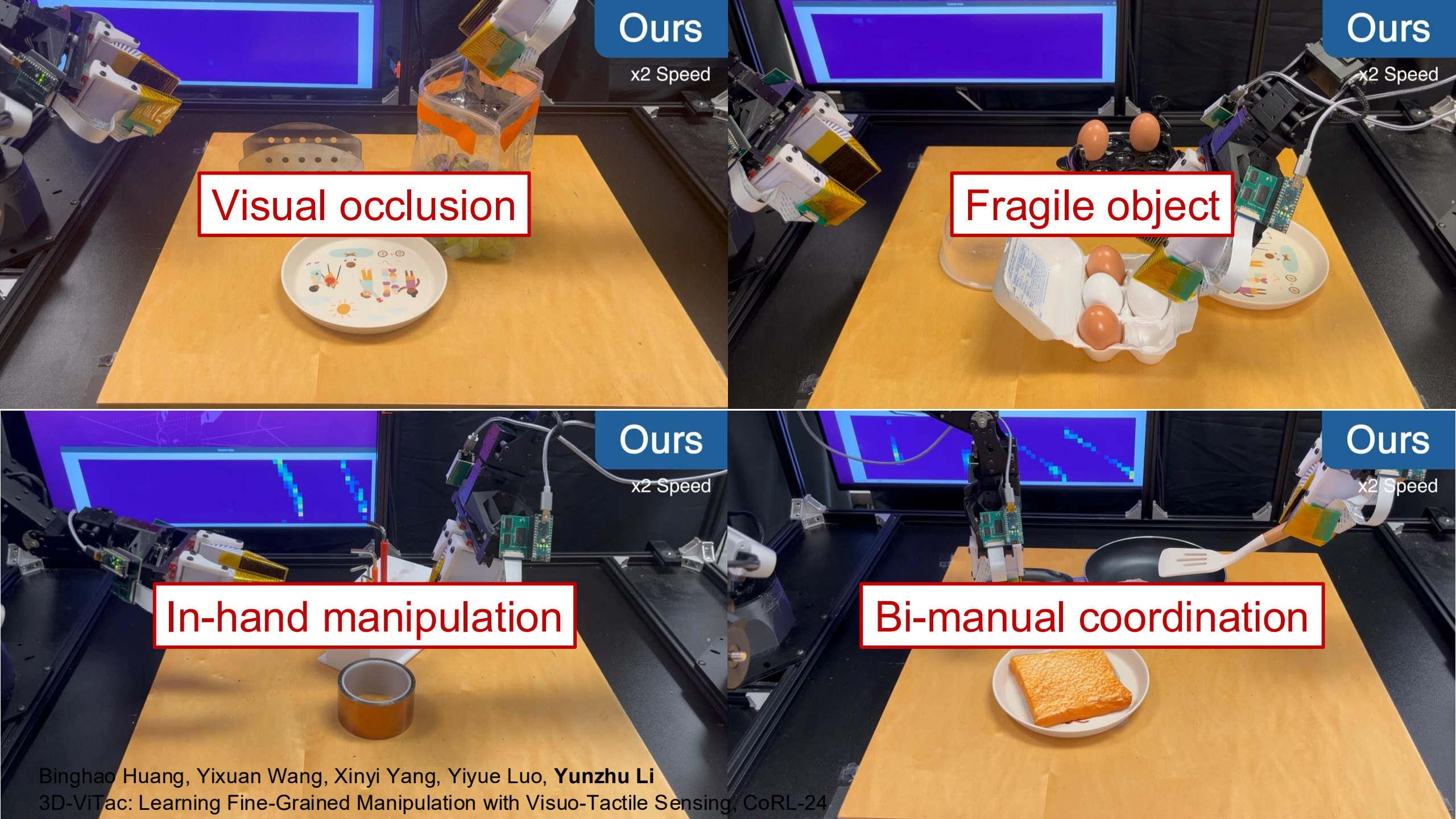
Huang et al., 3D-ViTac: Learning Fine-Grained Manipulation with Visuo-Tactile Sensing, CoRL 2024



Zhu et al., Touch in the Wild: Learning Fine-Grained Manipulation with a Portable Visuo-Tactile Gripper, NeurIPS 2025

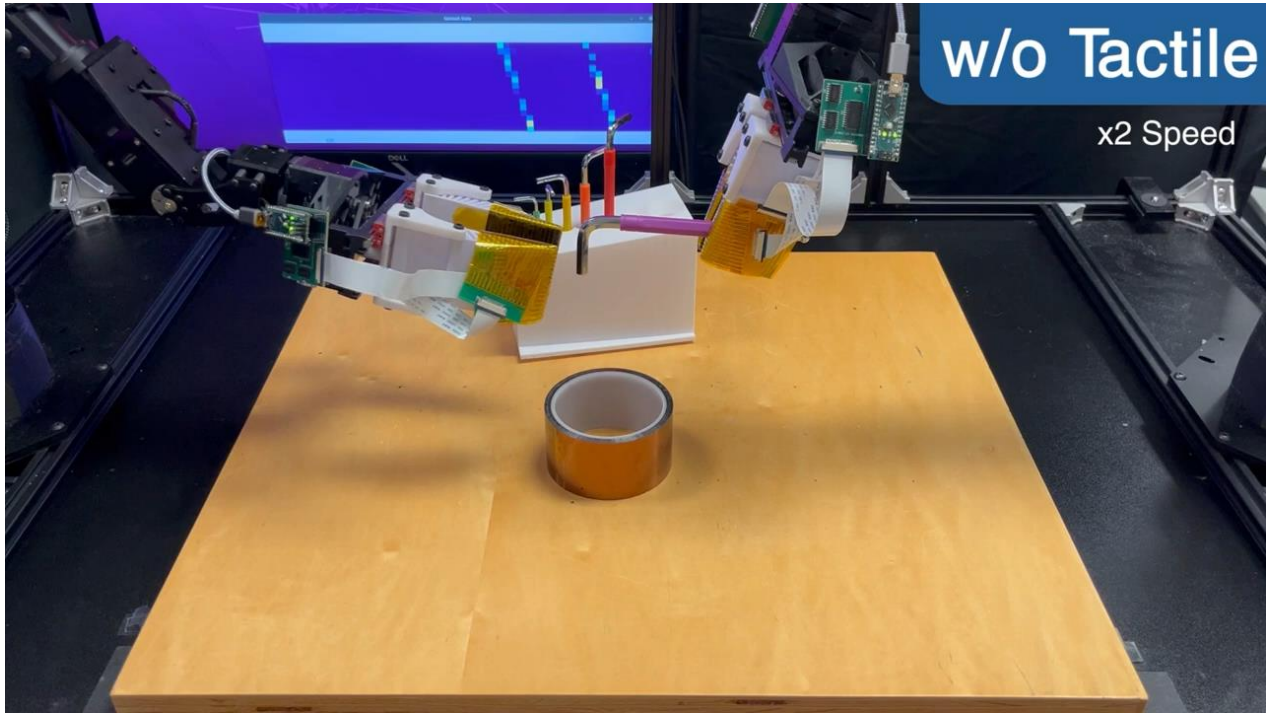
Our Flexible Tactile Sensor

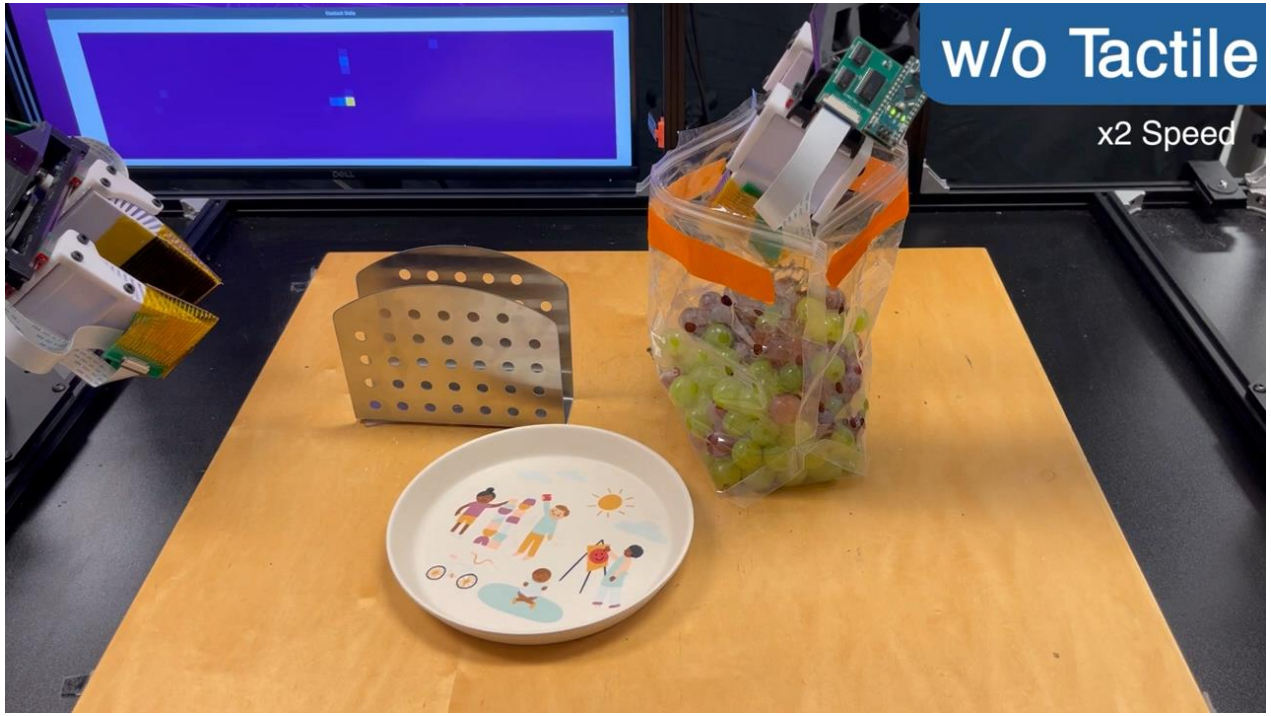




Binghao Huang, Yixuan Wang, Xinyi Yang, Yiyue Luo, **Yunzhu Li**
3D-ViTac: Learning Fine-Grained Manipulation with Visuo-Tactile Sensing, CoRL-24

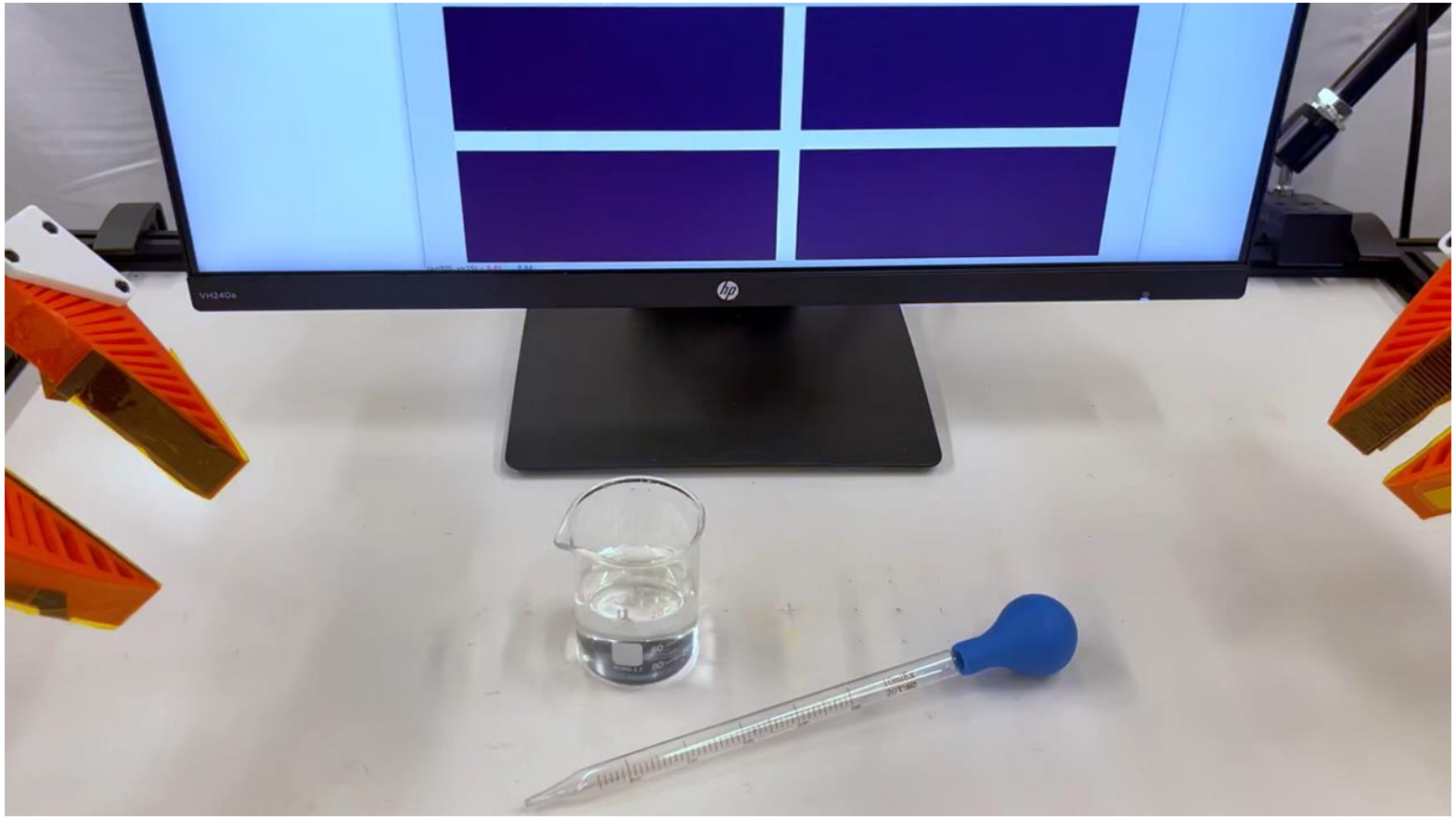
w/o Touch, Vision only

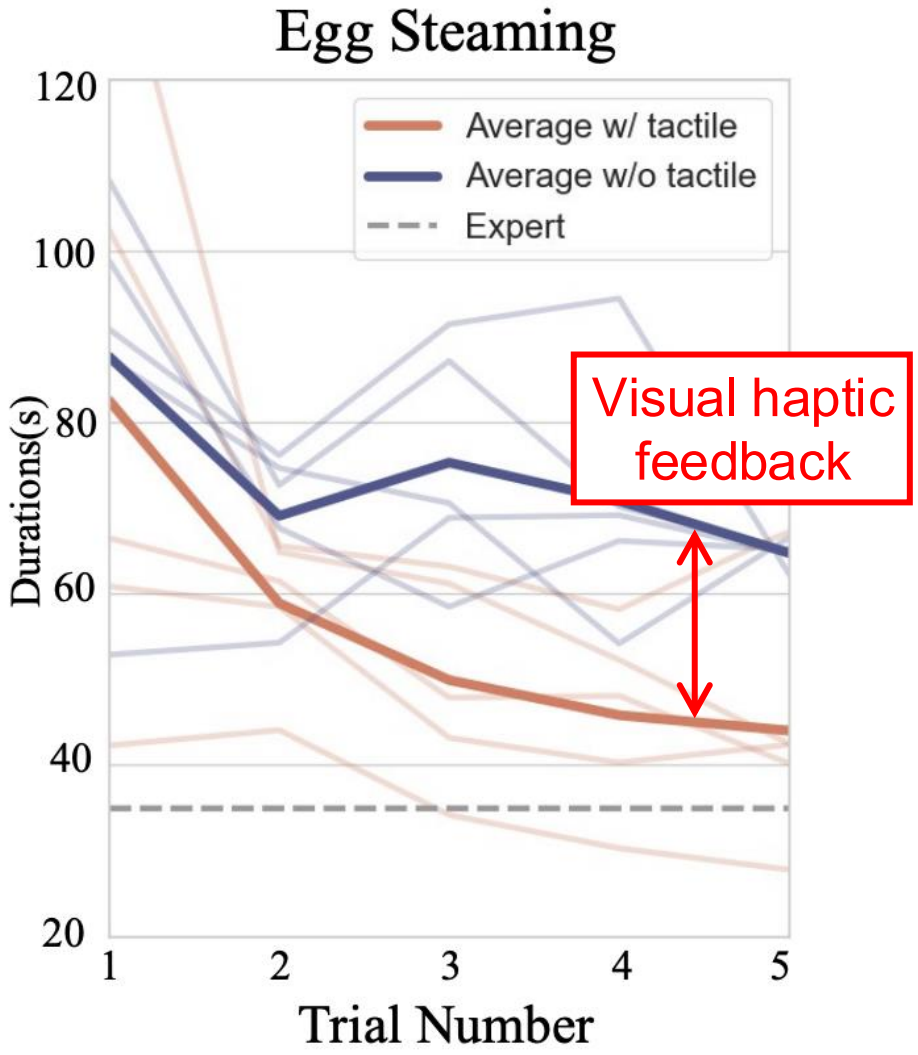




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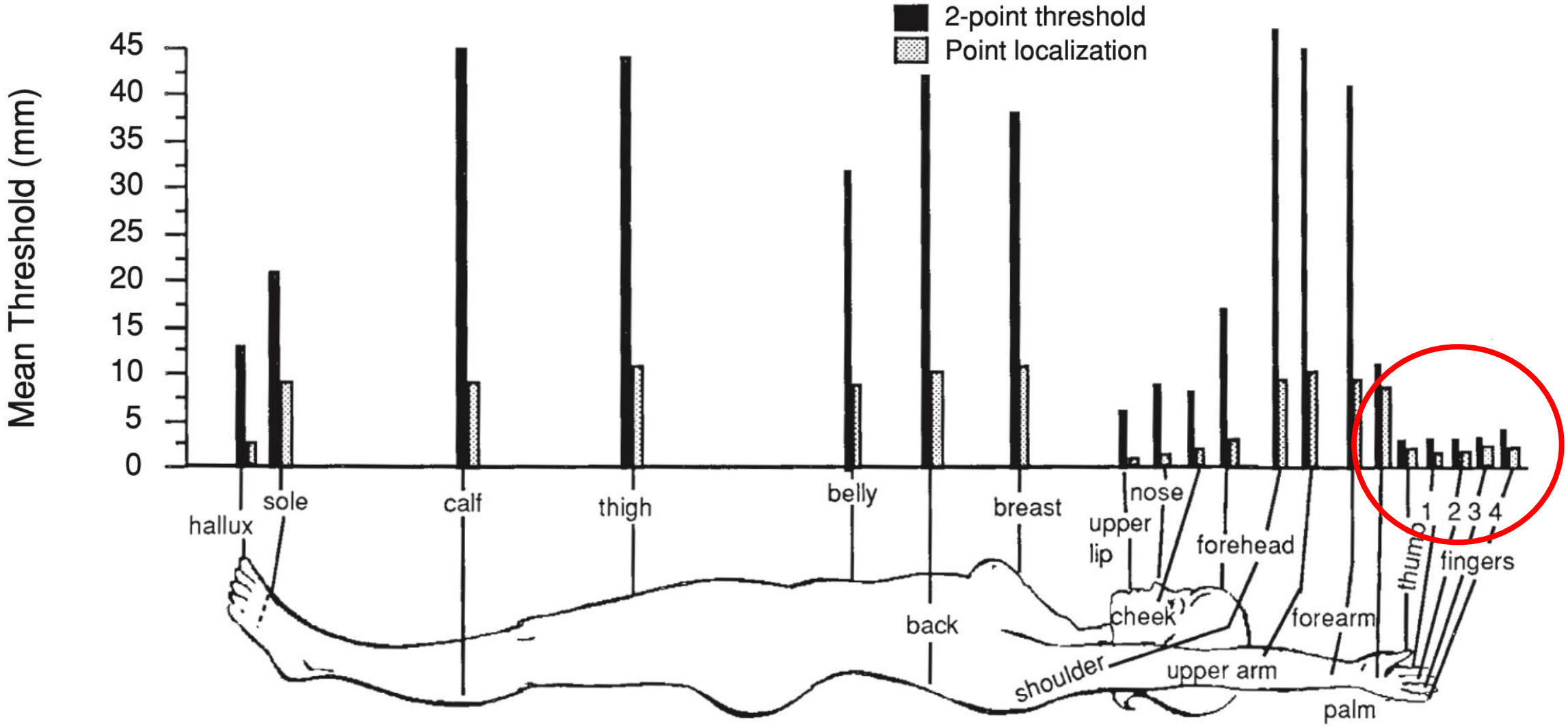
Haptic feedback? Visual haptic feedback!





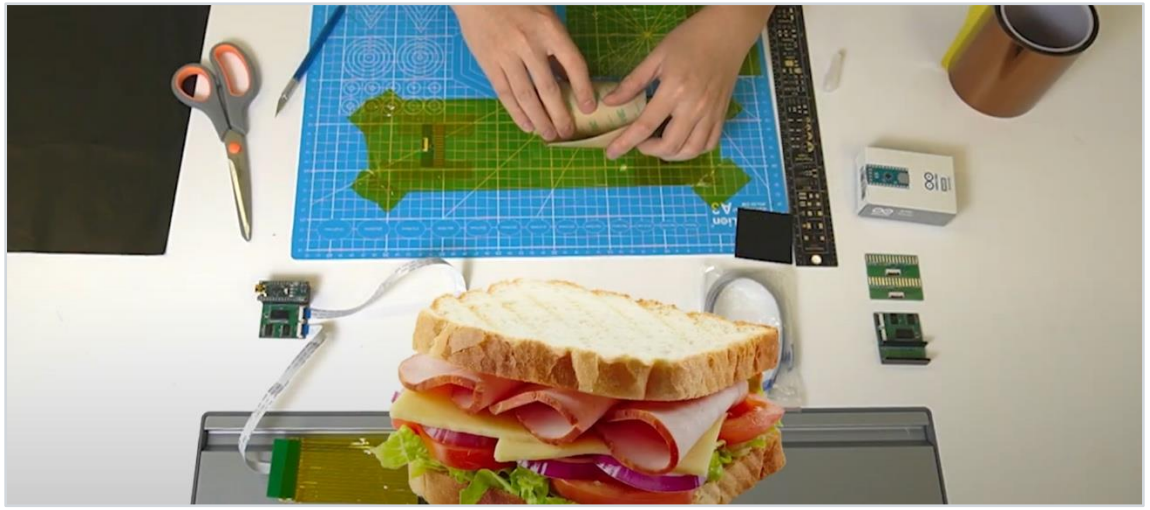
Binghao Huang, Yixuan Wang, Xinyi Yang, Yiyue Luo, **Yunzhu Li**
3D-ViTac: Learning Fine-Grained Manipulation with Visuo-Tactile Sensing, CoRL-24

Sensor resolution? Binary vs continuous?



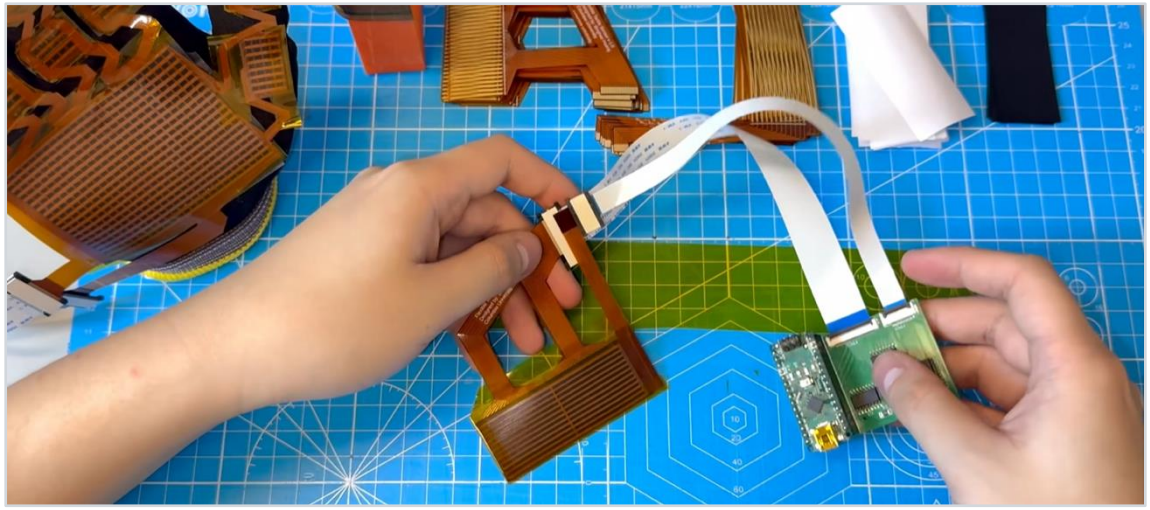
S. J. Lederman & R. L. Klatzky, Haptic perception: A tutorial

Our commitment: make tactile sensing accessible to everyone



FlexiTac-V1

Fabrication	~30 min
Price	~$5/piece
Customize	hand-made
Lead time	in-lab ~30 min



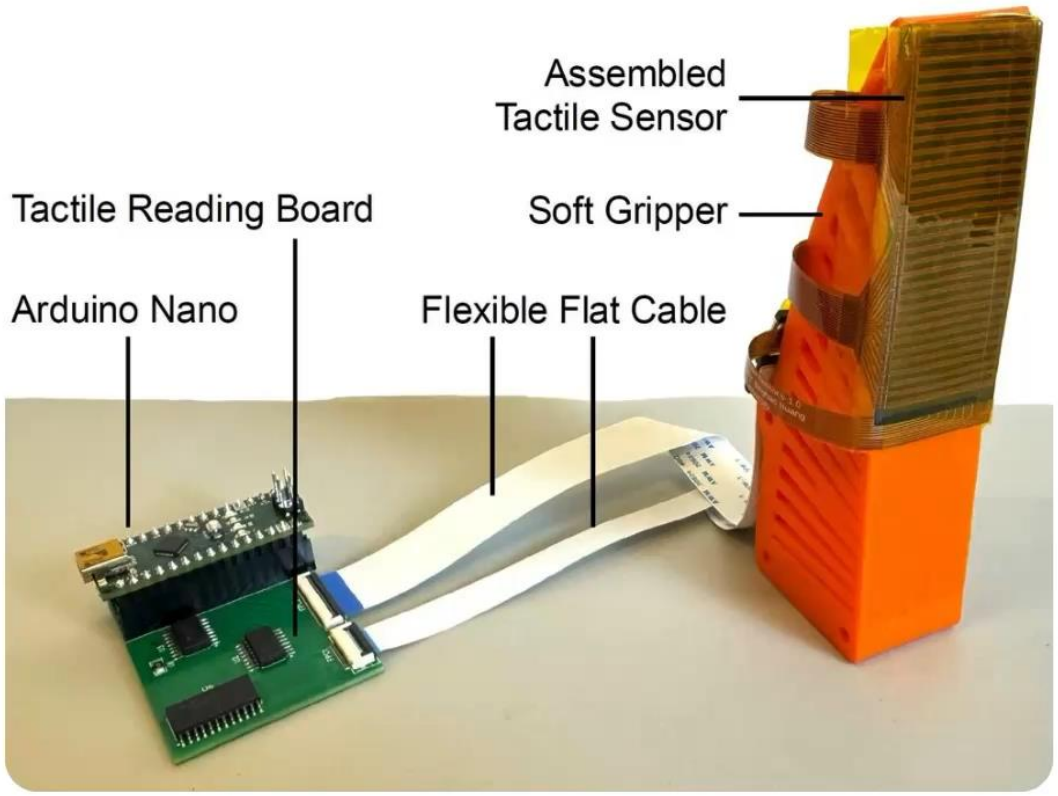
FlexiTac-V2

Fabrication	~3 min
Price	$4.19/piece (20 unit) $1.36/piece (1000 unit)
Customize	from PCB factory
Lead time	manufacture + ship < 1 week

Binghao Huang, **Yunzhu Li**
FlexiTac: An Open-Source, Scalable Tactile Solution for Robotic Systems

We have fully open-sourced FlexiTac

FlexiTac-V2 Hardware

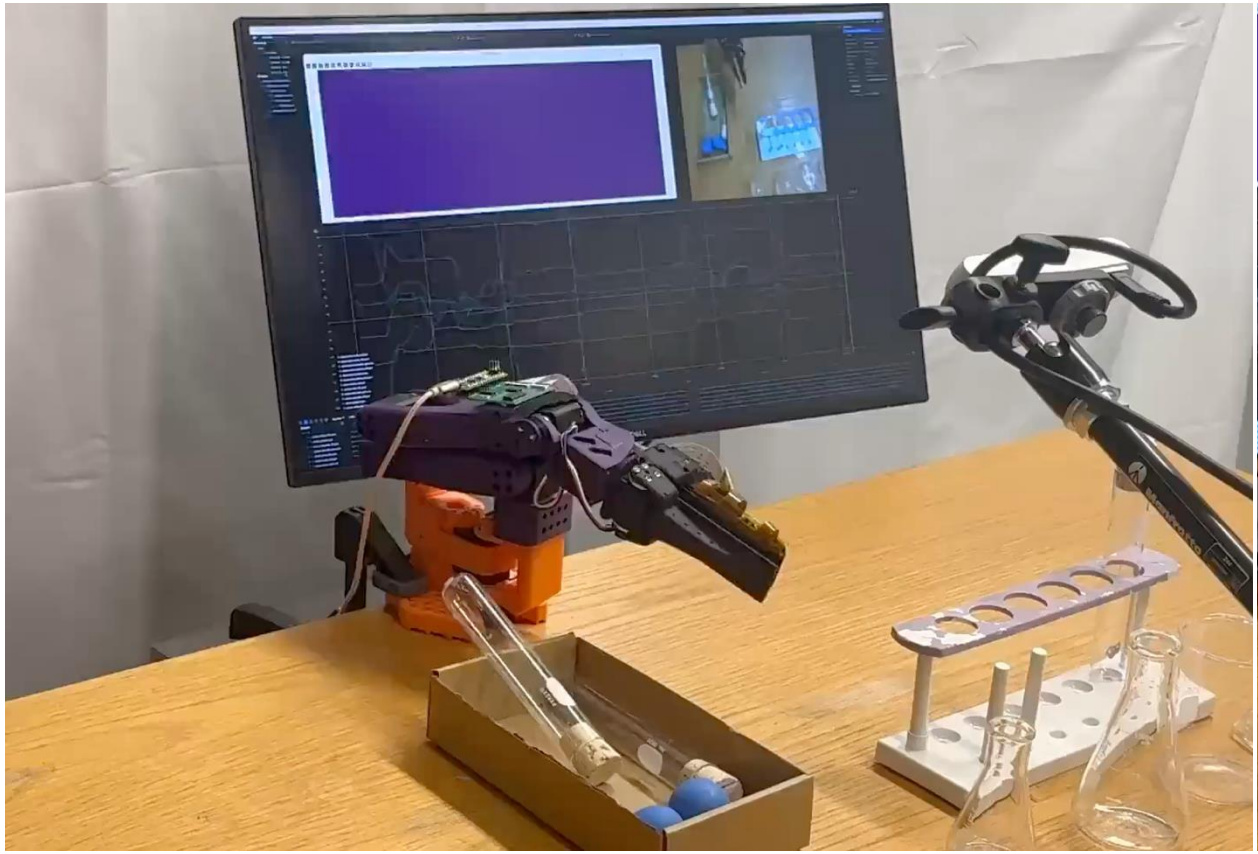


Hardware Tutorial Preview

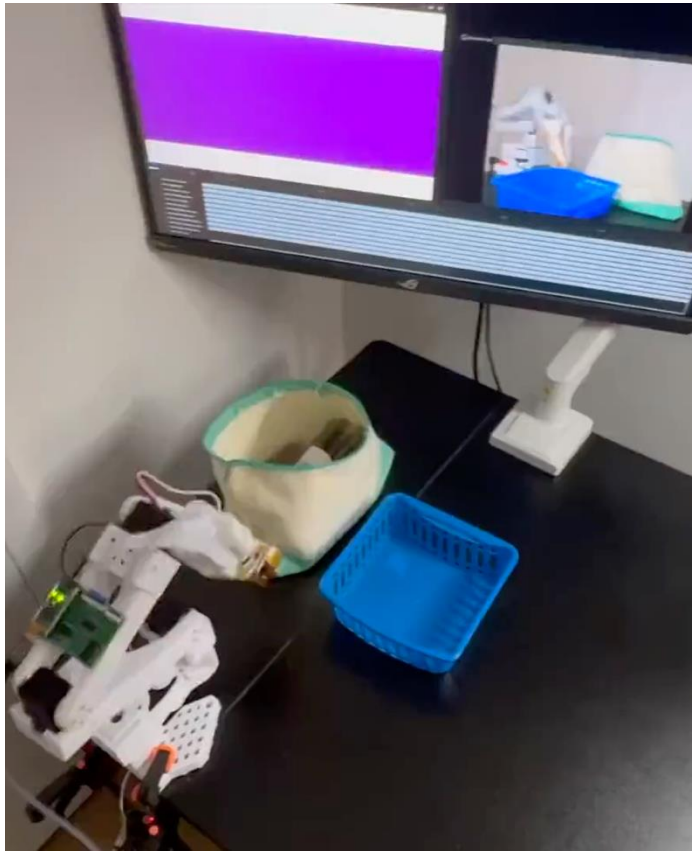


Binghao Huang, **Yunzhu Li**
FlexiTac: An Open-Source, Scalable Tactile Solution for Robotic Systems

… and integrated into the LeRobot ecosystem







Naian Tao, Yifan He*, Wesley Maa*, Binghao Huang, **Yunzhu Li**
LeFlexiTac: Giving Robots a Sense of Touch

A year later: a growing FlexiTac Ecosystem



Zhou et al., “Tactile-Reactive Gripper with an Active Palm for Dexterous Manipulation,” Nature Partner Journals Robotics 2026.



Chen et al., “Multi-Modal Manipulation via Multi-Modal Policy Consensus,” ICRA 2026.



Li et al., “Deform360: A Massive Multi-view Visuotactile Dataset for Deformable World Models,” ECCV 2026.



Zhou et al., “Learning Tactile-Aware Quadrupedal Loco-Manipulation Policies,” IROS 2026.



Ma et al., “TacCoRL: Integrating Tactile Feedback into VLA via Simulation,” 2026.



Lin & Zhao, “ART-Glove: Articulated Tactile Glove for Contact-Grounded Dexterous Interaction Capture,” 2026.

Hardware **Perception** **Data**

Acknowledgement

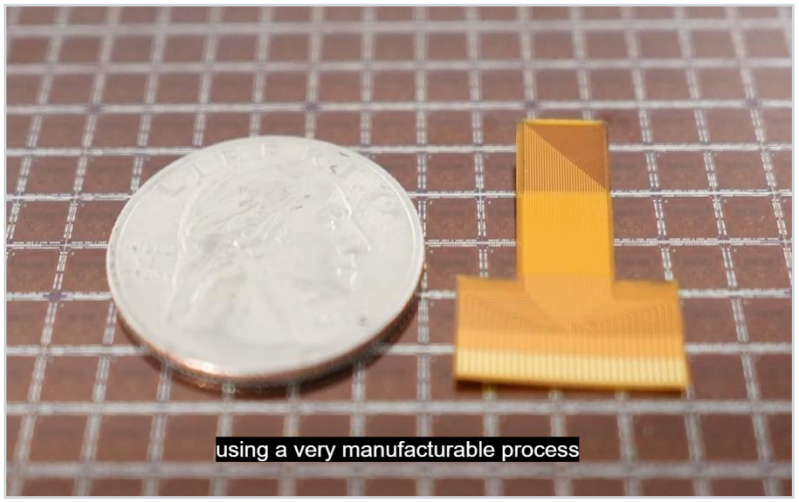
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COLUMBIA UNIVERSITY IN THE CITY OF NEW YORK
ANALOG DEVICES
SAMSUNG RESEARCH AMERICA
NVIDIA

A year later: industry adoption

NEW PRODUCT
Tactile sensing at 5x the resolution of a human finger

ANALOG DEVICES

ADVANCED PRODUCTION
Built on semiconductor fabrication technology









[1] Redefining Tactile Sensing. https://www.analog.com/en/resources/media-center/videos/6379738445112.html
[2] THE TOUCH THAT WILL TRANSFORM ROBOTICS. https://www.analog.com/en/signals/articles/the-touch-that-will-transform-robotics.html

A year later: a growing FlexiTac Ecosystem

Awesome FlexiTac FlexiTac Homepage

Join Discord

Categories
Filter Research

- All Works (32)
- Hardware Design (16)
- Glove/Hand (7)
- Multi-Modal Learning (15)
- Simulation (4)
- Locomanipulation (3)
- Industry (2)
- Dataset (2)

AWESOME LIST

Awesome FlexiTac

The growing ecosystem of research building on FlexiTac, together with the prior works.

Maintained by FlexiTac Team

Have a relevant work? Open a PR or issue to add it.

Papers built on FlexiTac have been published at **CoRL**, **NeurIPS**, **ICRA**, **IROS**, **ECCV** and **npj Robotics**.

All Works | Hardware Design | Glove/Hand | Simulation | Multi-Modal Learning | Locomanipulation | Industry | Dataset



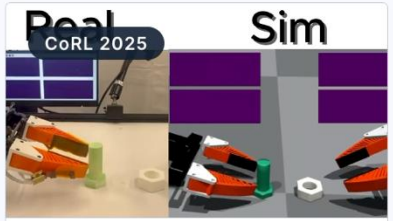
• OUR OPEN-SOURCE SENSOR

FlexiTac: An Open-Source, Scalable Tactile Solution for Robotic Systems



• BUILT WITH FLEXITAC

3D-ViTac: Learning Fine-Grained Manipulation with Visuo-Tactile Sensing



• BUILT WITH FLEXITAC

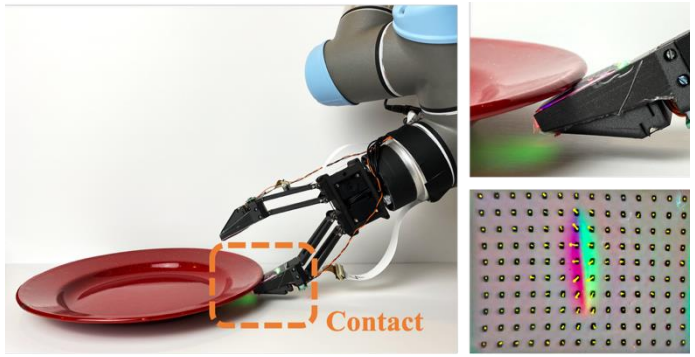
VT-Refine: Learning Bimanual Assembly with Visuo-Tactile Feedback via Simulation Fine-

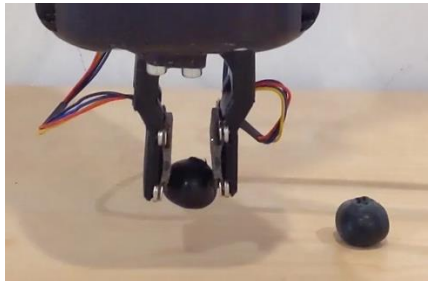


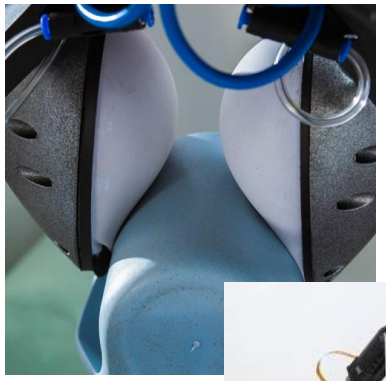
• BUILT WITH FLEXITAC

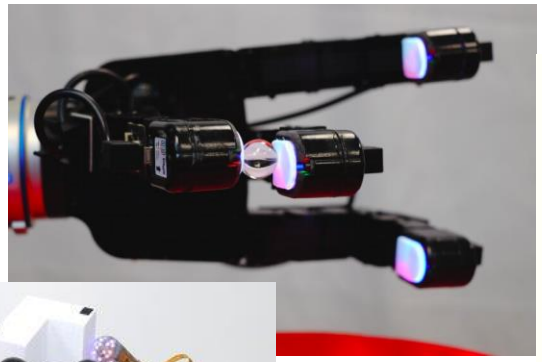
Touch in the Wild: Learning Fine-Grained Manipulation with a Portable Visuo-Tactile

Comparison with other tactile sensors

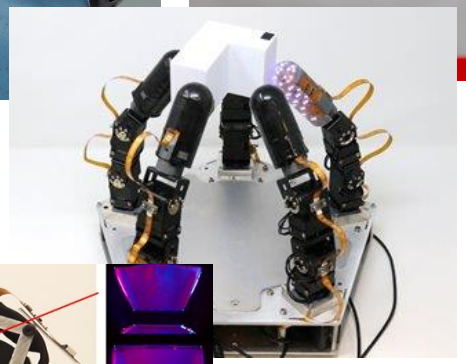


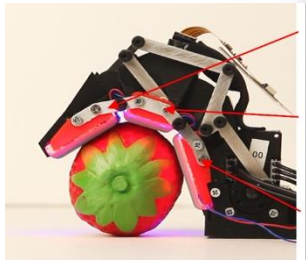


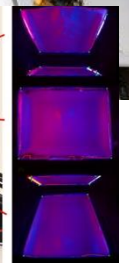


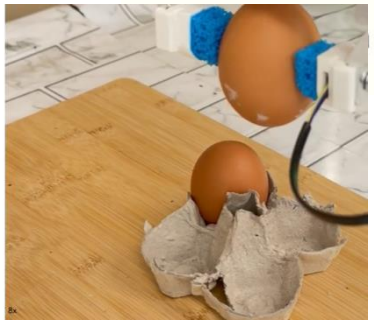






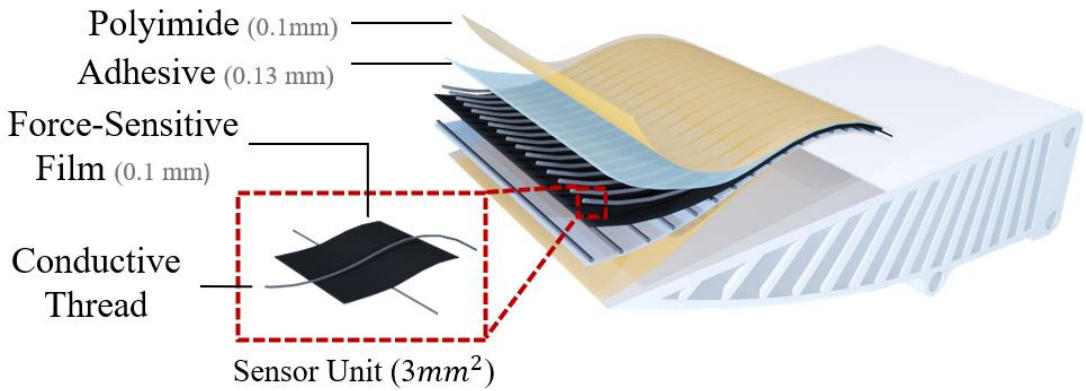


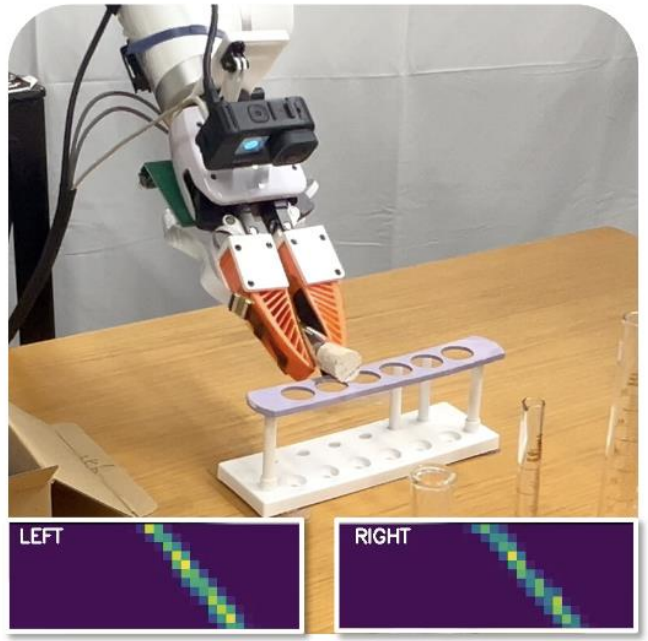




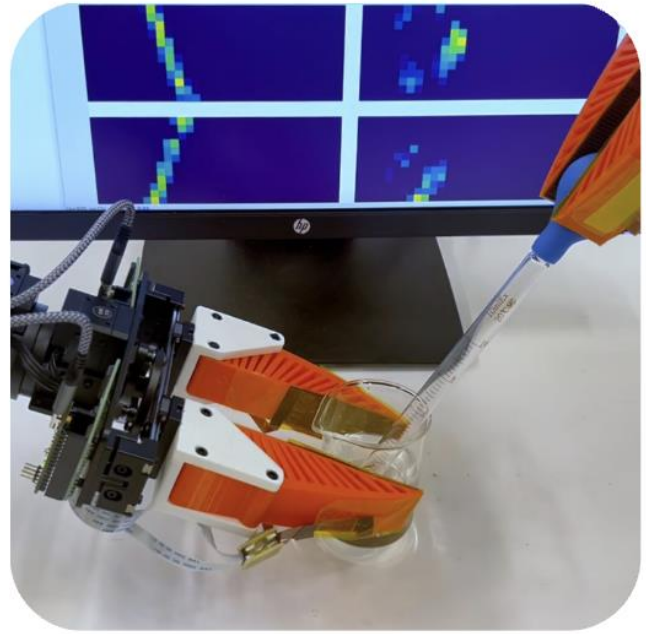
Benefit of our sensor

- Scalable
- Flexible
- Customizable





(i) Xarm with Transparent Tube



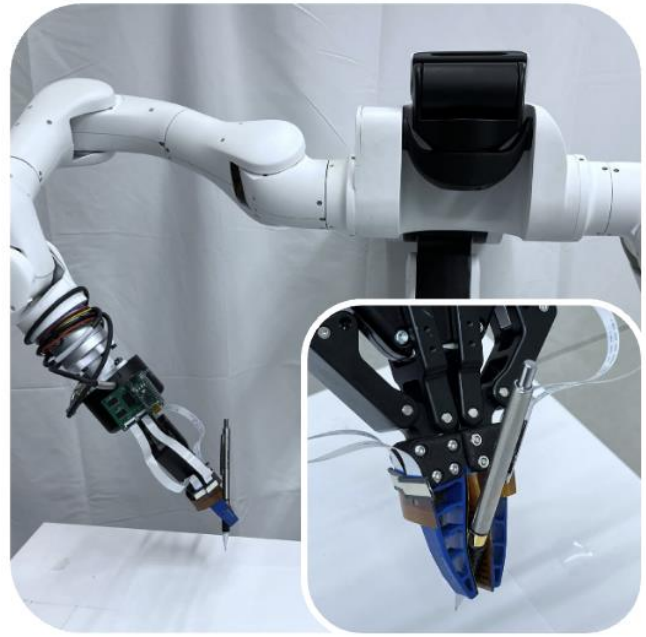
(ii) Aloha with Beaker and Dripper



(iii) LeRobot SO-ARM with Hex Key



(iv) Franka with Tactile Skin

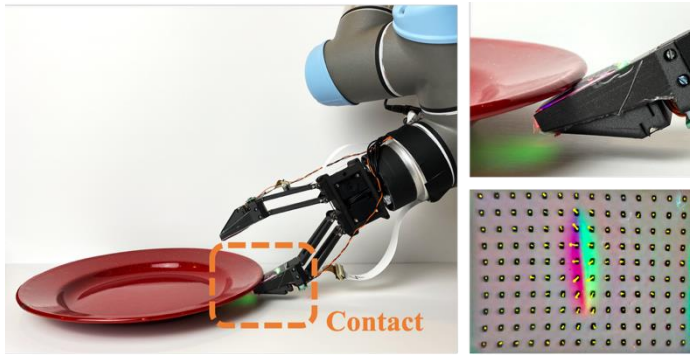


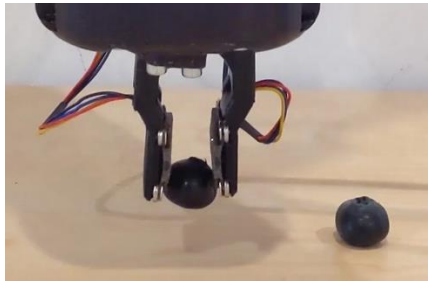
(v) Mobile Bimanual Robot with Pen

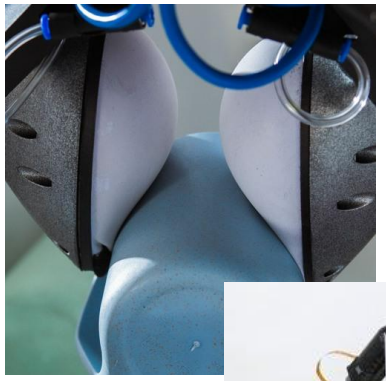


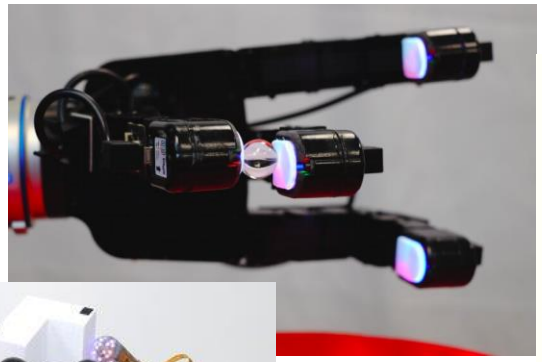
(vi) Tactile Glove

Comparison with other tactile sensors

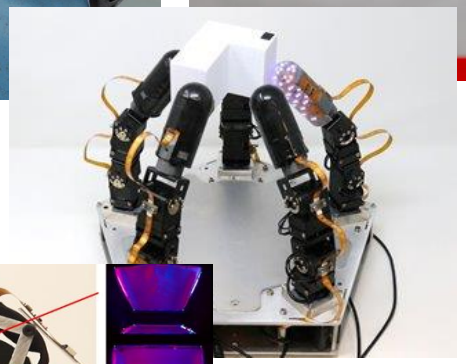


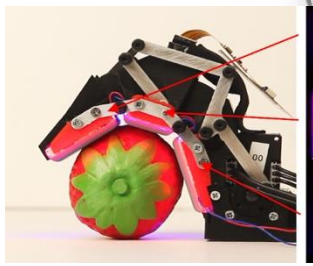


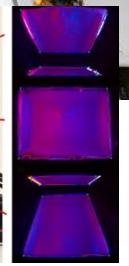


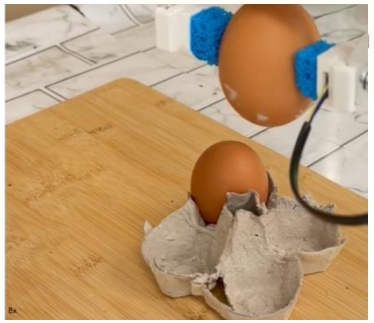






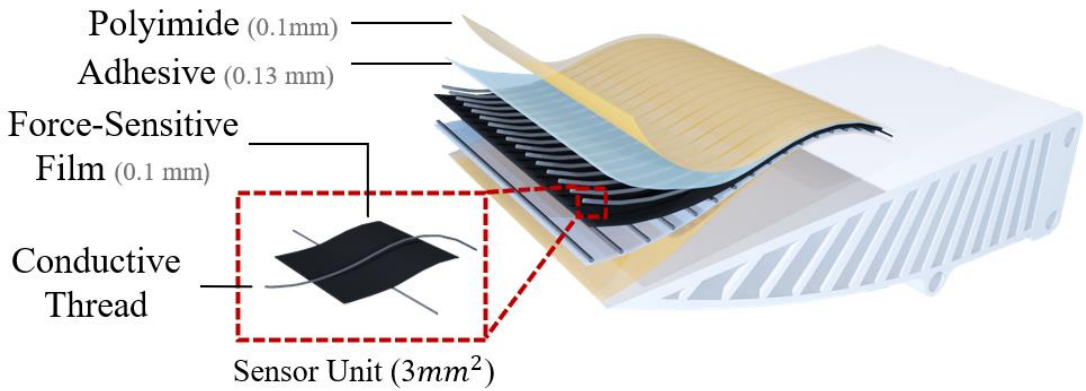






Benefit of our sensor

- Scalable
- Flexible
- Customizable
- Robustness / Consistency
- Easier to simulate

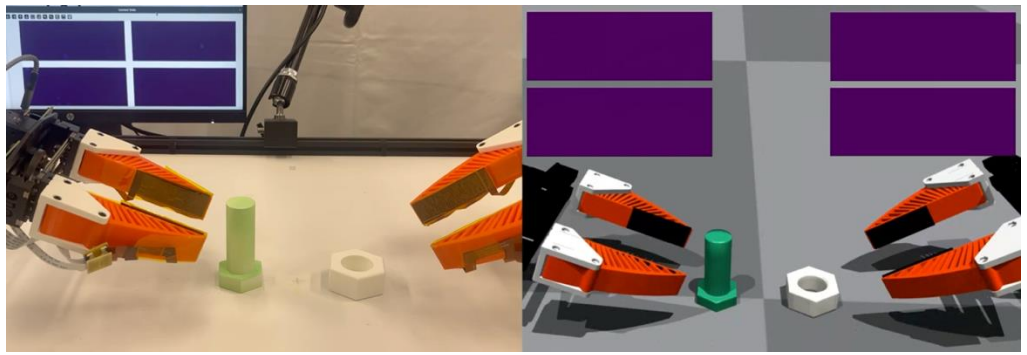


Scalable Tactile Sensing for Robotic Manipulation





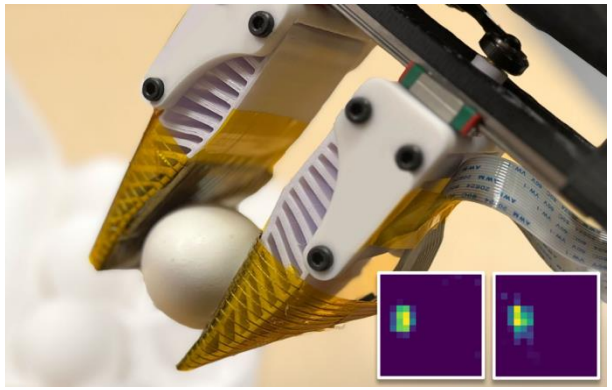
Sundaram et al., Learning the Signatures of the Human Grasp Using a Scalable Tactile Glove, Nature 2019



Huang et al., VT-Refine: Learning Bimanual Assembly with Visuo-Tactile Feedback via Simulation Fine-Tuning, CoRL 2025

Scale up **sim data**

Scale up **real data**

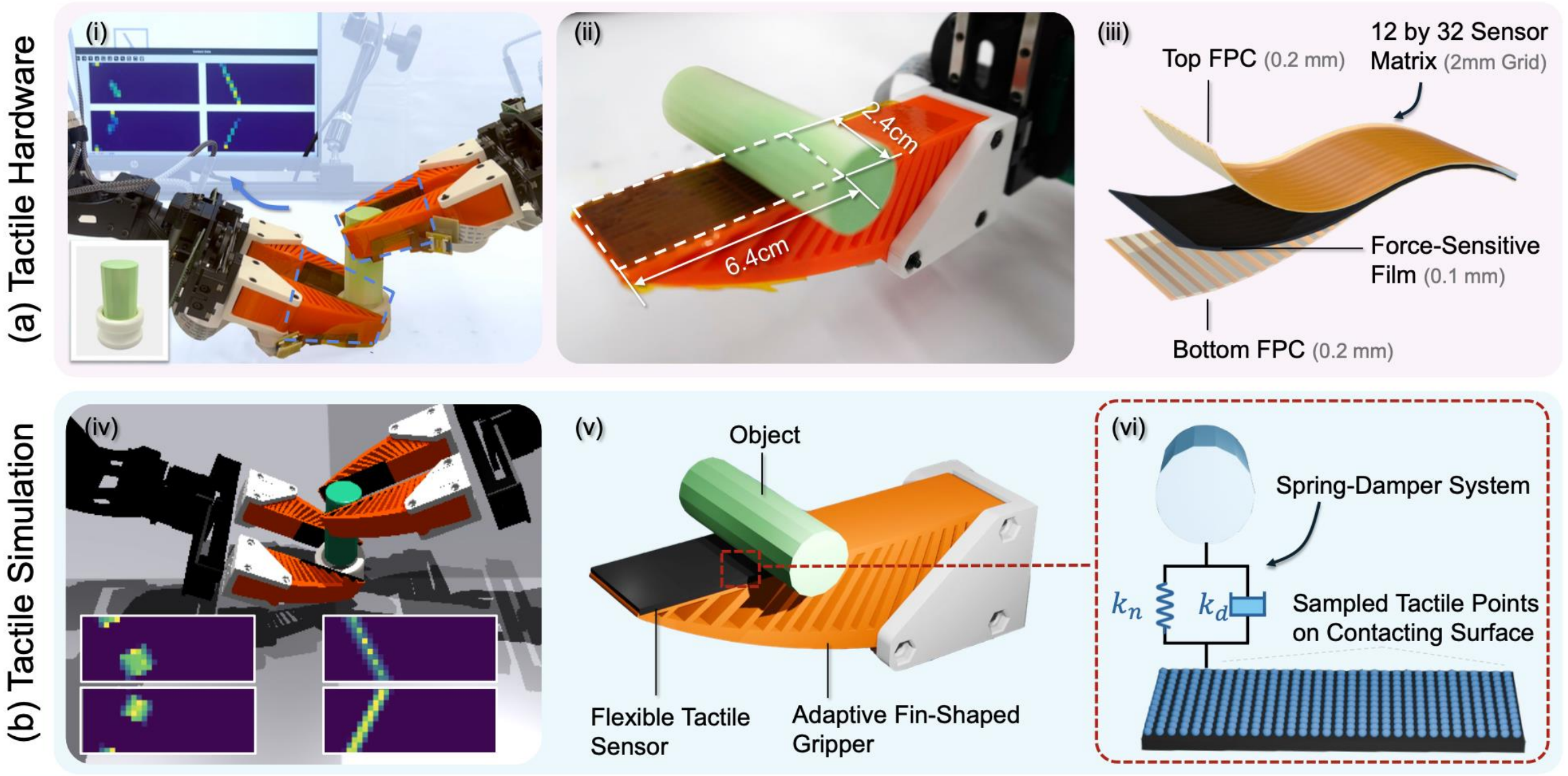


Huang et al., 3D-ViTac: Learning Fine-Grained Manipulation with Visuo-Tactile Sensing, CoRL 2024



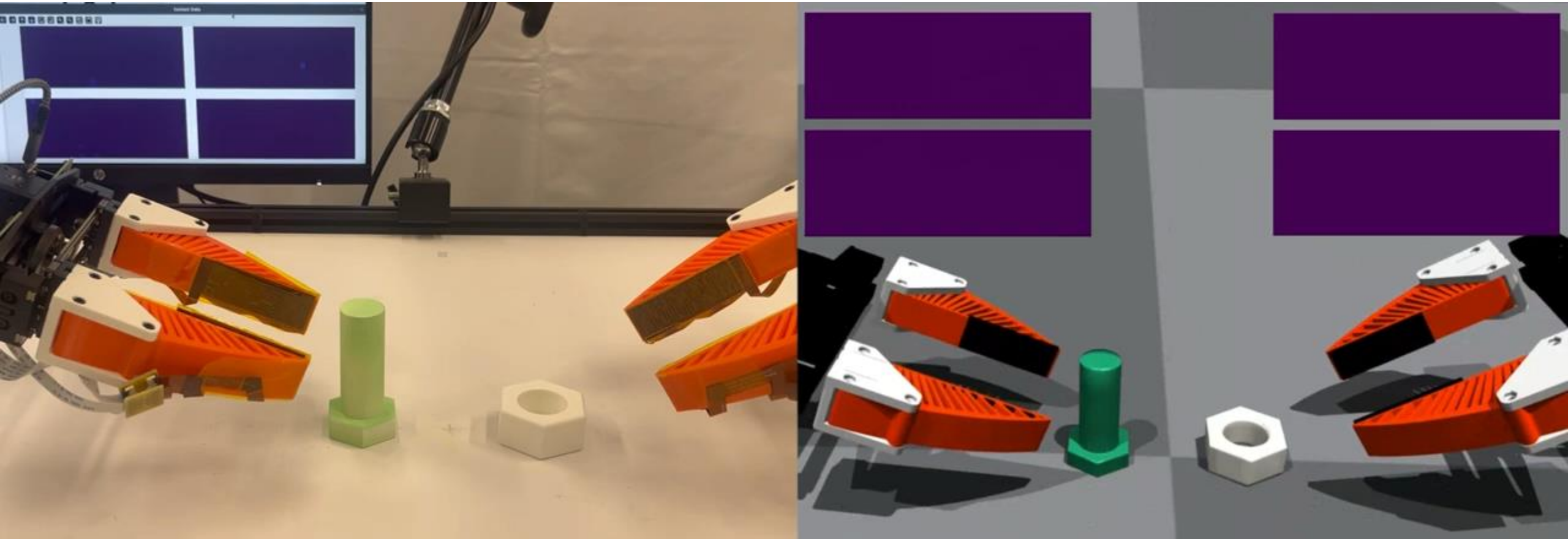
Zhu et al., Touch in the Wild: Learning Fine-Grained Manipulation with a Portable Visuo-Tactile Gripper, NeurIPS 2025

Tactile simulation in Isaac using a simple spring-damper system



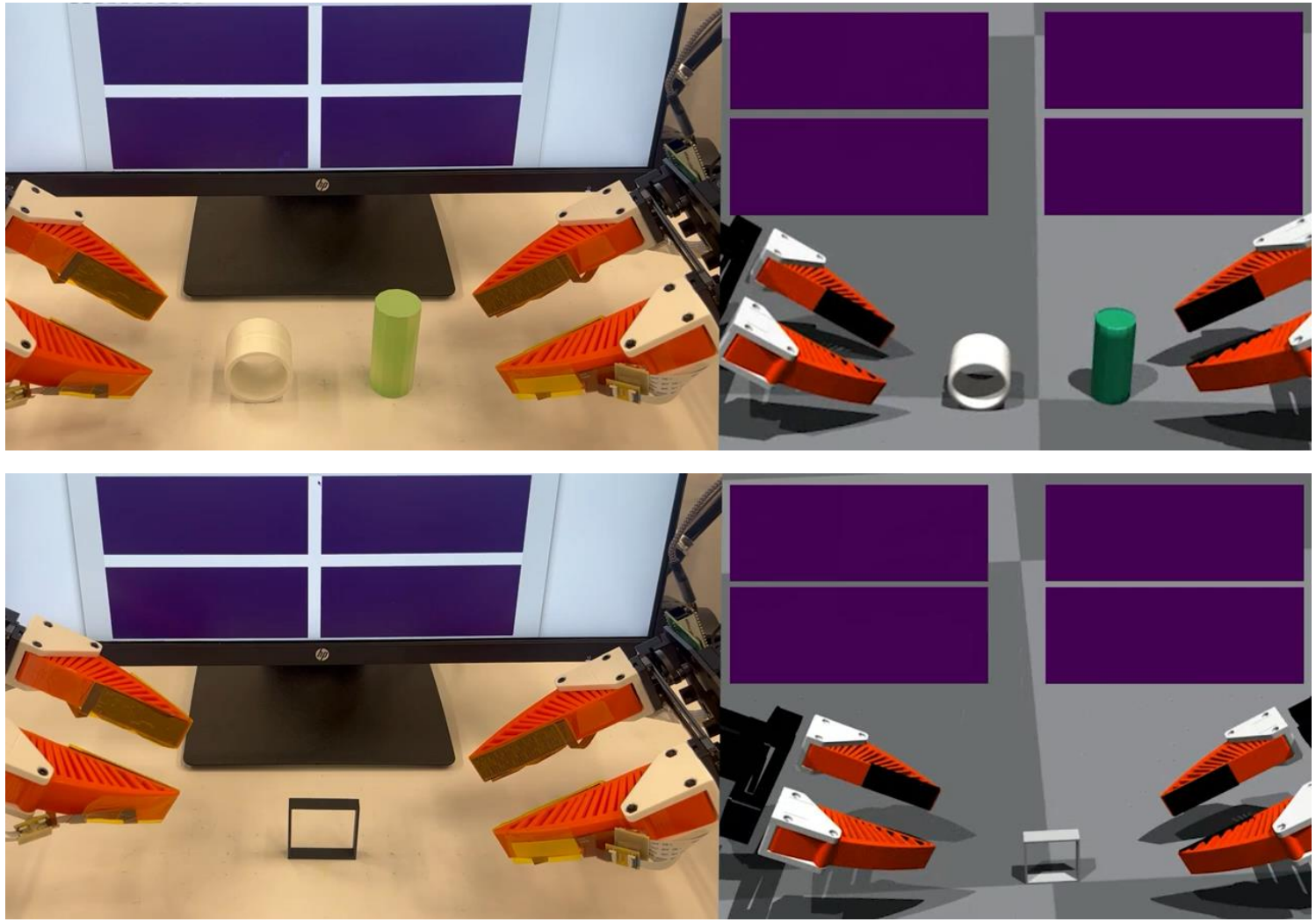
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CoRL-25, **Best Paper Award** at IROS 2025 Workshop on Advanced Haptic Tech for Healthcare Robots

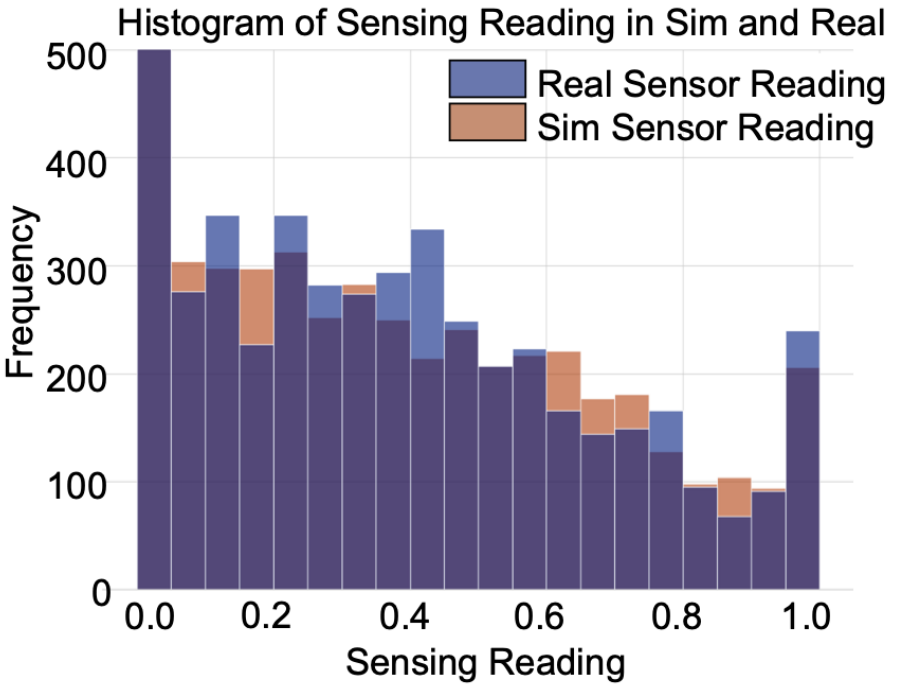
Tactile simulation in Isaac Gym using a simple spring-damper system



Huang et al., VT-Refine: Learning Bimanual Assembly with Visuo-Tactile Feedback via Simulation Fine-Tuning
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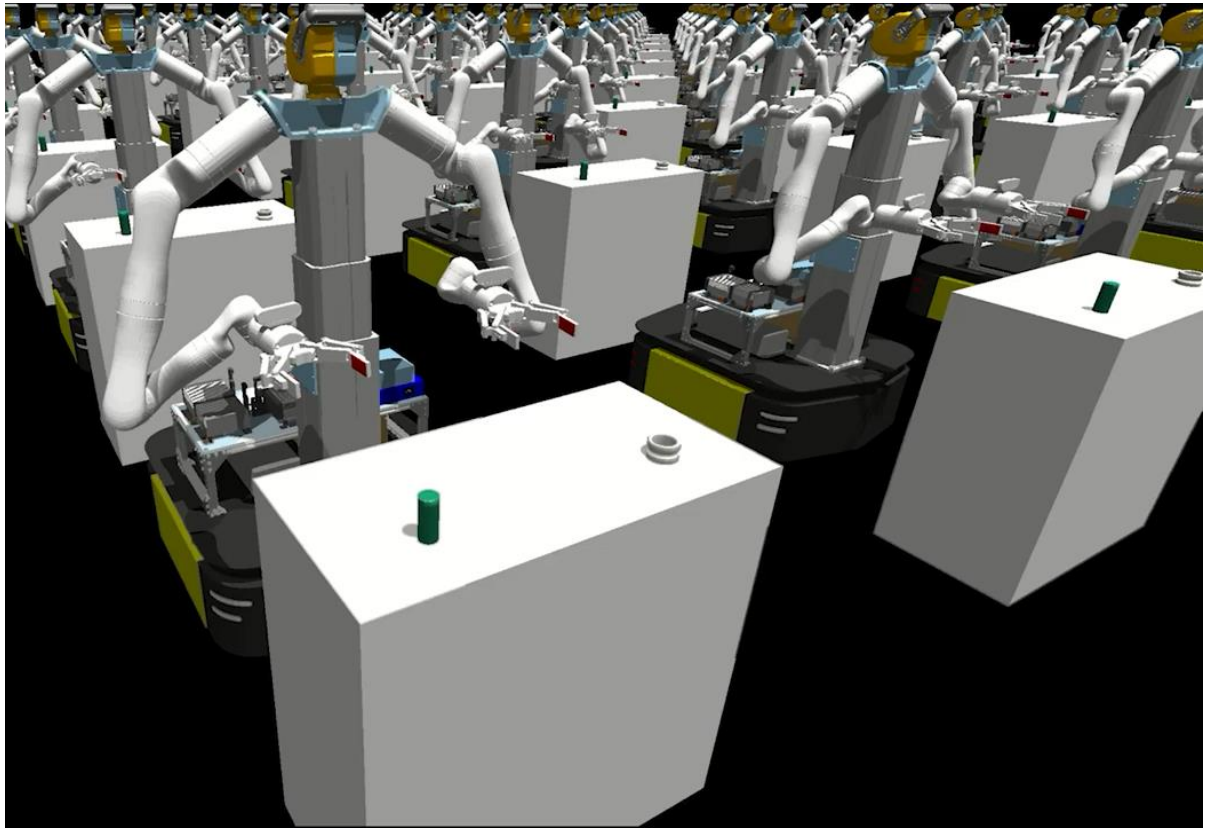
How accurate is the tactile simulation?

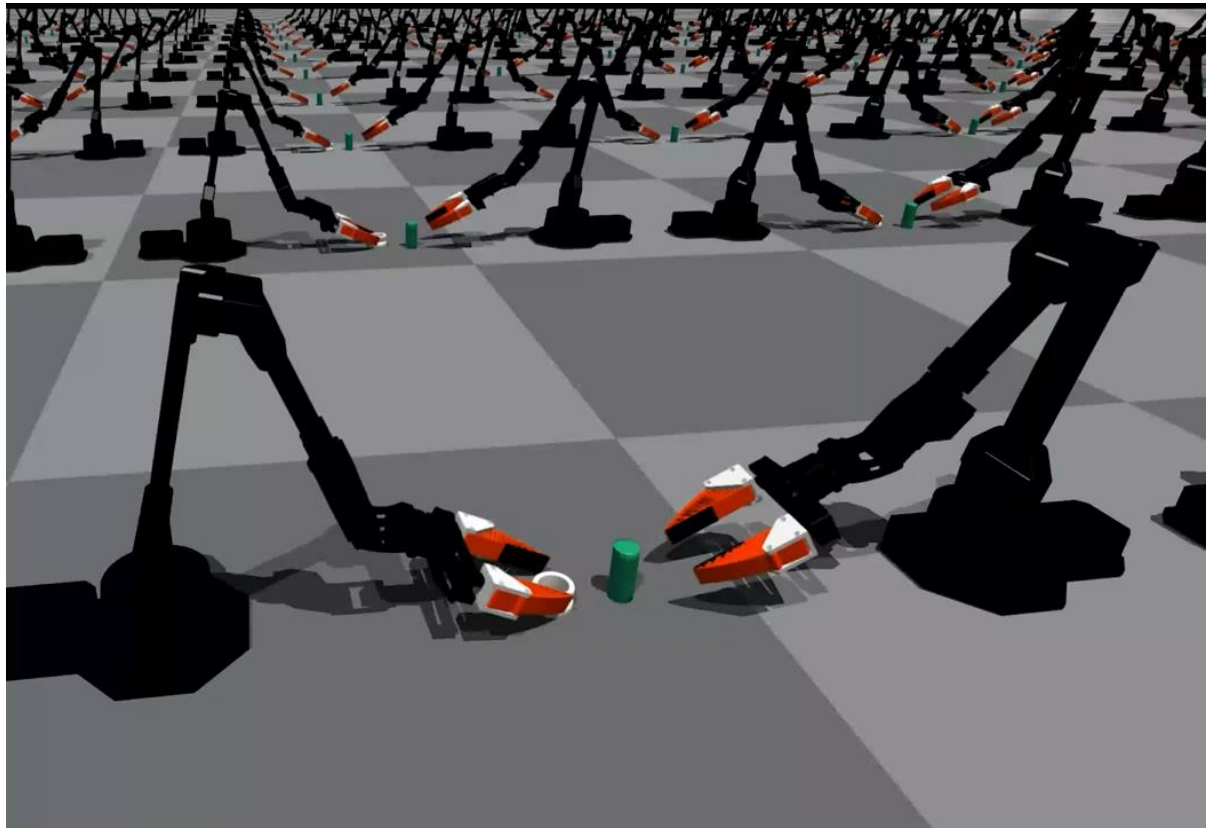




Huang et al., VT-Refine: Learning Bimanual Assembly with Visuo-Tactile Feedback via Simulation Fine-Tuning
CoRL-25, **Best Paper Award** at IROS 2025 Workshop on Advanced Haptic Tech for Healthcare Robots

Real2sim2real with scalable RL training





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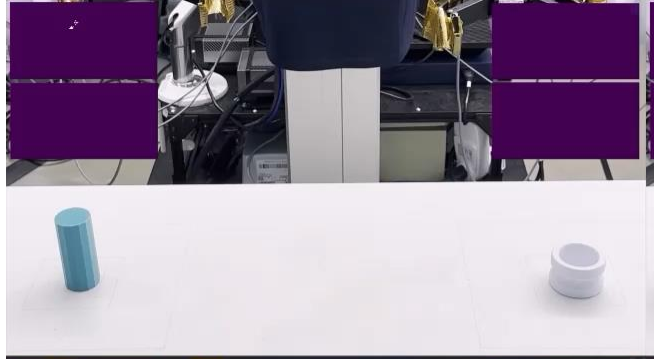




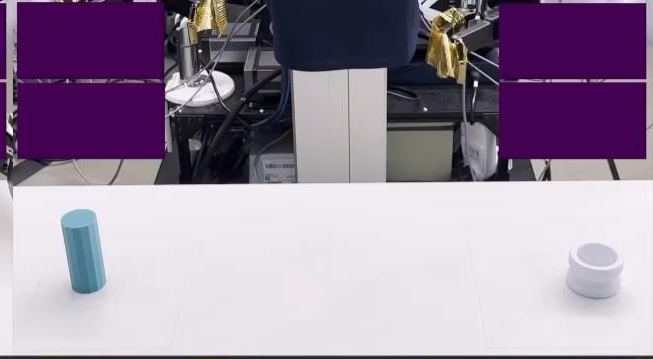


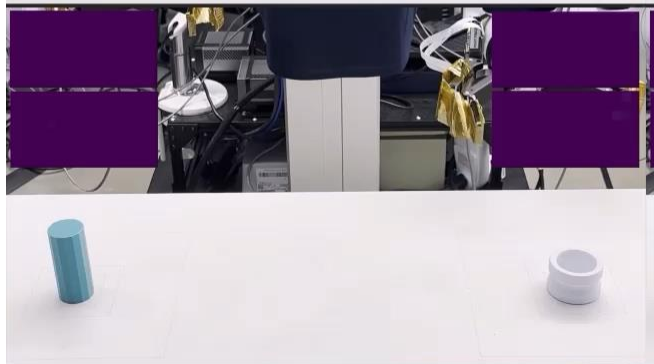


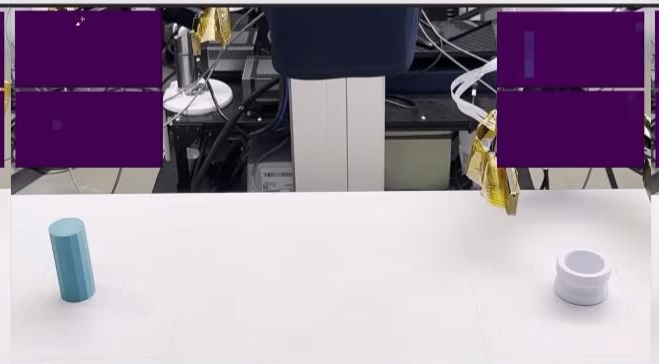


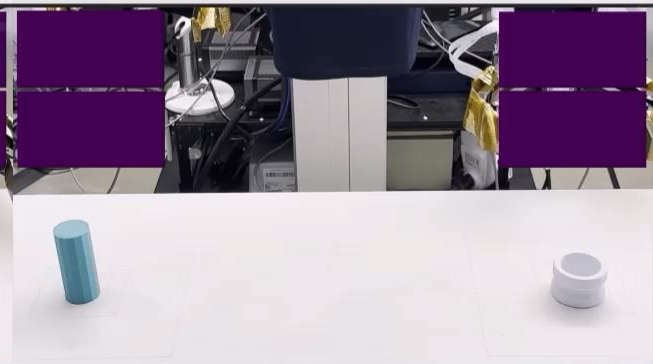


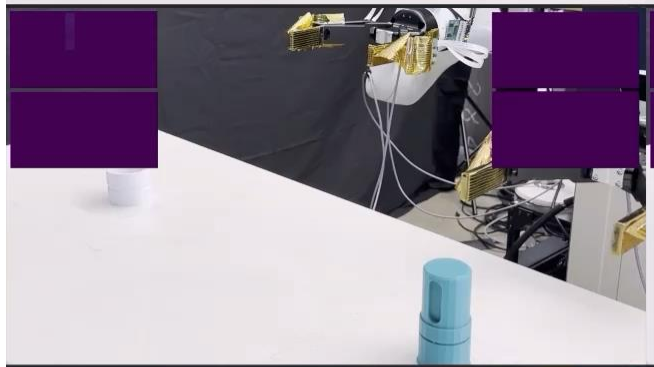


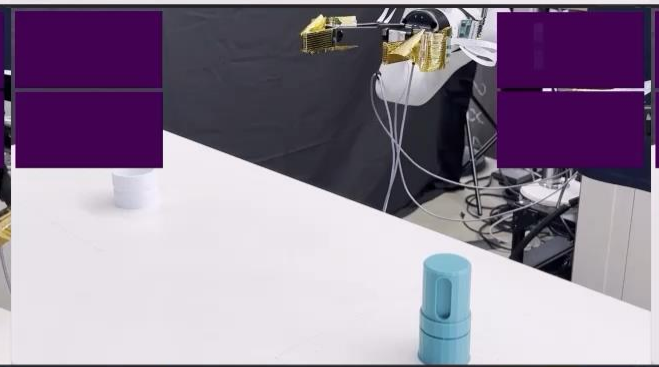


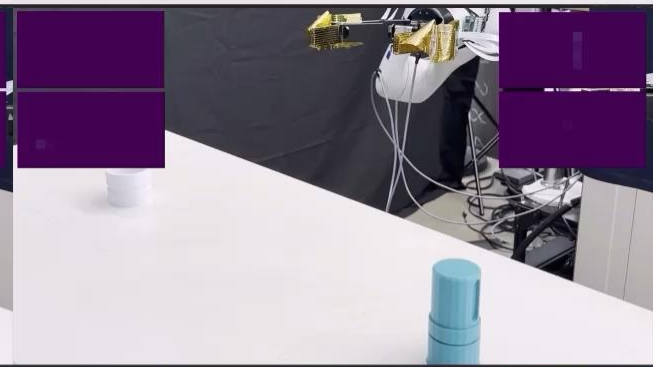


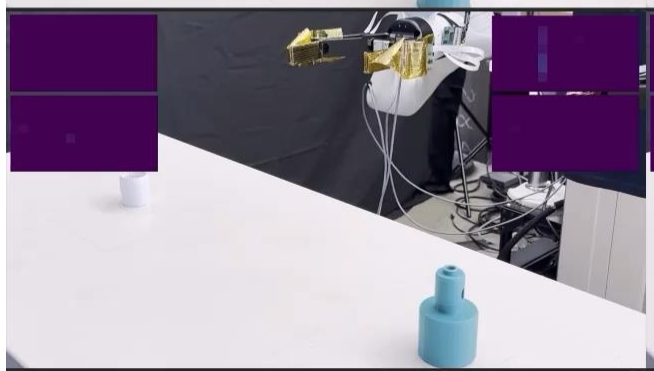


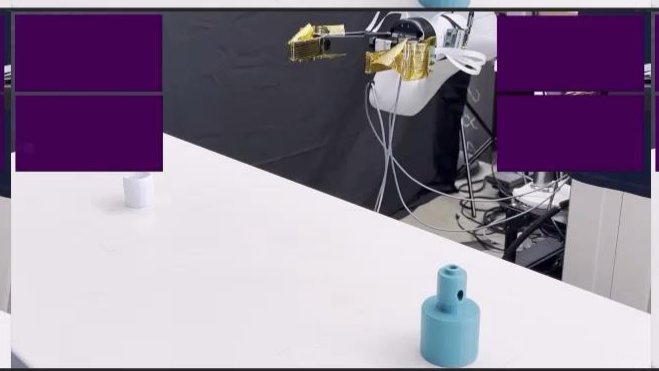


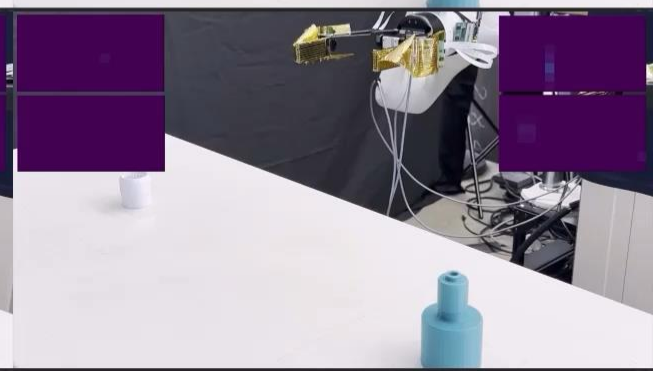




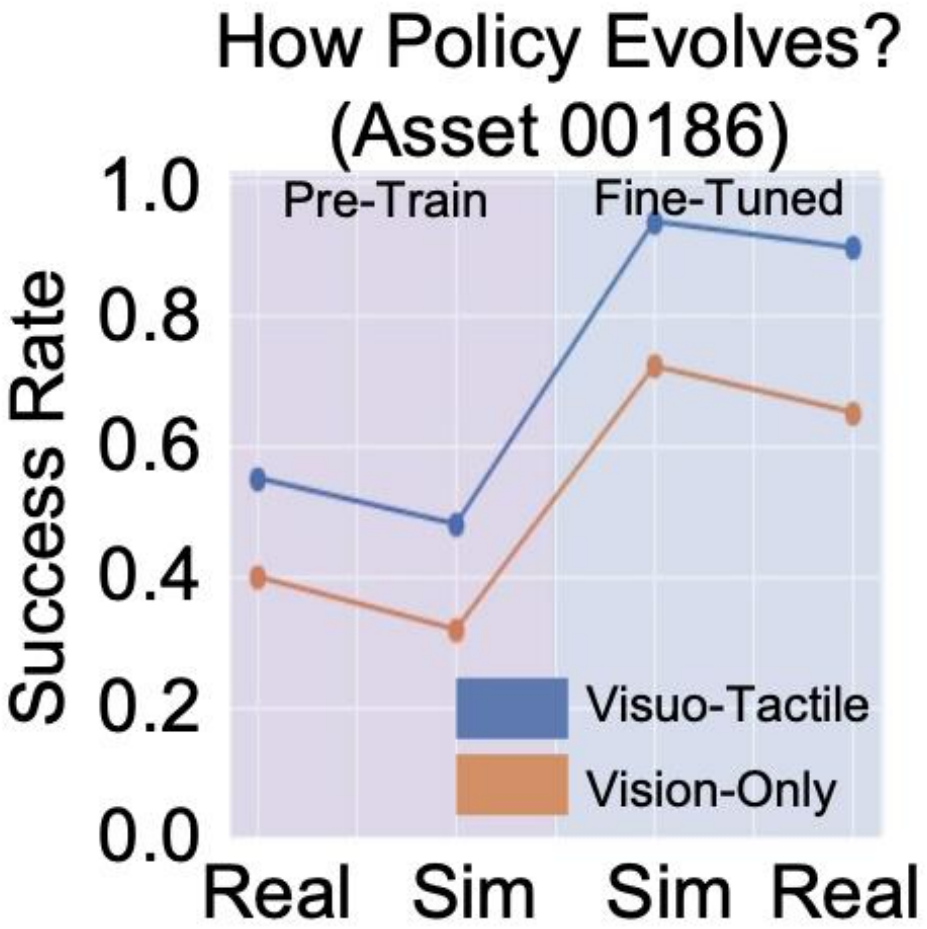








Does the simulation RL finetuning help?



- Sim to real gap still exists and can degrade performance
- RL finetuning significantly boost the performance
- Touch is essential to reach the best performance

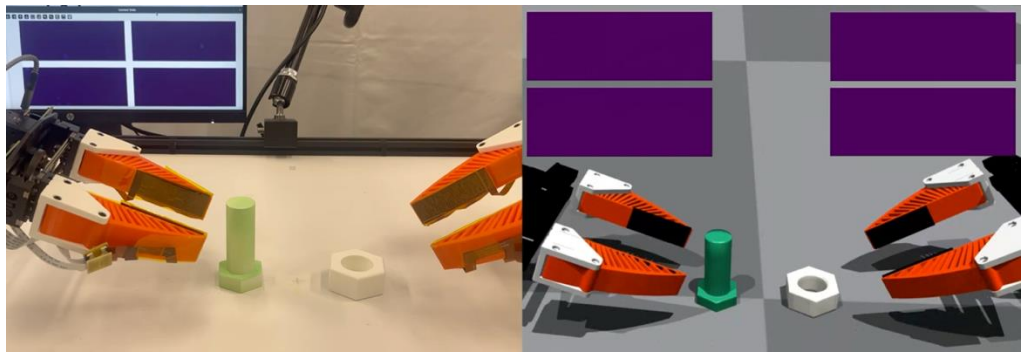
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Scalable Tactile Sensing for Robotic Manipulation





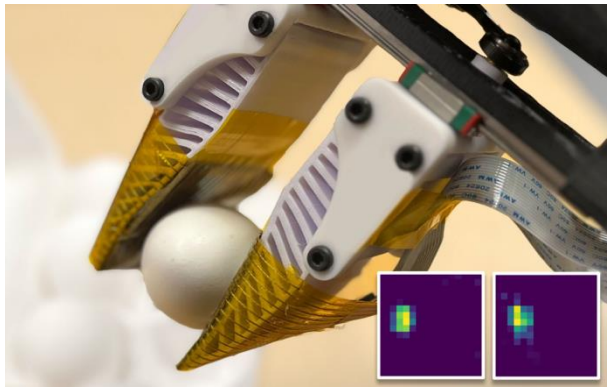
Sundaram et al., Learning the Signatures of the Human Grasp Using a Scalable Tactile Glove, Nature 2019



Huang et al., VT-Refine: Learning Bimanual Assembly with Visuo-Tactile Feedback via Simulation Fine-Tuning, CoRL 2025

Scale up **sim data**

Scale up **real data**



Huang et al., 3D-ViTac: Learning Fine-Grained Manipulation with Visuo-Tactile Sensing, CoRL 2024

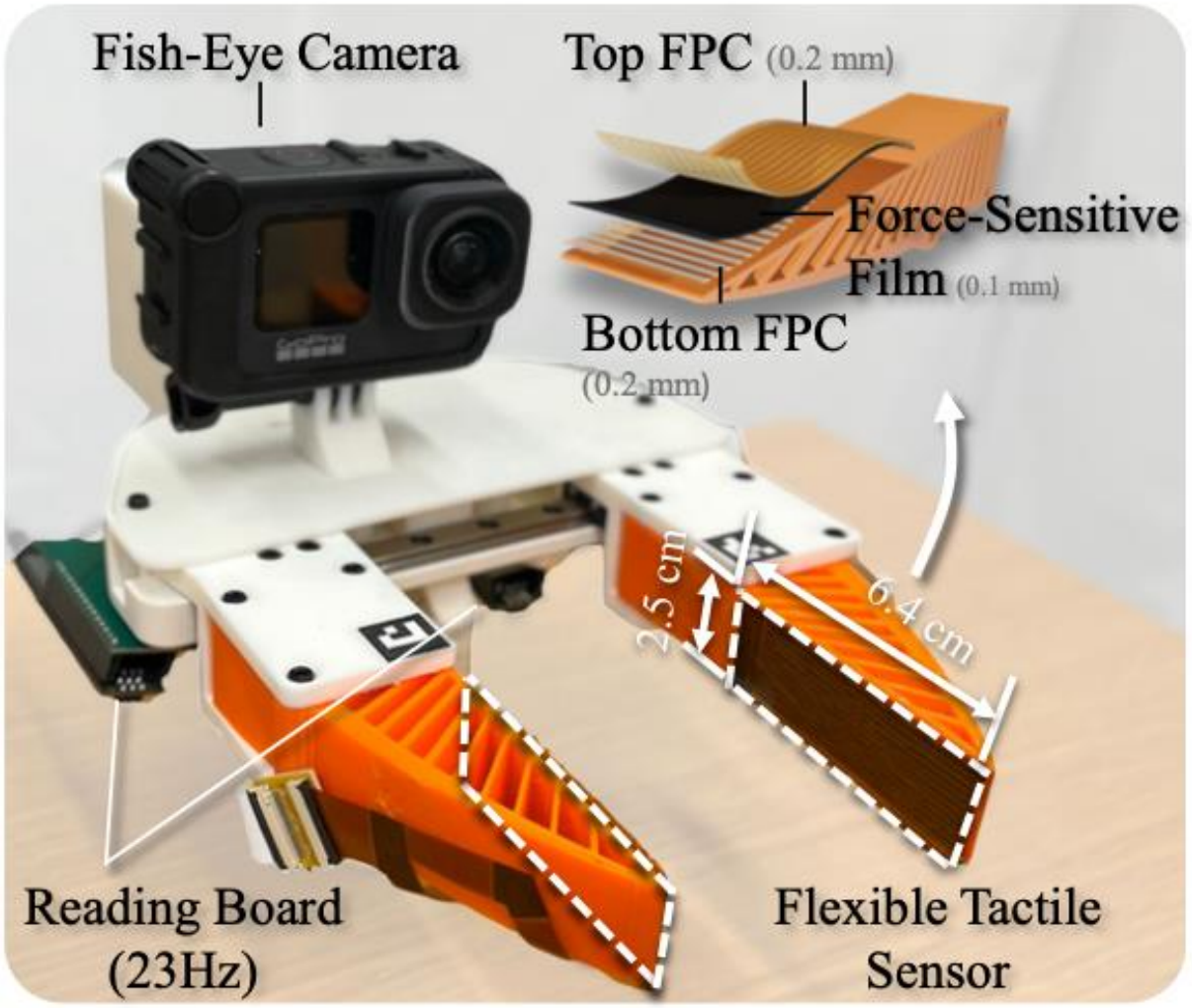


Zhu et al., Touch in the Wild: Learning Fine-Grained Manipulation with a Portable Visuo-Tactile Gripper, NeurIPS 2025

Scale up the real-world data with portable tactile gripper



(a) Data Collection in the Wild



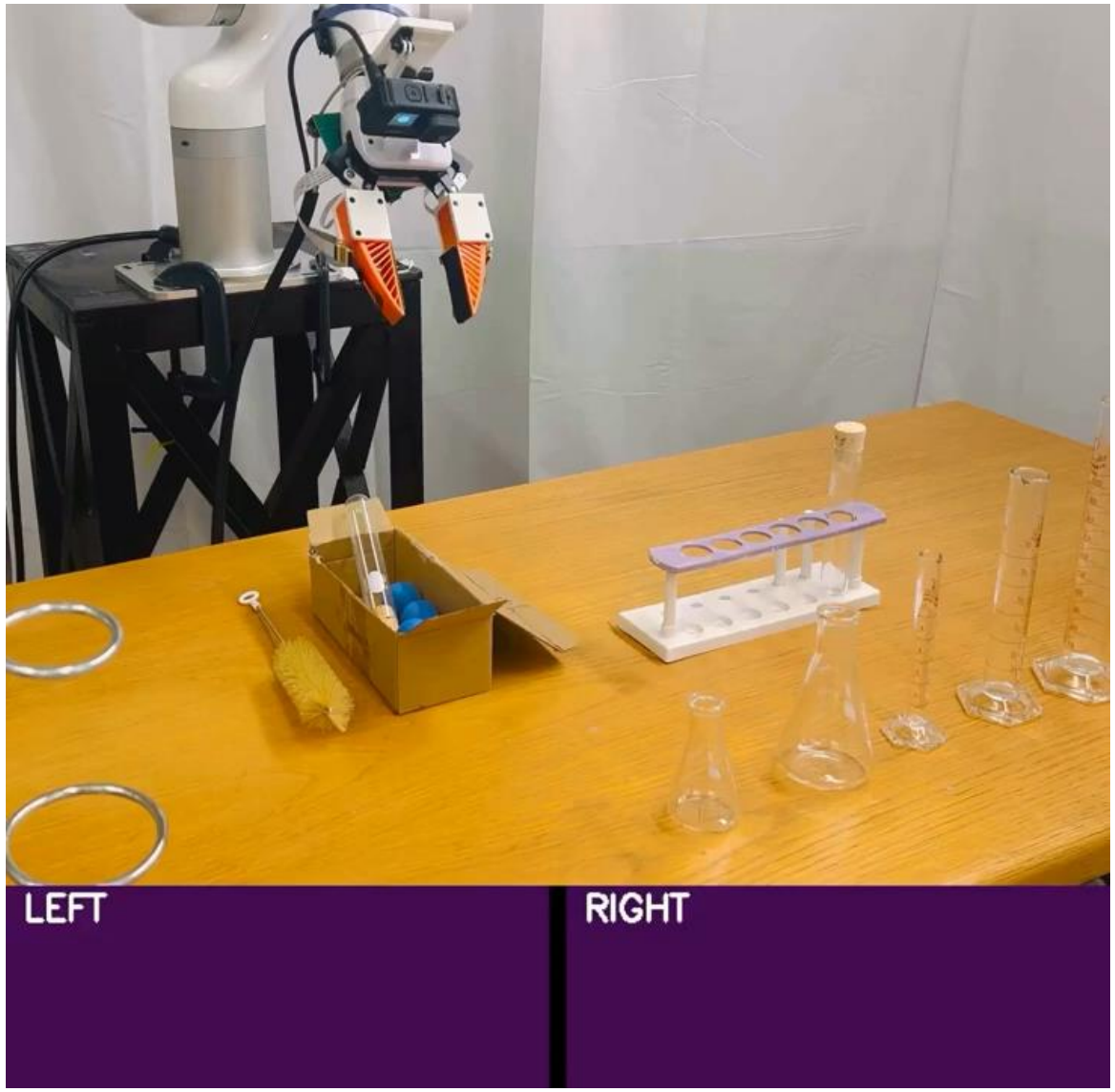
(b) Portable Visuo-Tactile Data Collection Gripper

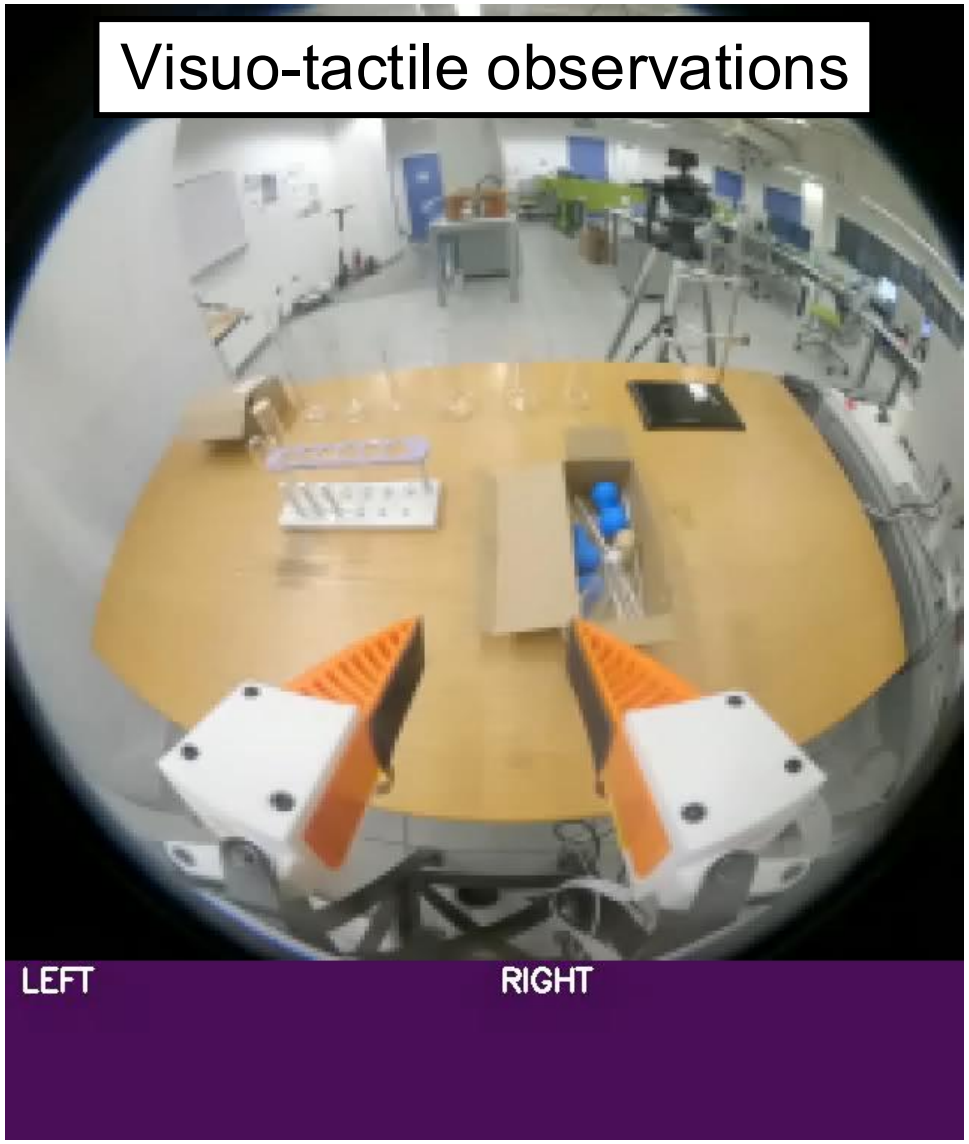
Xinyue Zhu*, Binghao Huang*, **Yunzhu Li,** Touch in the Wild: Learning Fine-Grained Manipulation with a Portable Visuo-Tactile Gripper NeurIPS-25, **Best Demo Award** at RSS-25 Workshop on Robot Hardware-Aware Intelligence




LEFT
RIGHT

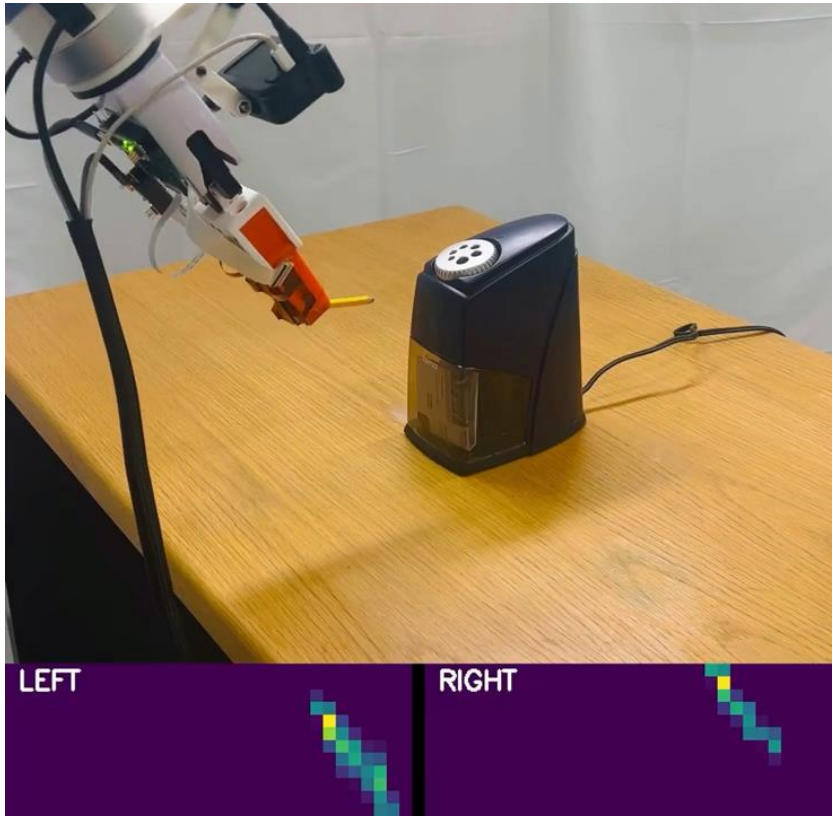
How well does the downstream policy work?

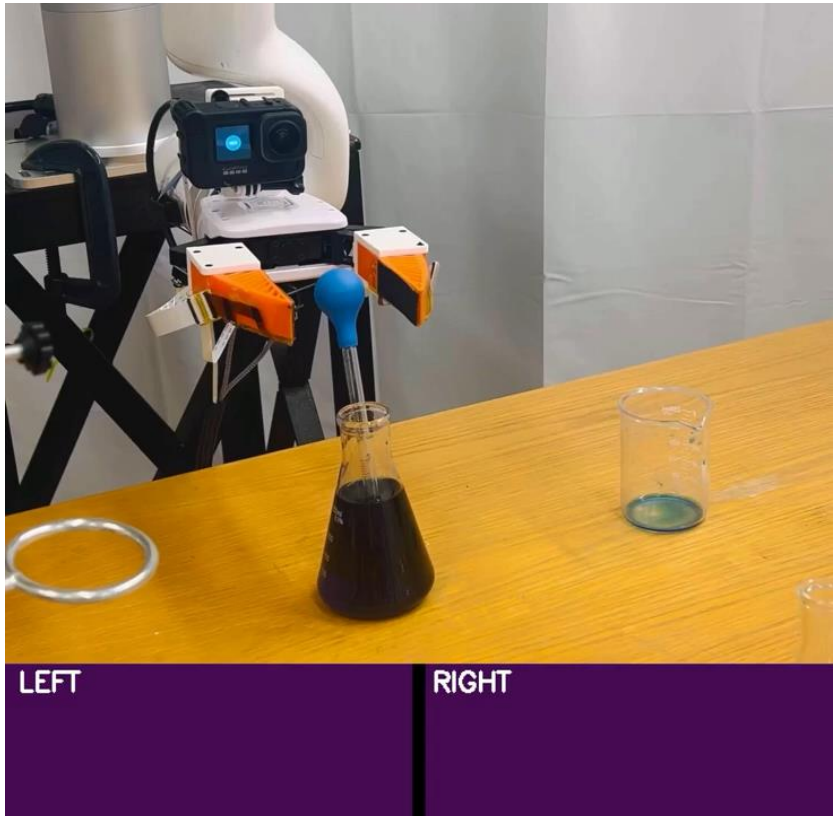


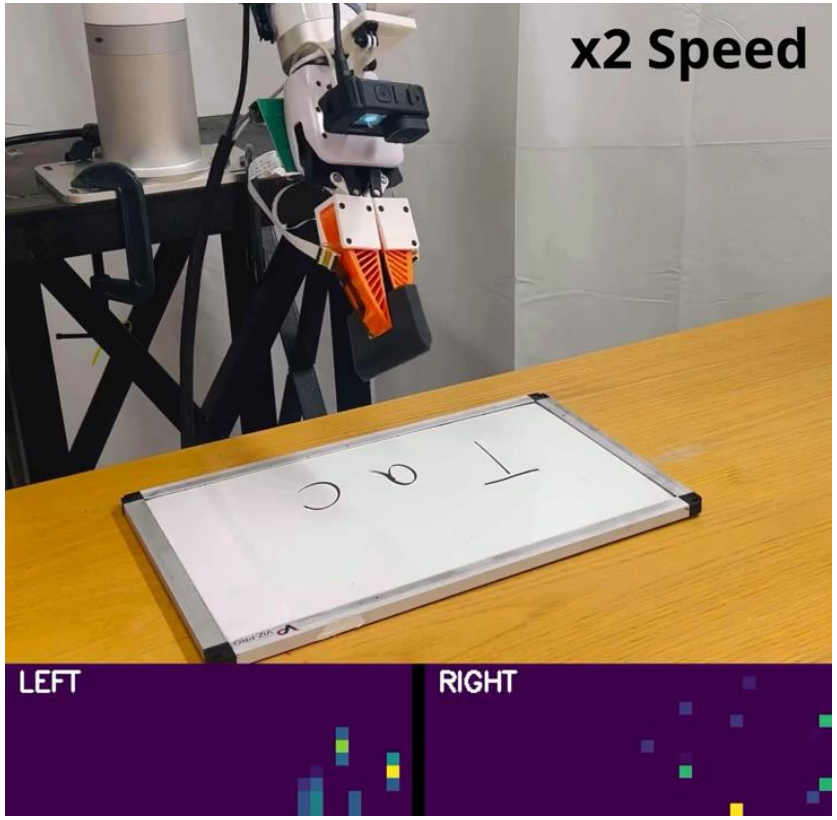


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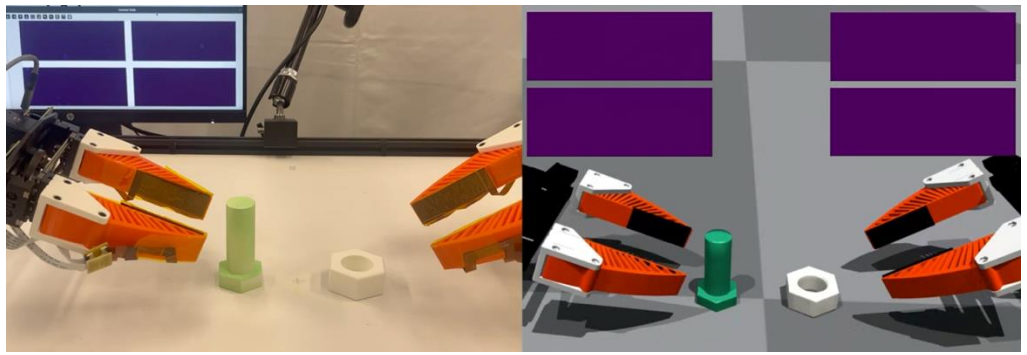
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Scalable Tactile Sensing for Robotic Manipulation





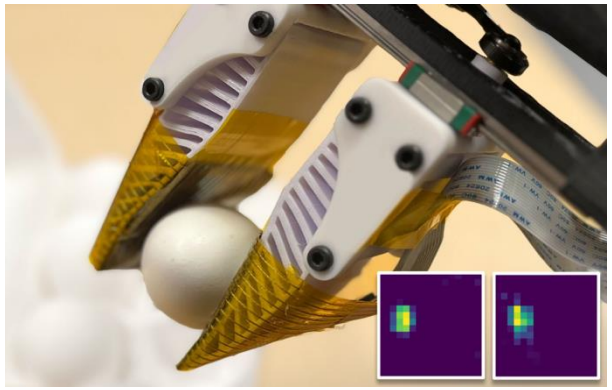
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Scale up **sim data**

Scale up **real data**

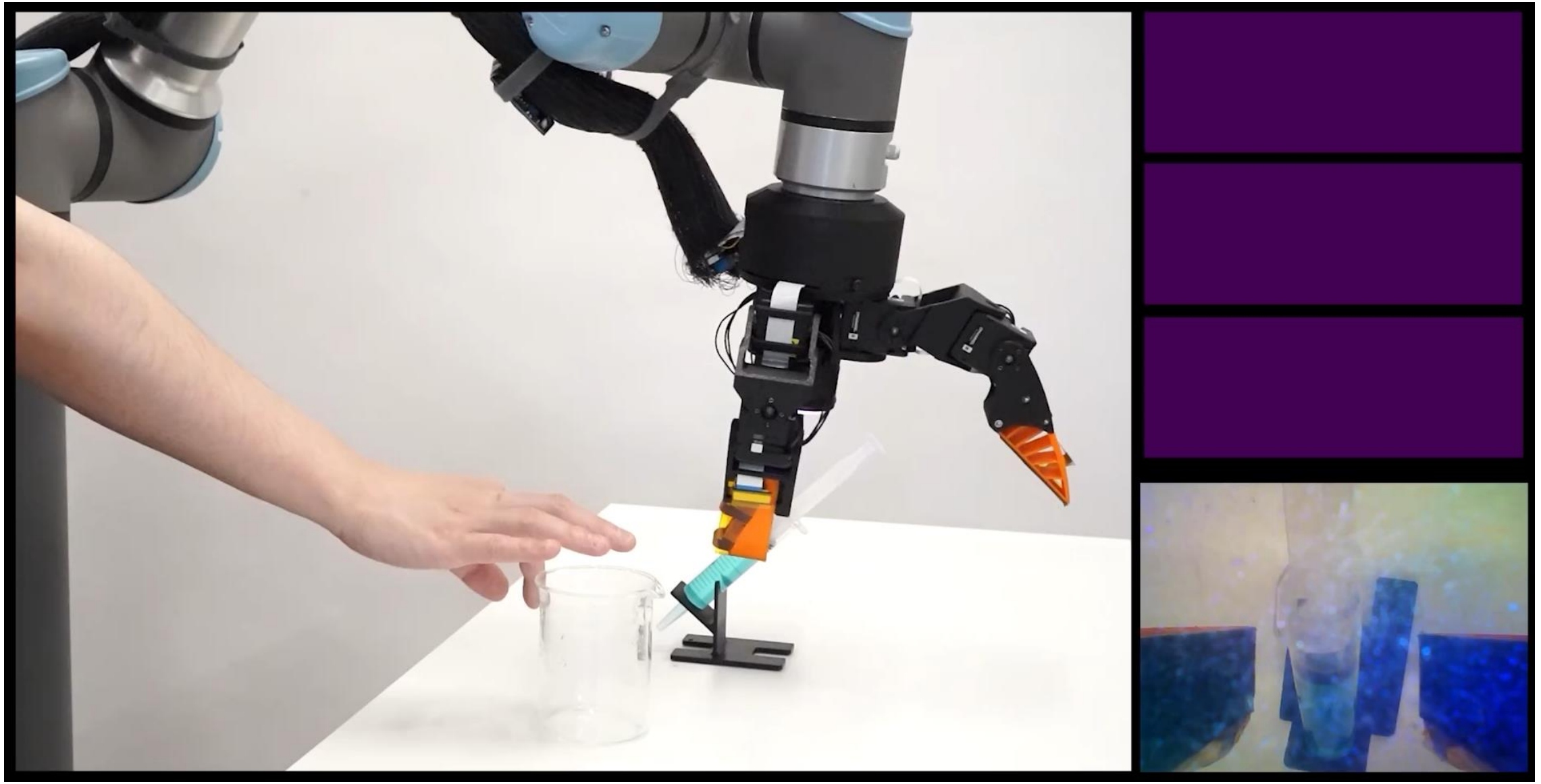


Huang et al., 3D-ViTac: Learning Fine-Grained Manipulation with Visuo-Tactile Sensing, CoRL 2024



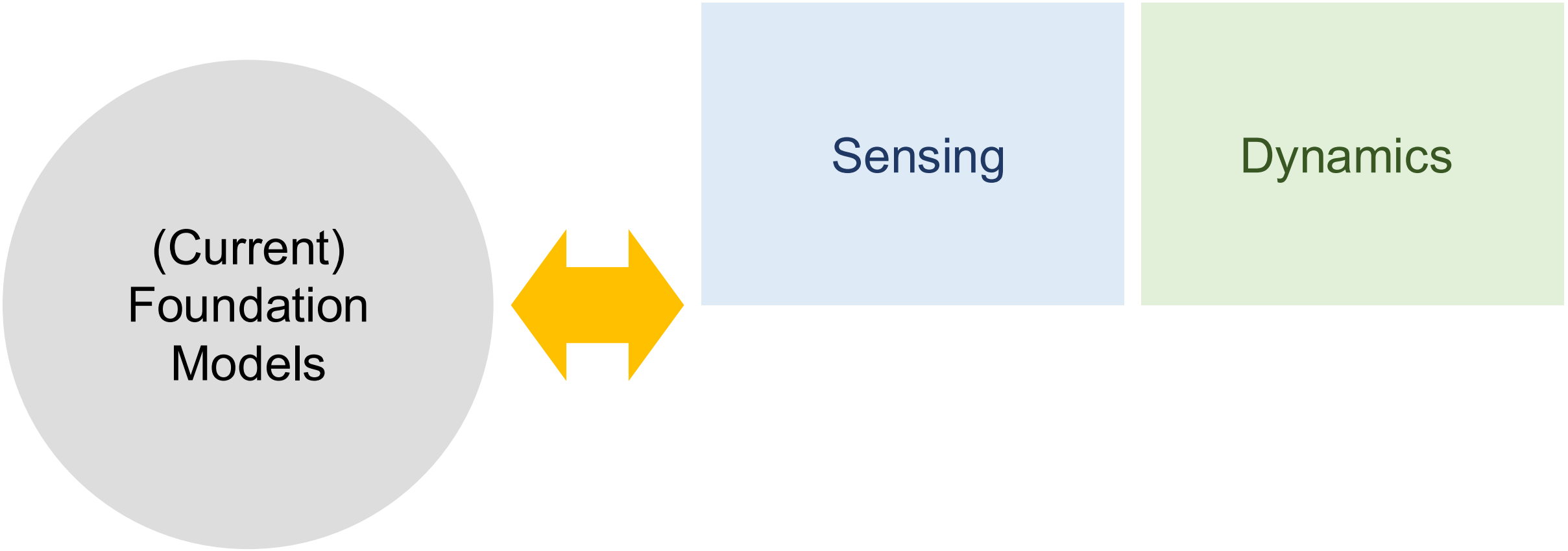
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Multi-modal tactile sensing with a multi-finger gripper



Yuhao Zhou, Sheeraz Athar*, Zhixian Hu*, Binghao Huang, Yunzhu Li, Juan Wachs, Yu She, VTAP Gripper: Synergizing Fingertip Sensing and a Visuo-Tactile Active Palm for Dexterous In-Hand Manipulation, IROS-26, **Best Paper Award Finalist**

Towards **foundational robotic intelligence**



Key: intuitive models of the world



Making Dough



Manipulating Onion Pieces

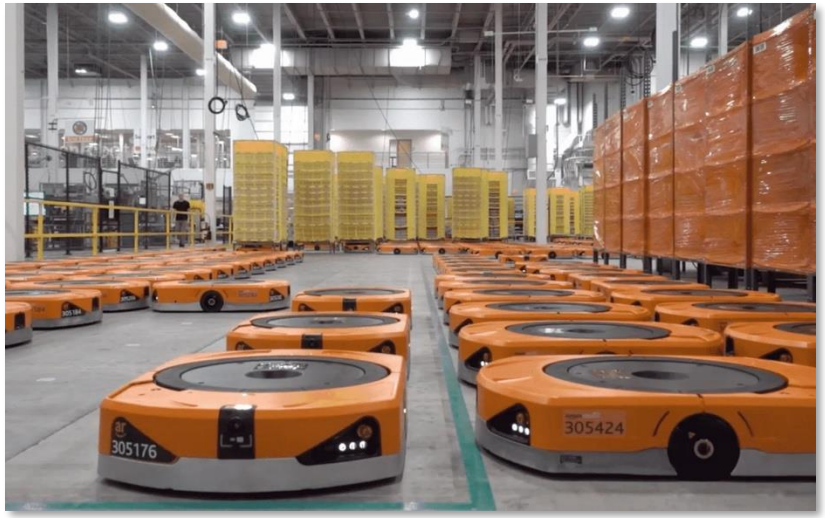
Model has been the key enabling power for existing robotic applications







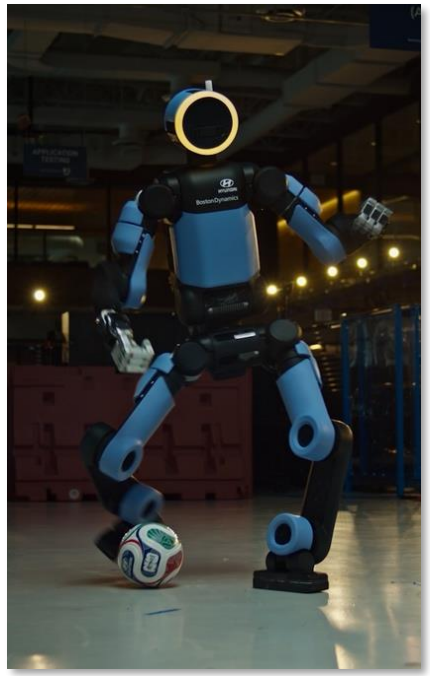












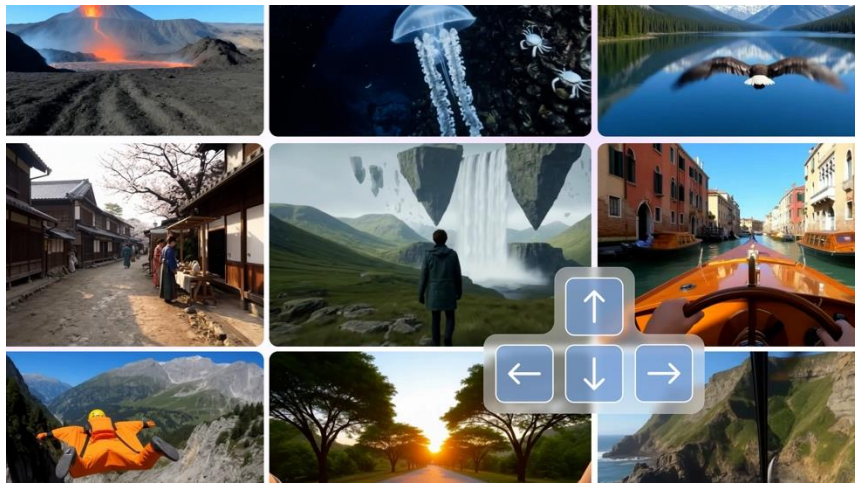
Learning structured world models from and for physical interactions

?

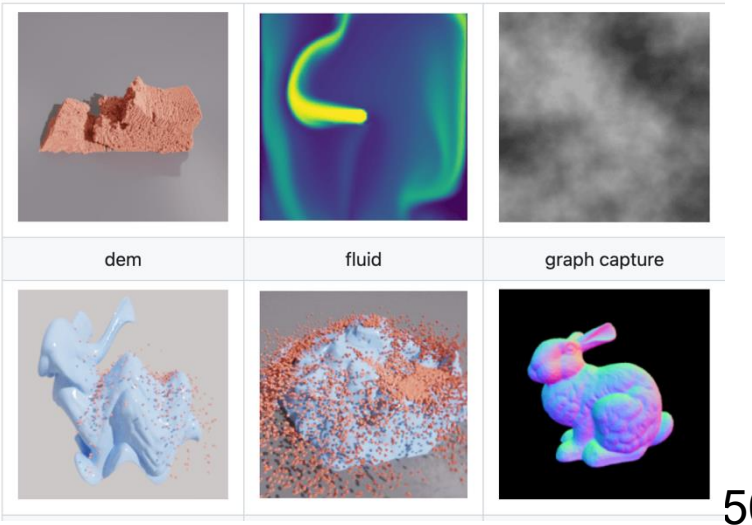
More Learning

More Physics

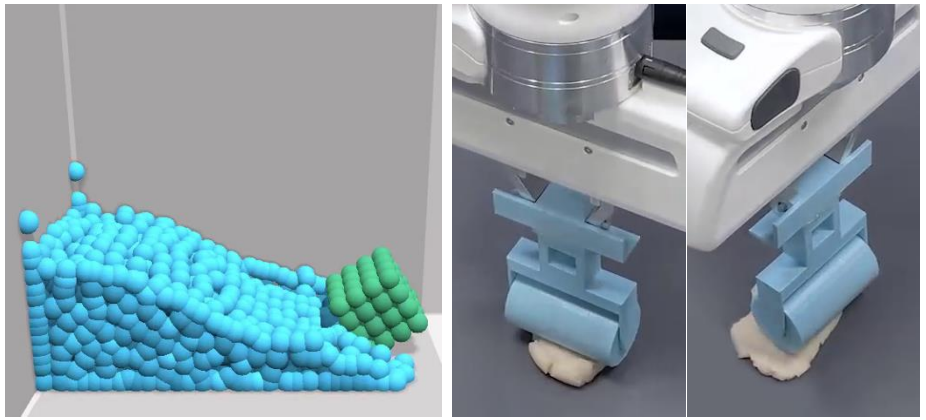
Genie 3



Nvidia Warp



Learning structured world models from and for physical interactions

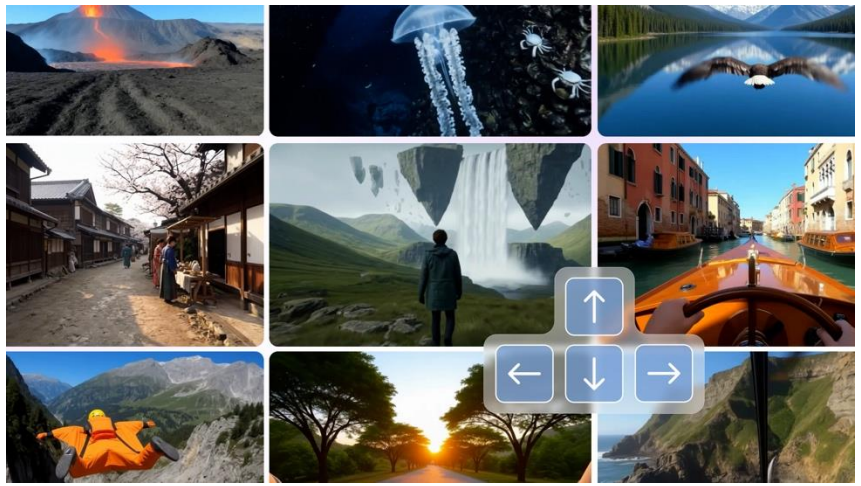


More Learning

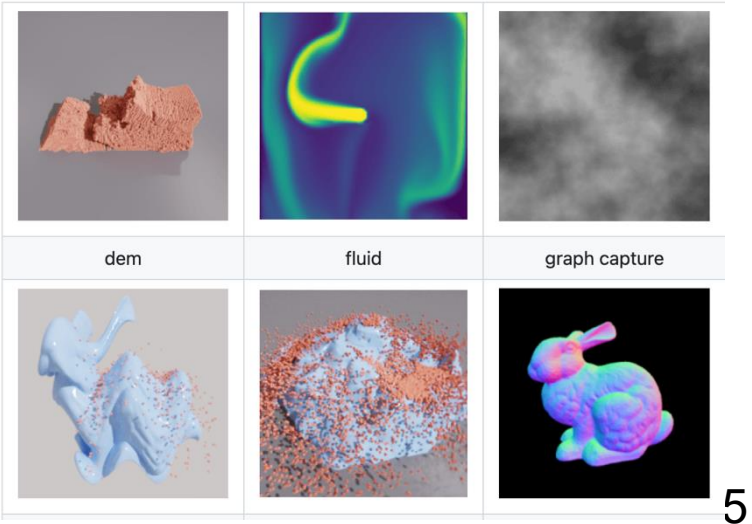
Graph-Based Neural Dynamics (GBND)

More Physics

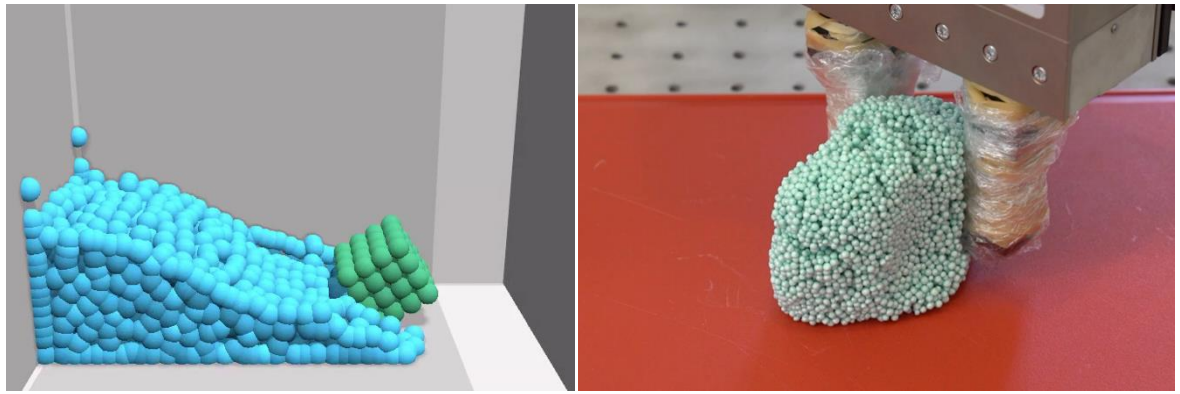
Genie 3



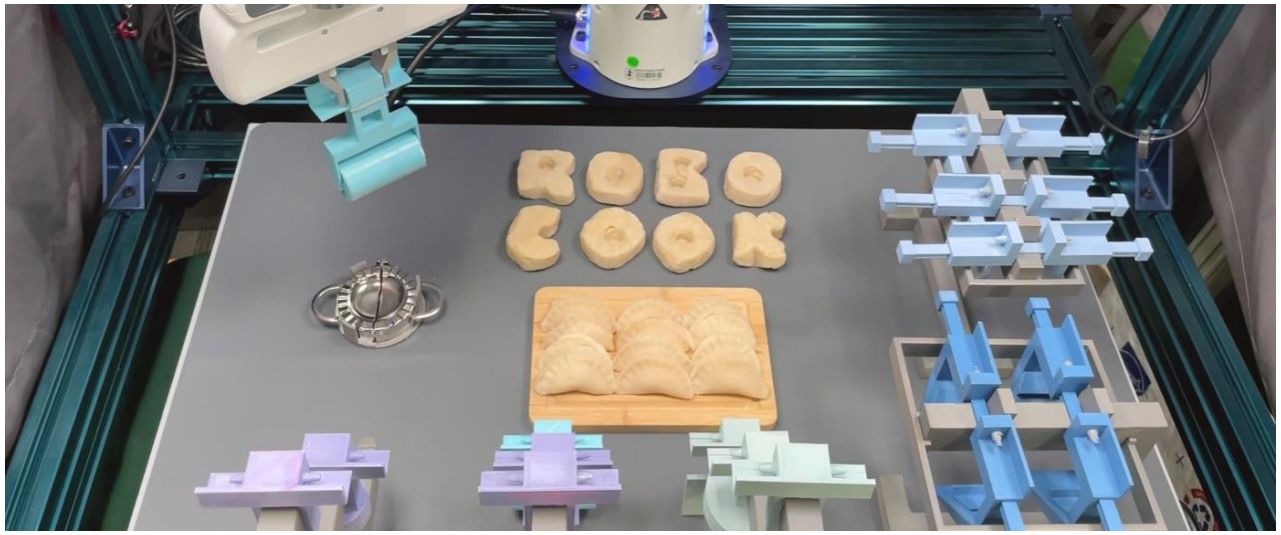
Nvidia Warp



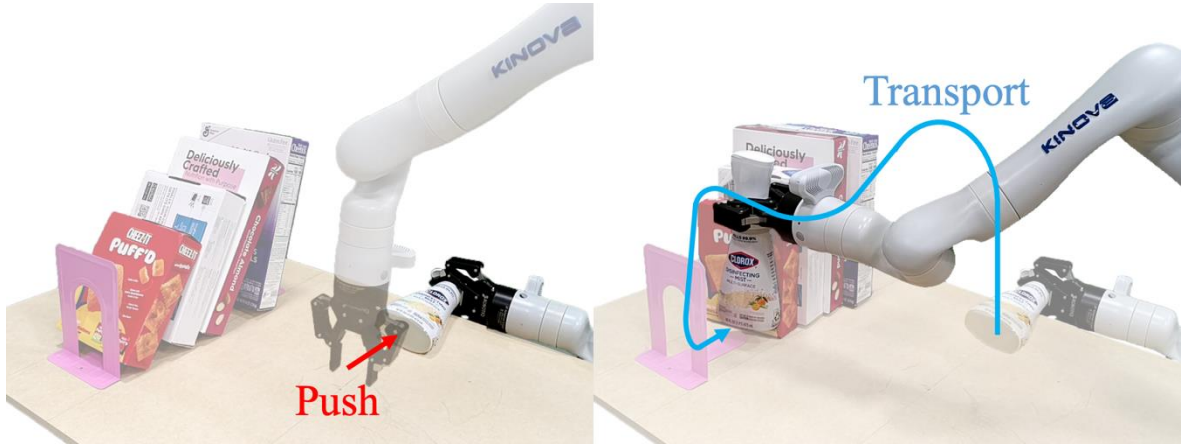
Structured world models for simulating and manipulating deformable objects



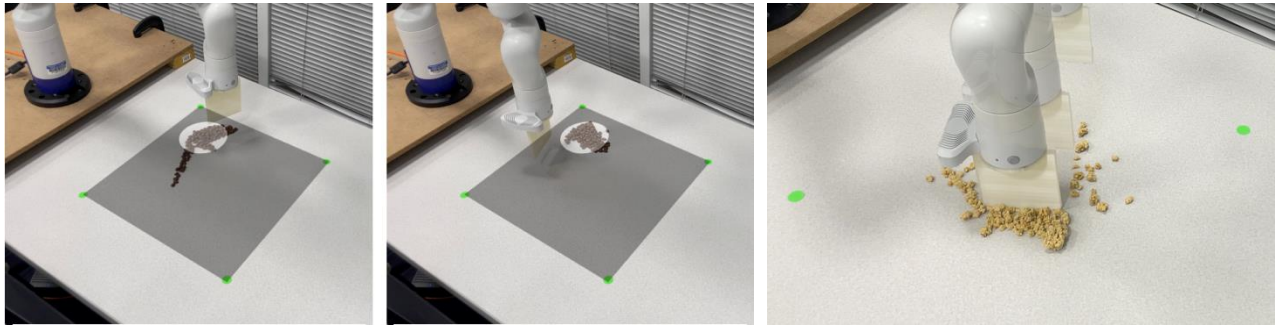
Yunzhu Li, Jiajun Wu, Russ Tedrake, Joshua B. Tenenbaum, Antonio Torralba, Learning Particle Dynamics for Manipulating Rigid Bodies, Deformable Objects, and Fluids, **ICLR-19**



Haochen Shi, Huazhe Xu, Samuel Clarke, **Yunzhu Li**, Jiajun Wu, RoboCook: Long-Horizon Elasto-Plastic Object Manipulation with Diverse Tools, **CoRL-23**
Best Systems Papers Award



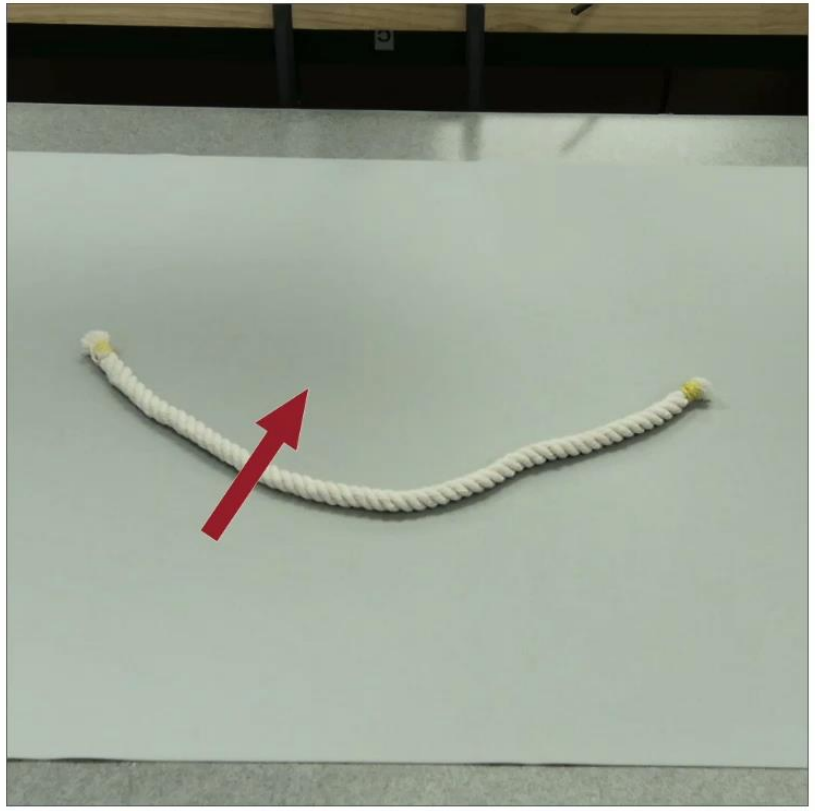
Haonan Chen, Yilong Niu, Kaiwen Hong, Shuijing Liu, Yixuan Wang, **Yunzhu Li**, Katherine Driggs-Campbell, Predicting Object Interactions with Behavior Primitives: An Application in Stowing Tasks, **CoRL-23**
Finalist - Best Paper/Best Student Paper Awards



Yixuan Wang*, **Yunzhu Li***, Katherine Driggs-Campbell, Li Fei-Fei, and Jiajun Wu, Dynamic-Resolution Model Learning for Object Pile Manipulation, **RSS 2023**, **Best Paper Award** at **IROS 2023 Workshop**

Representing a wide variety of real objects as graphs

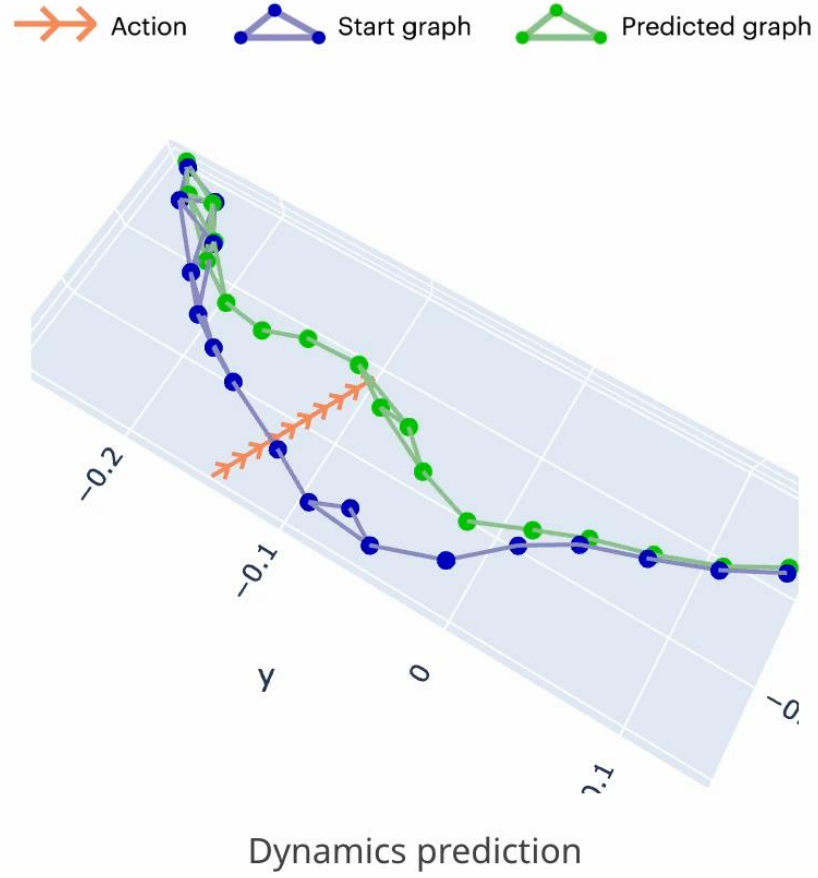
Box | Rope | Granular | Cloth



Before push



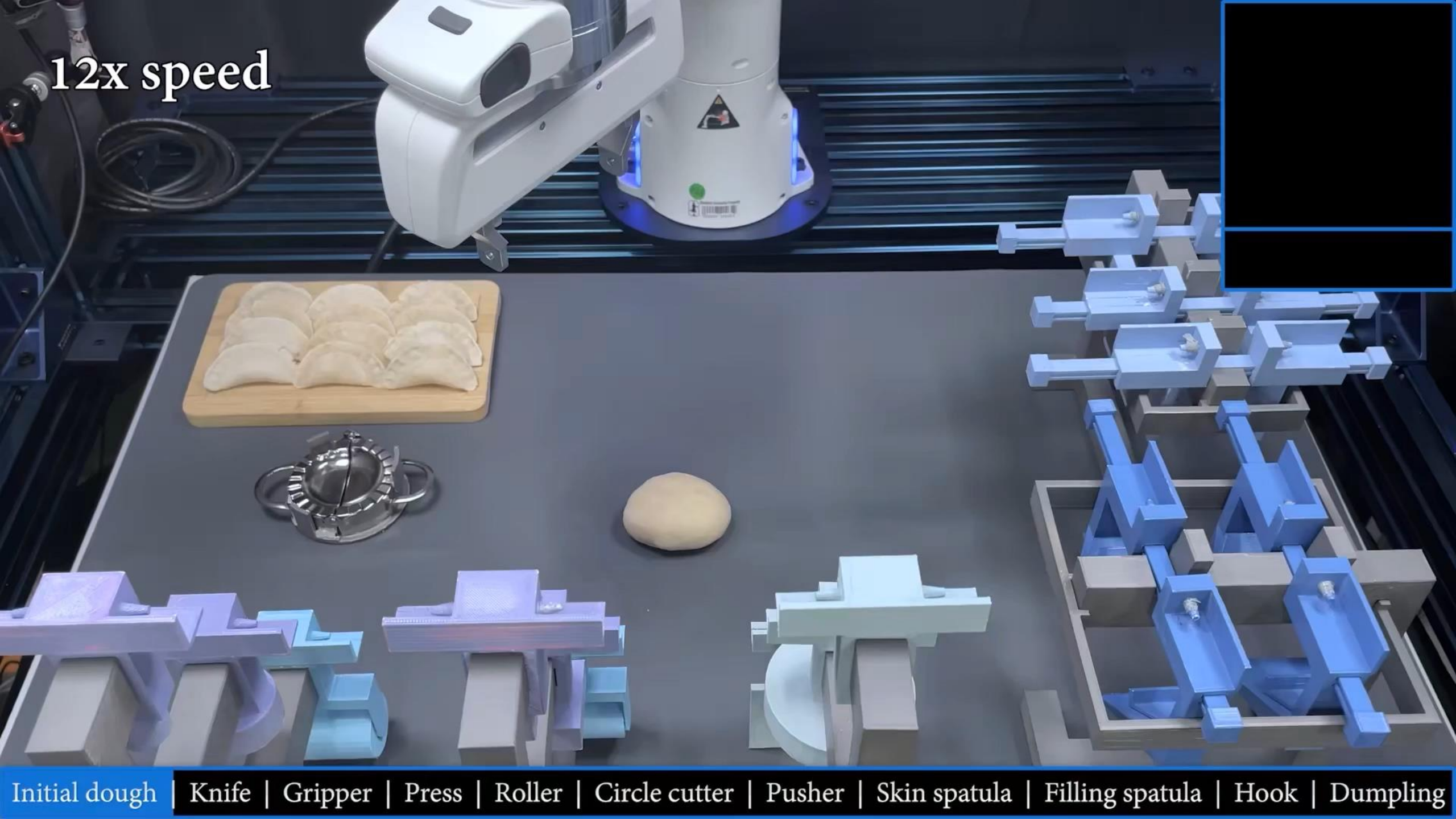
After push



Dynamics prediction

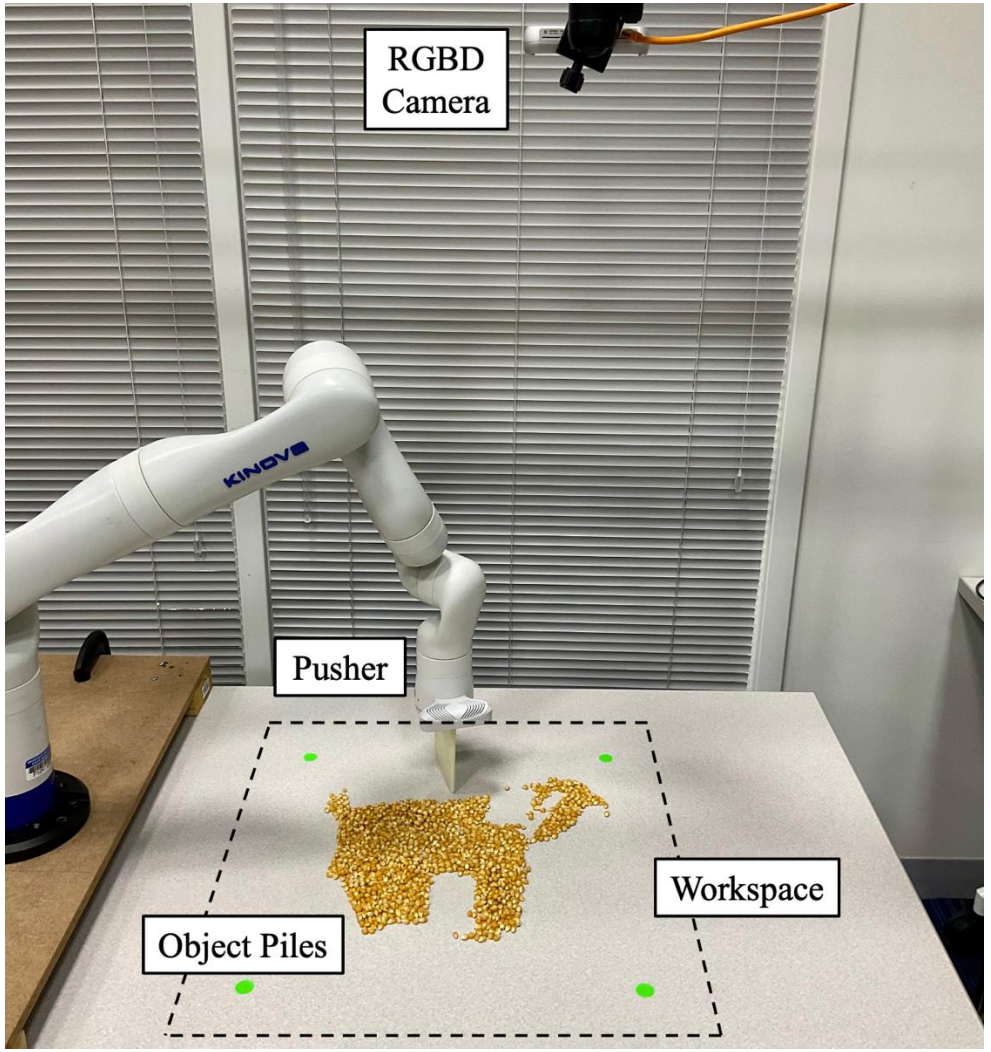
Kaifeng Zhang*, Baoyu Li*, Kris Hauser, and **Yunzhu Li**
AdaptiGraph: Material-Adaptive Graph-Based Neural Dynamics for Robotic Manipulation
RSS 2024, **Best Abstract Award** at ICRA 2024 Workshop on Representing and Manipulating Deformable Objects



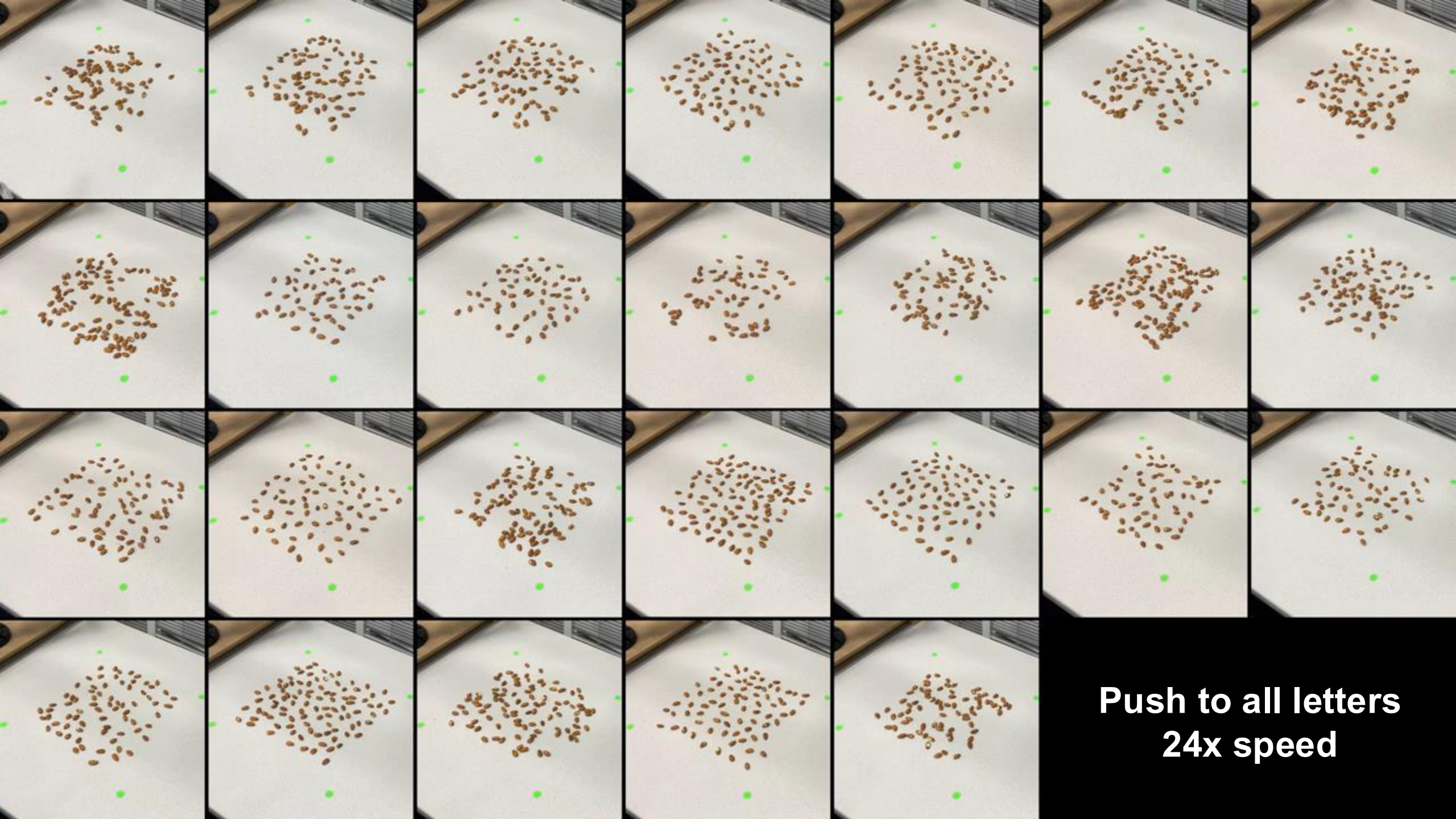

12x speed
Initial dough | Knife | Gripper | Press | Roller | Circle cutter | Pusher | Skin spatula | Filling spatula | Hook | Dumpling

Application to granular object manipulation



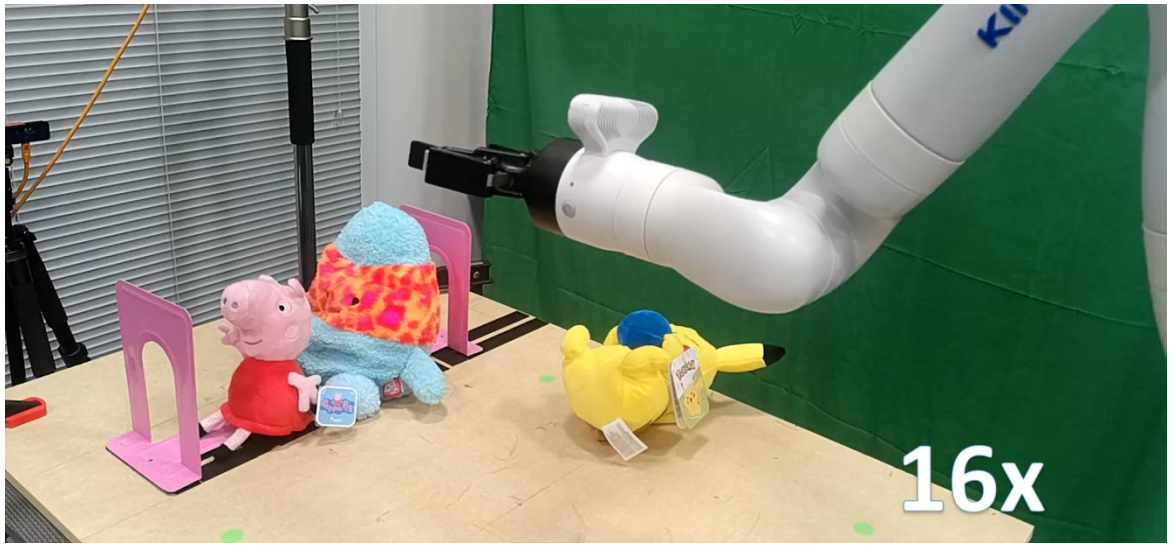


Yixuan Wang*, **Yunzhu Li***, Katherine Driggs-Campbell, Li Fei-Fei, and Jiajun Wu
Dynamic-Resolution Model Learning for Object Pile Manipulation
RSS 2023, **Best Paper Award** at IROS 2023 Workshop on Learning Meets Model-based Methods

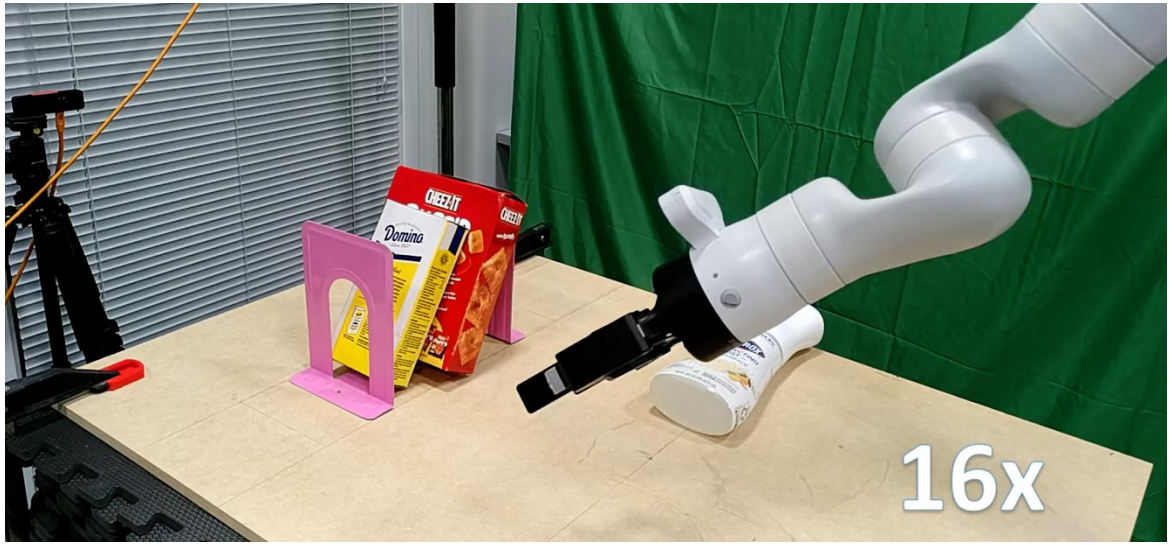

Push to all letters
24x speed

Application to object stowing









Haonan Chen, Yilong Niu*, Kaiwen Hong*, Shuijing Liu, Yixuan Wang, **Yunzhu Li**, Katherine Driggs-Campbell
Predicting Object Interactions with Behavior Primitives: An Application in Stowing Tasks
Conference on Robot Learning (CoRL) 2023 - **Finalist - Best Paper/Best Student Paper Awards**

A *Science Robotics* paper on learning-based dynamics models

SCIENCE ROBOTICS | REVIEW

MANIPULATION

A review of learning-based dynamics models for robotic manipulation

Bo Ai[1]*, Stephen Tian[2], Haochen Shi[2], Yixuan Wang[3], Tobias Pfaff[4], Cheston Tan[5], Henrik I. Christensen[1], Hao Su[1,6], Jiajun Wu[2], Yunzhu Li[3]*

Dynamics models that predict the effects of physical interactions are essential for planning and control in robotic manipulation. Although models based on physical principles often generalize well, they typically require full-state information, which can be difficult or impossible to extract from perception data in complex, real-world scenarios. Learning-based dynamics models provide an alternative by deriving state transition functions purely from perceived interaction data, enabling the capture of complex, hard-to-model factors and predictive uncertainty and accelerating simulations that are often too slow for real-time control. Recent successes in this field have demonstrated notable advancements in robot capabilities, including long-horizon manipulation of deformable objects, granular materials, and complex multiobject interactions such as stowing and packing. A crucial aspect of these investigations is the choice of state representation, which determines the inductive biases in the learning system for reduced-order modeling of scene dynamics. This article provides a timely and comprehensive review of current techniques and trade-offs in designing learned dynamics models, highlighting their role in advancing robot capabilities through integration with state estimation and control and identifying critical research gaps for future exploration.

What's next?

- More complicated tasks/objects
- Heterogeneous interactions
- Scale up to in-the-wild videos (e.g., Open X-Embodiment, DROID)
- Towards scalable data/evaluation engines

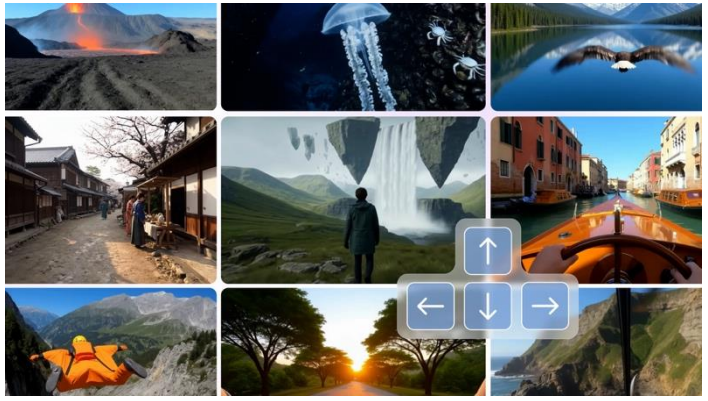
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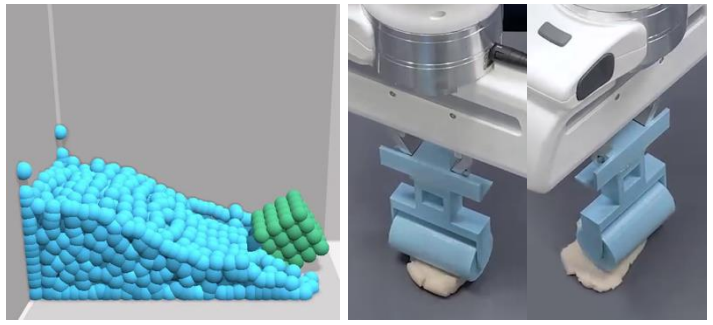
More Learning

More Physics

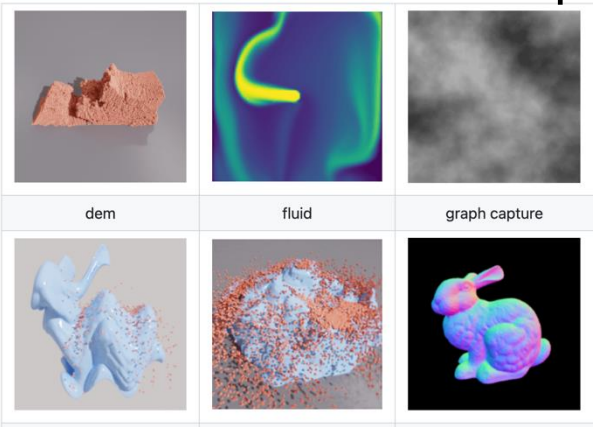
Genie 3



Graph-Based Neural Dynamics (GBND)



Nvidia Warp



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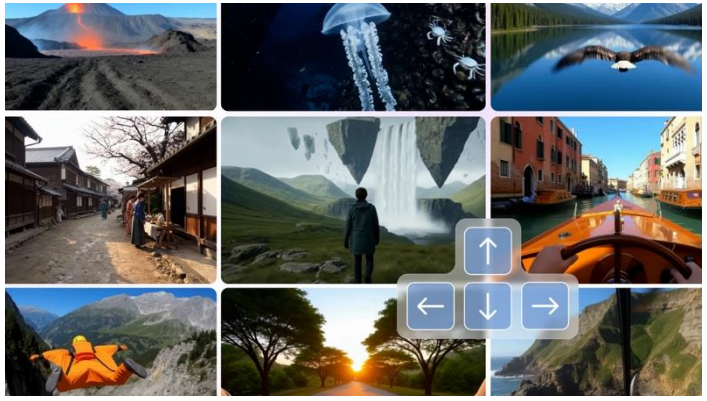
PhysTwin

?

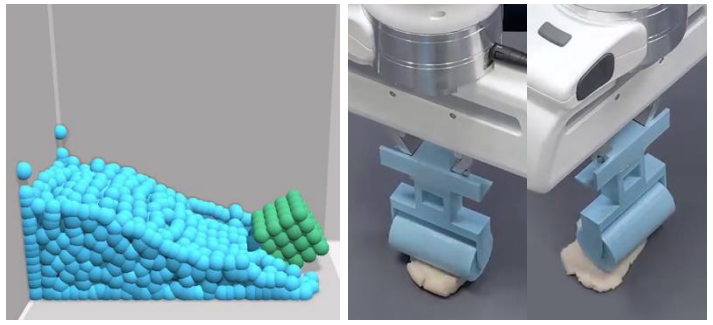
More Learning

More Physics

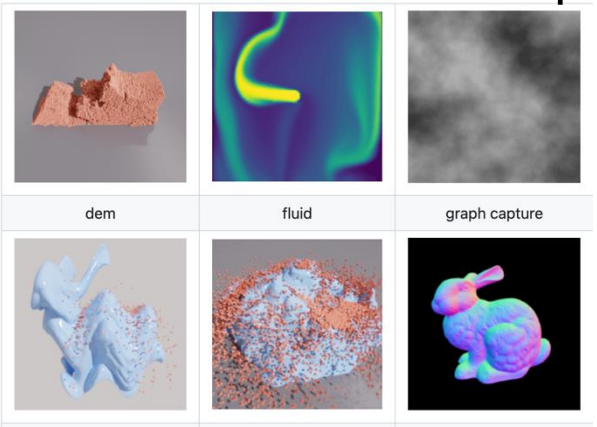
Genie 3

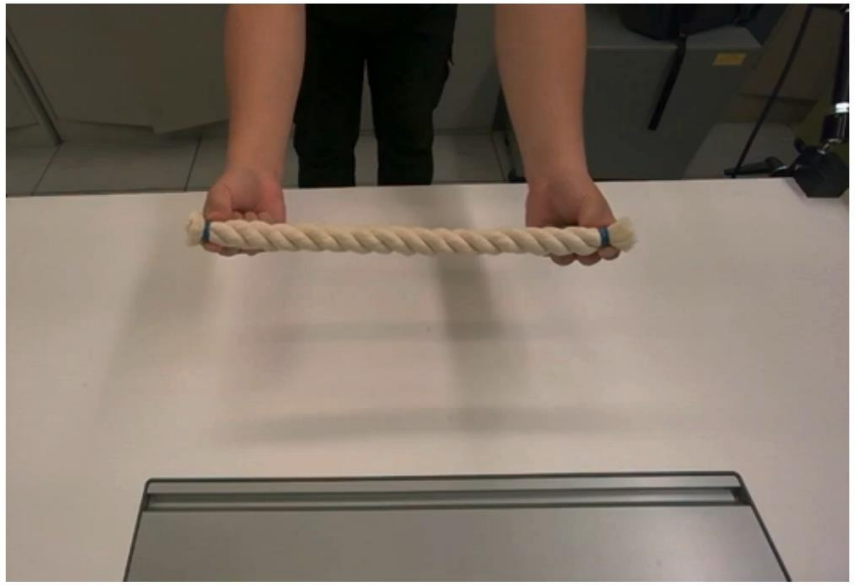


Graph-Based Neural Dynamics (GBND)

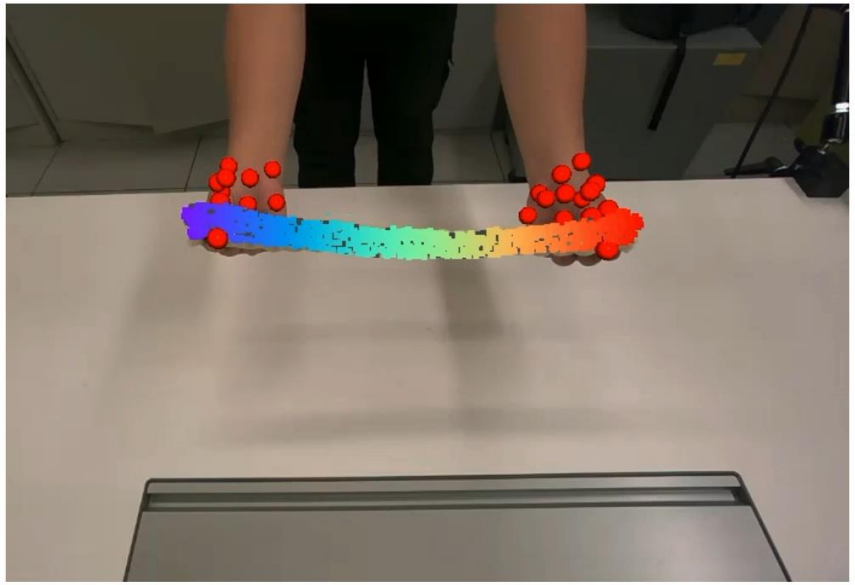


Nvidia Warp





Observation



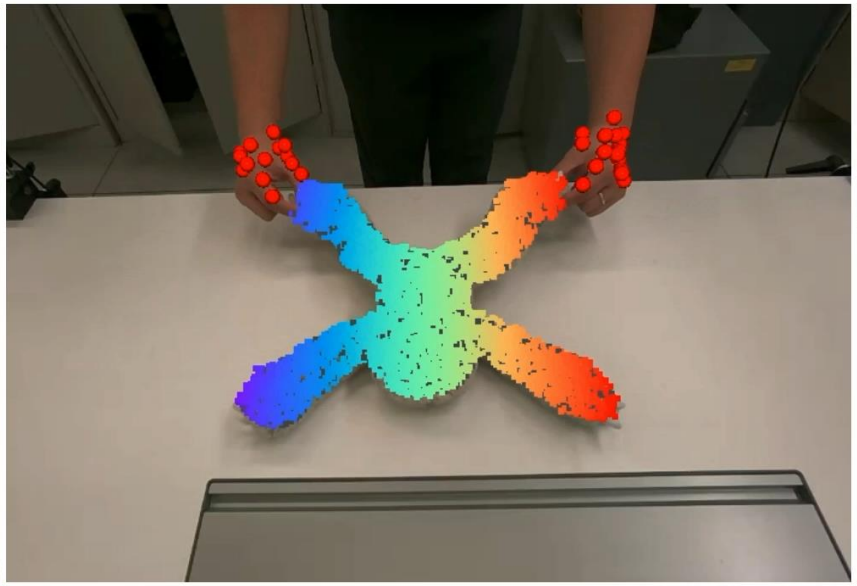
Reconstructed state/action



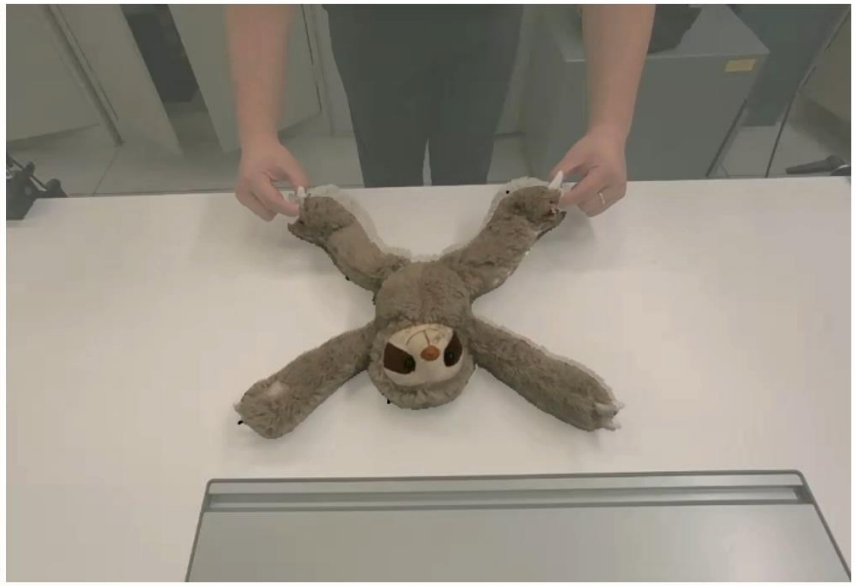
Interactable digital twin



Observation



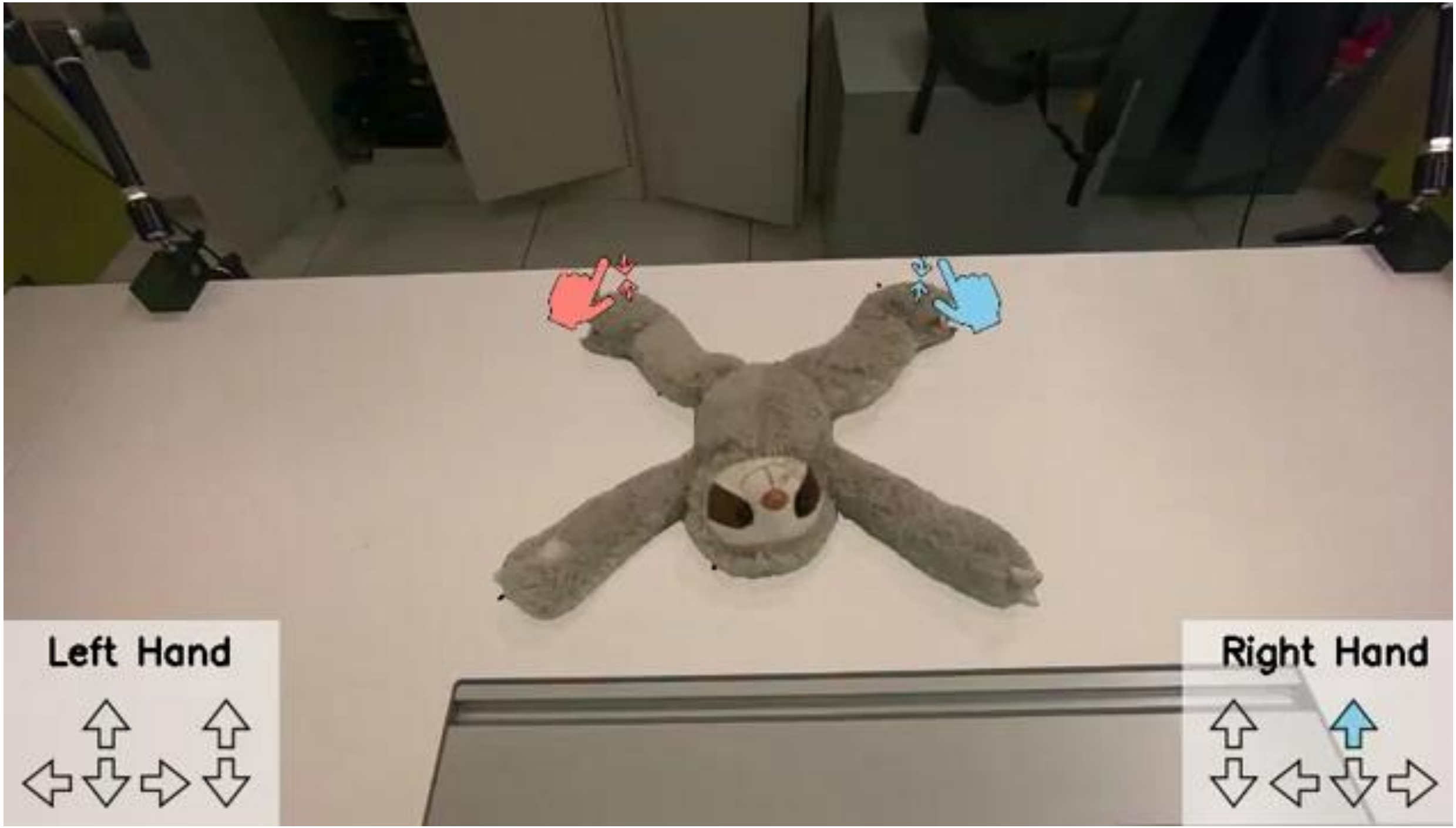
Reconstructed state/action

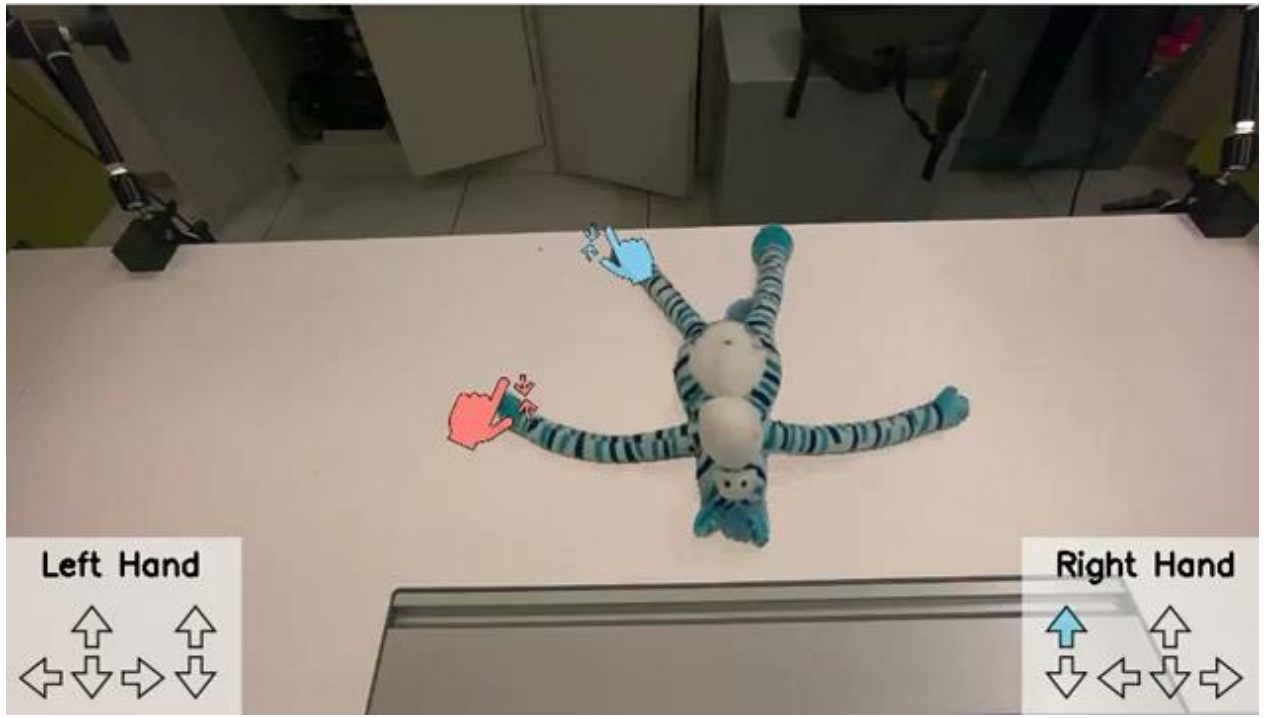


Interactable digital twin

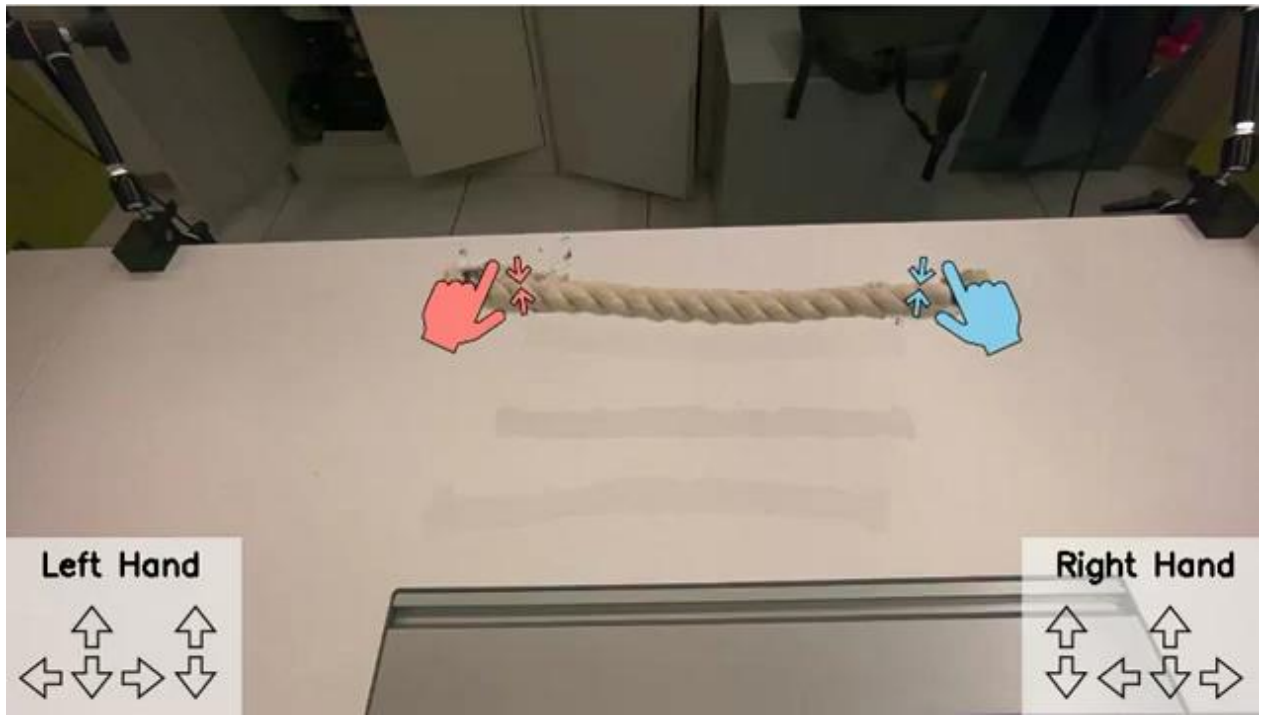
Hanxiao Jiang, Hao-Yu Hsu, Kaifeng Zhang, Hsin-Ni Yu, Shenlong Wang, **Yunzhu Li**

PhysTwin: Physics-Informed Reconstruction and Simulation of Deformable Objects from Videos, ICCV-25


Left Hand
Right Hand


Left Hand
Right Hand


COLUMBIA
UNIVERSITY
Left Hand
Right Hand


Left Hand
Right Hand

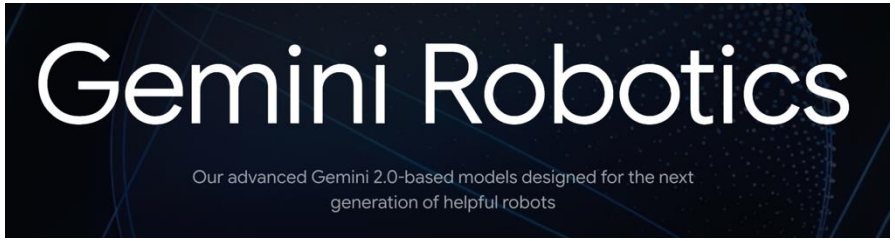

Left Hand
Right Hand

PhysTwin as a scalable evaluation engine



We have large enough budget such that we can still make progress.





Gaussian Splatting Construction

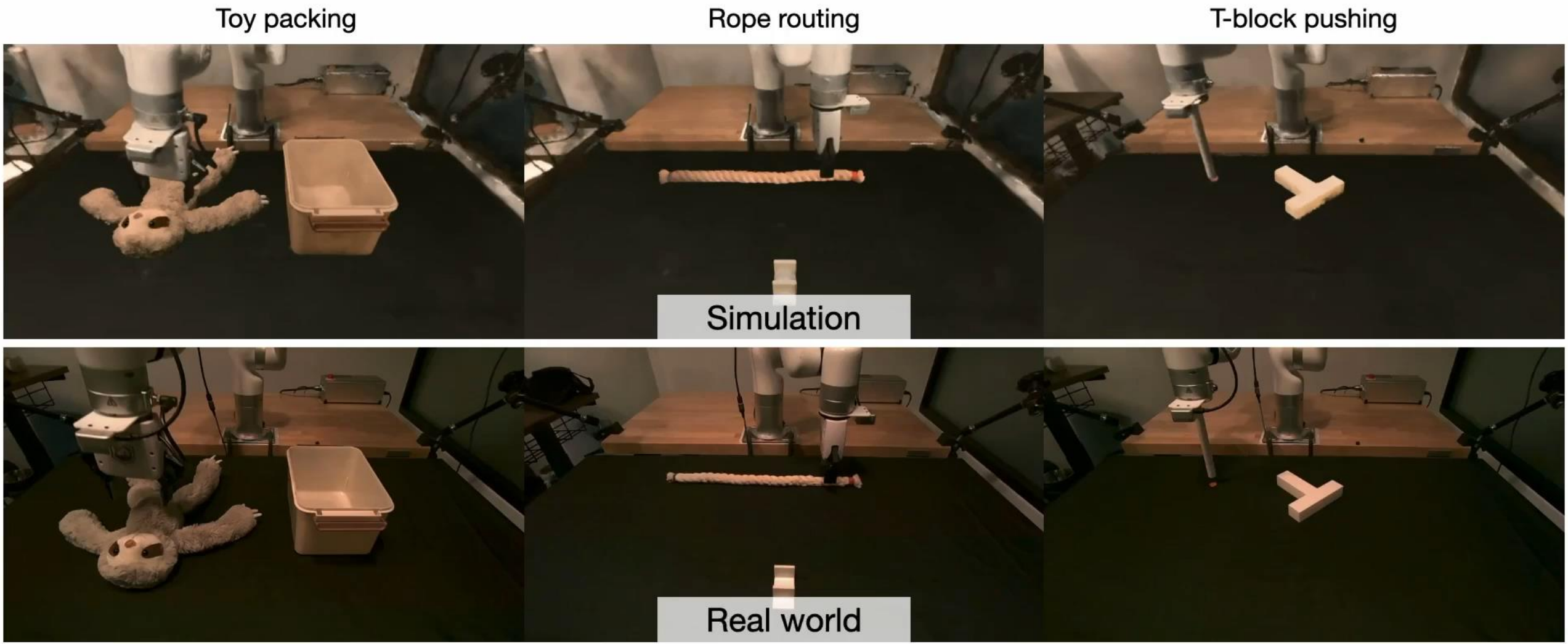
Positional Alignment

Color Alignment

PhysTwin

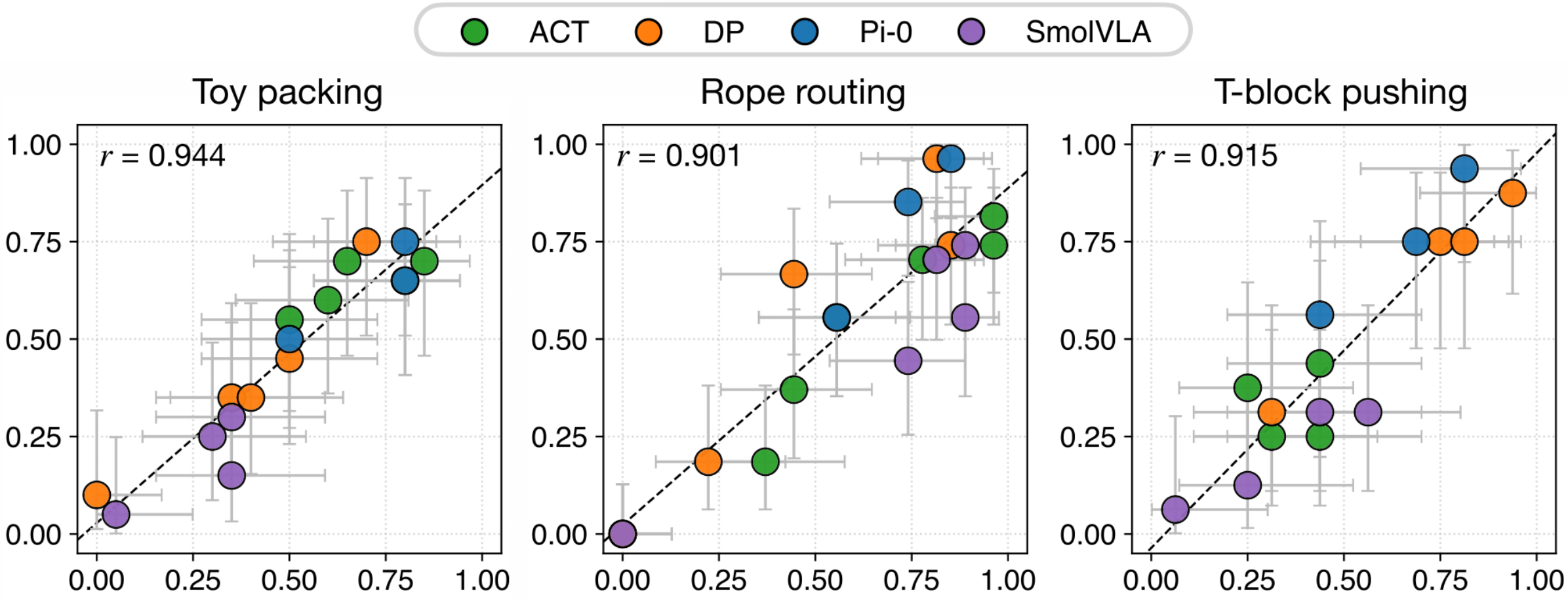
Physics & Deformation

Side-by-side comparison of running the **same policy** in simulation and the real world.



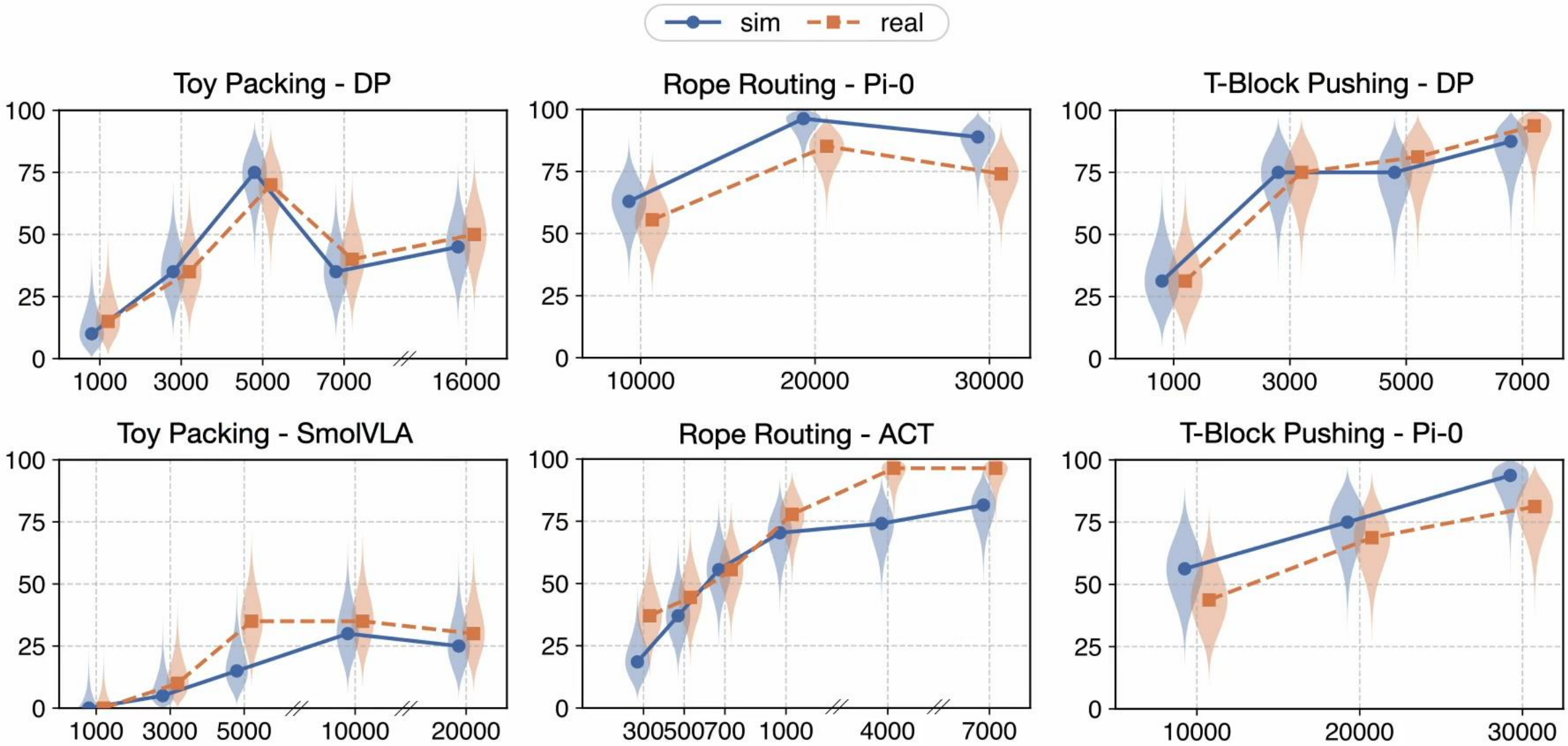
Kaifeng Zhang*, Shuo Sha*, Hanxiao Jiang, Matthew Loper, Hyunjong Song, Guangyan Cai, Zhuo Xu, Xiaochen Hu, Changxi Zheng, **Yunzhu Li**
Real-to-Sim Robot Policy Evaluation with Gaussian Splatting Simulation of Soft-Body Interactions, ICRA 2026

State-of-the-art correlation between simulation and the real world



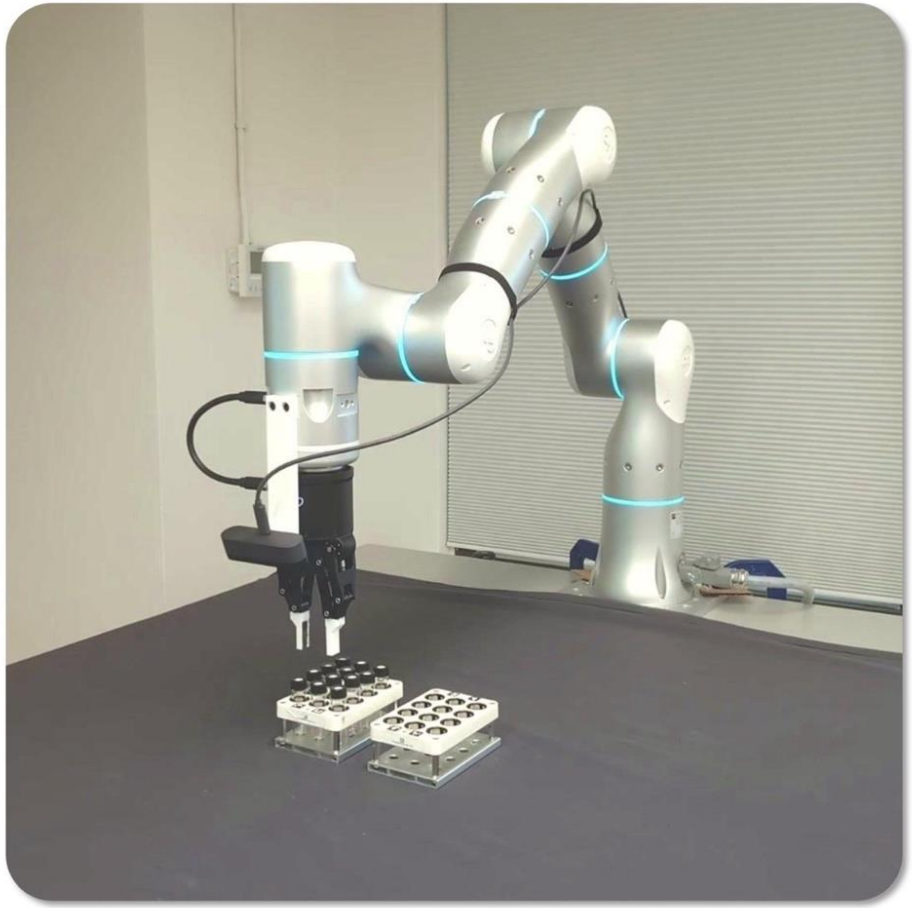
Kaifeng Zhang*, Shuo Sha*, Hanxiao Jiang, Matthew Loper, Hyunjong Song, Guangyan Cai, Zhuo Xu, Xiaochen Hu, Changxi Zheng, **Yunzhu Li**
Real-to-Sim Robot Policy Evaluation with Gaussian Splatting Simulation of Soft-Body Interactions, ICRA 2026

Per-policy, per-task performance across iterations showing **sim-and-real correlation.**



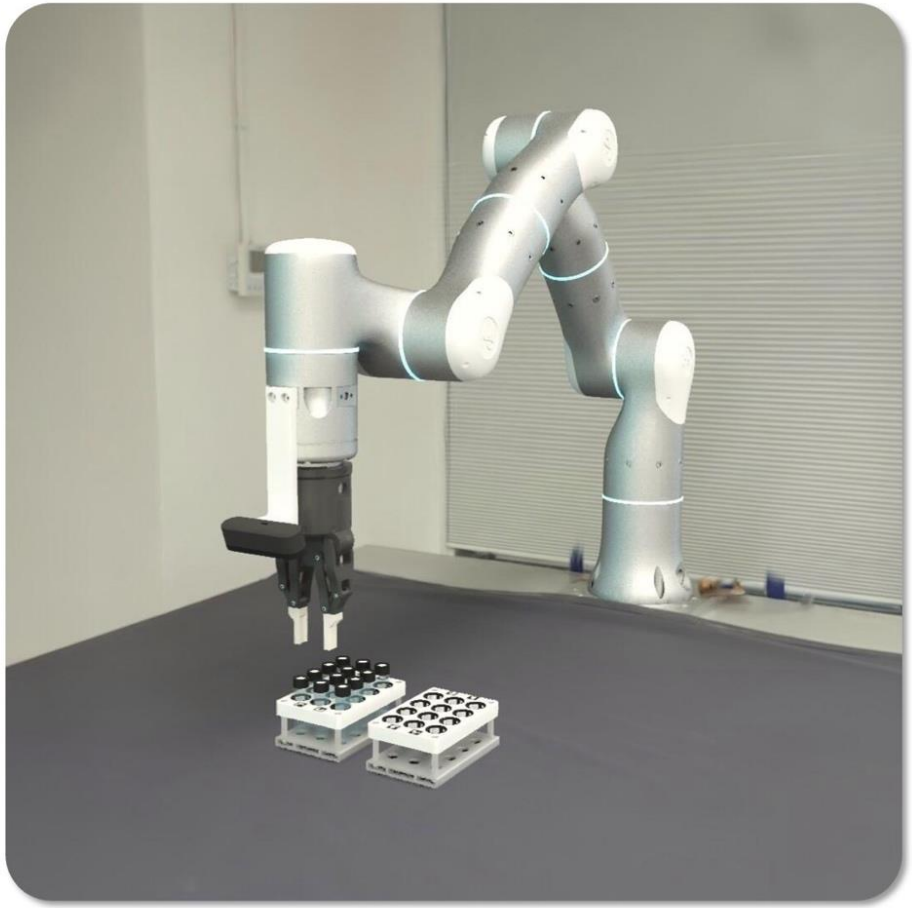
At World Labs, we take this from prototypes to real production systems

Real Task
Captured in the physical world

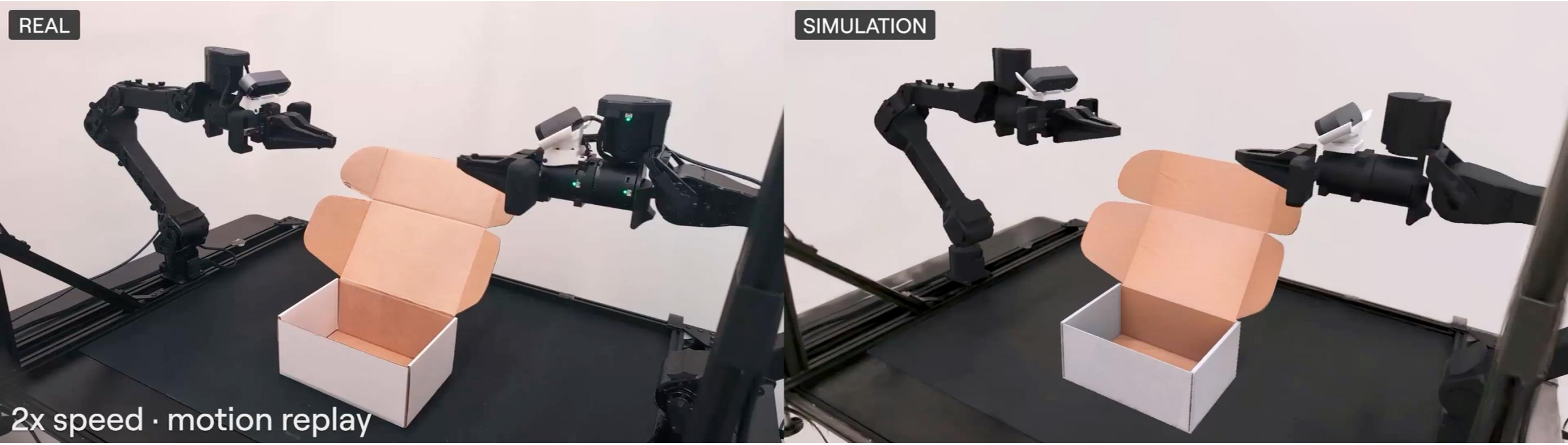


Real
-to-
Sim

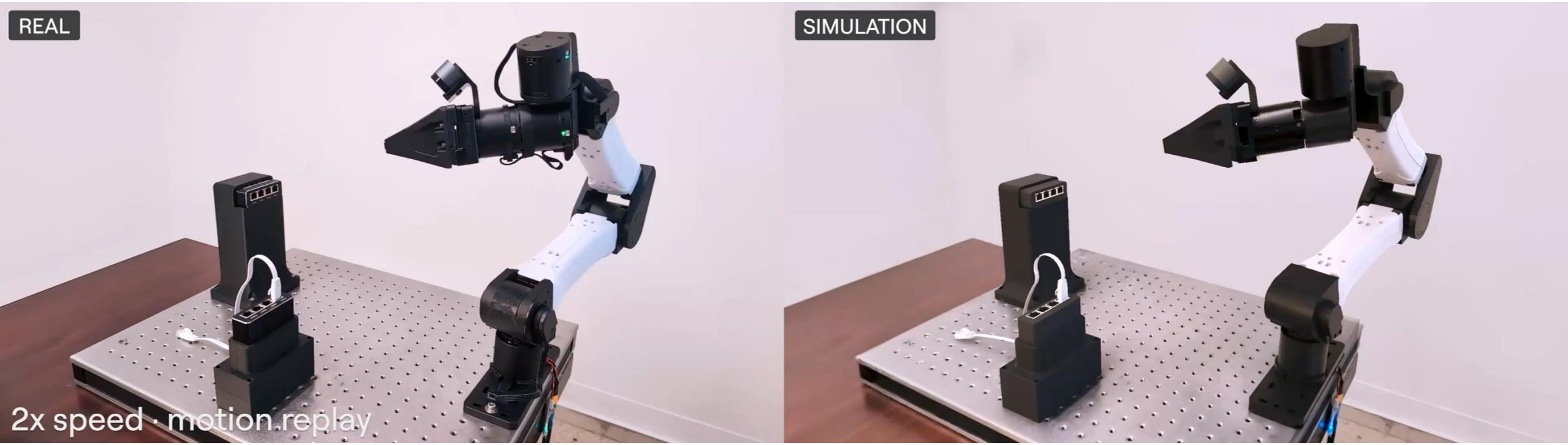
Aligned World
Interactive, faithful simulation



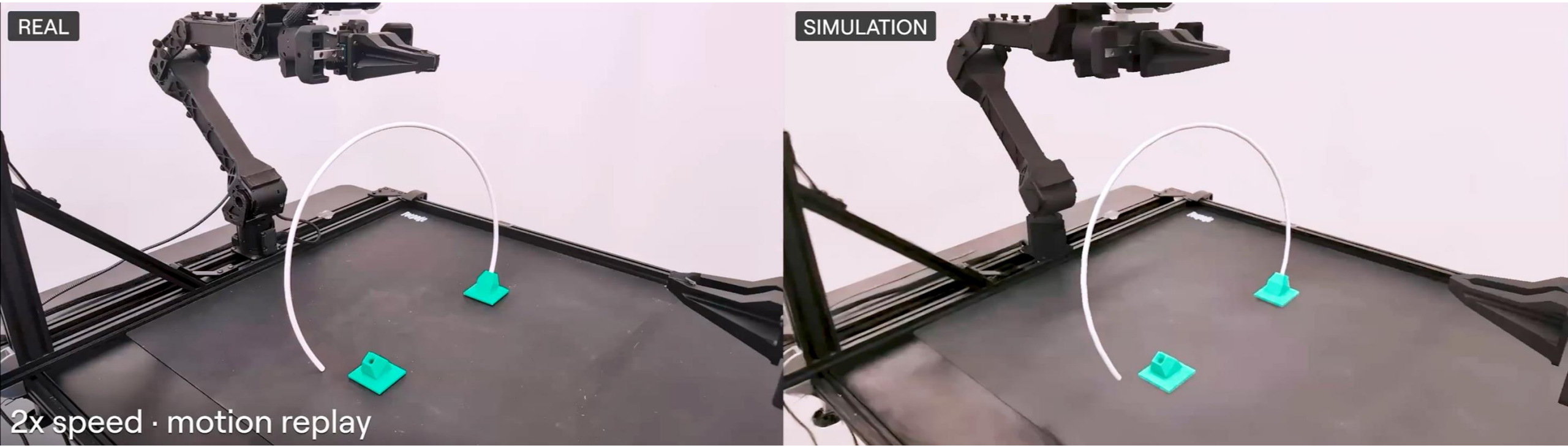
Sim-and-real alignment for elasto-plastic boxes



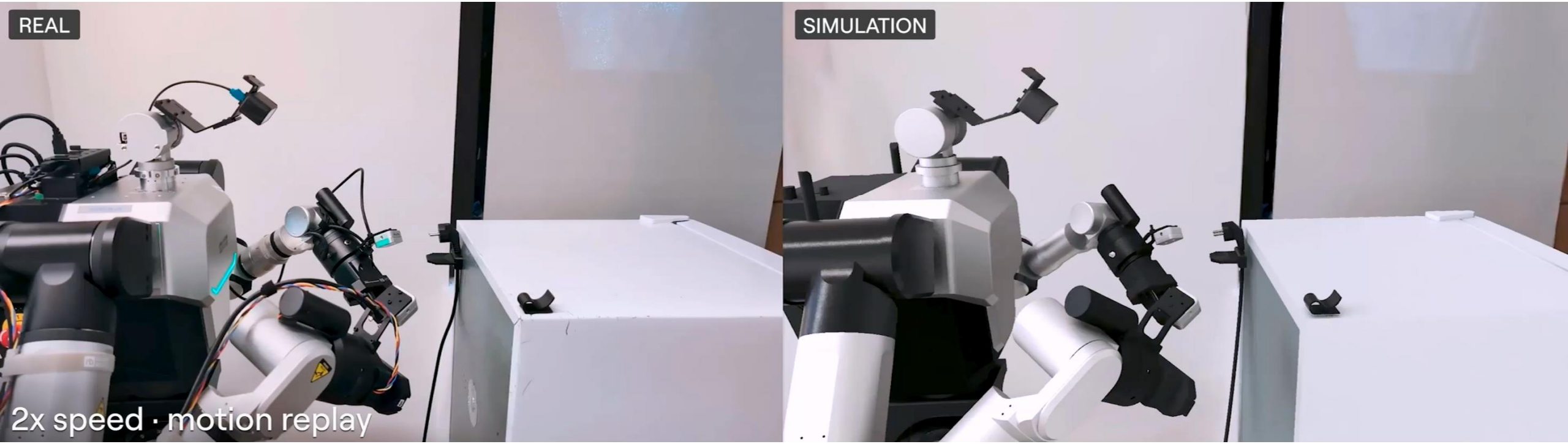
Sim-and-real alignment for linear deformable objects



Sim-and-real alignment for linear deformable objects

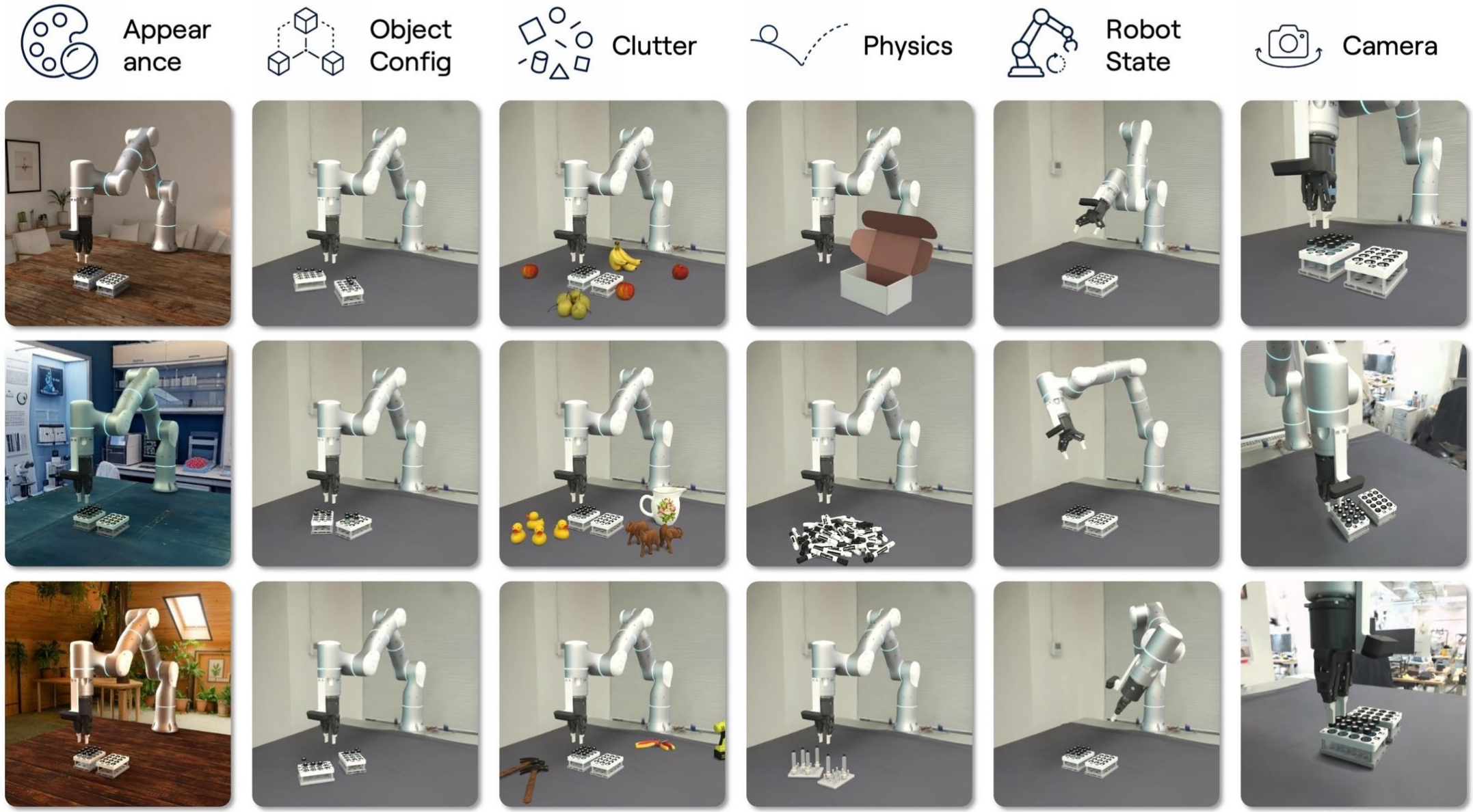


Sim-and-real alignment for linear deformable objects



Controllable Variations

One aligned world, systematic coverage of state possibilities



Sim-to-real transfer with zero real world data




Sparse real-world data capture

What's next?

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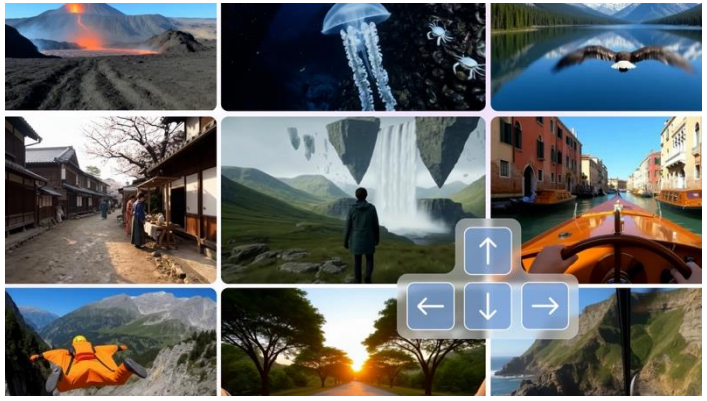
PhysTwin

?

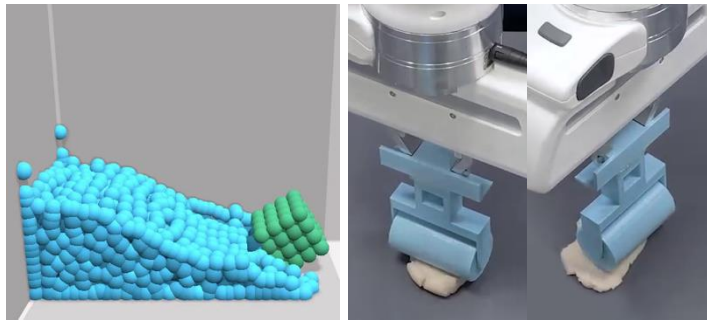
More Learning

More Physics

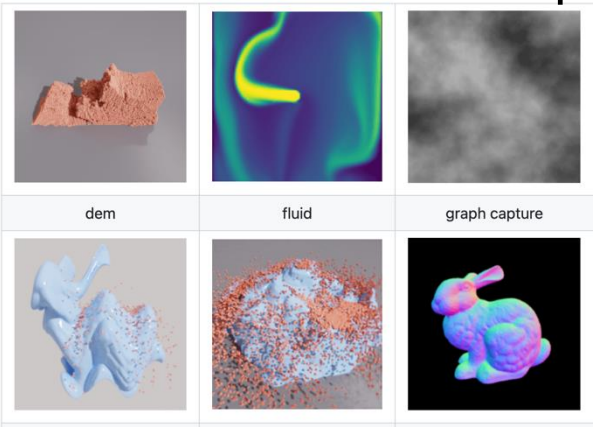
Genie 3



Graph-Based Neural Dynamics (GBND)

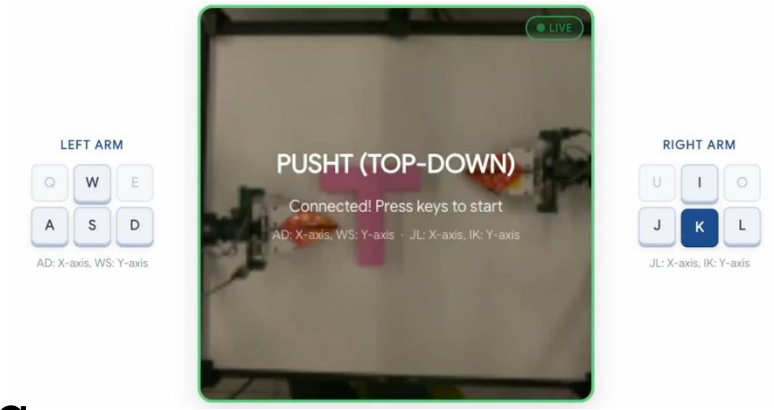


Nvidia Warp



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- Towards scalable data/evaluation engines



Interactive WS

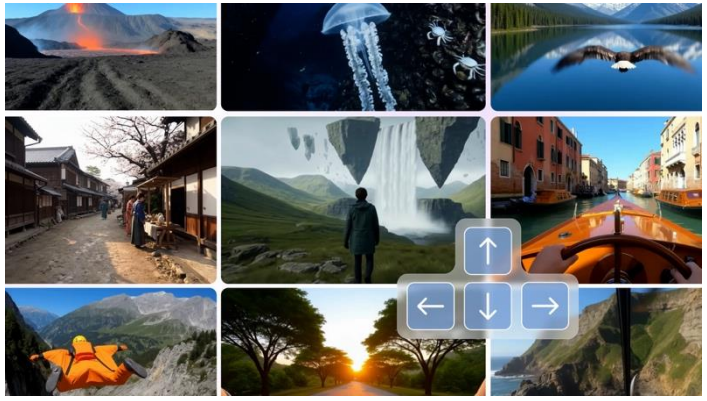


PhysTwin

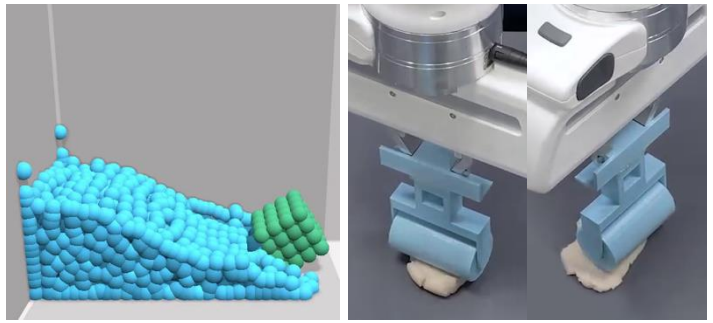
More Learning

More Physics

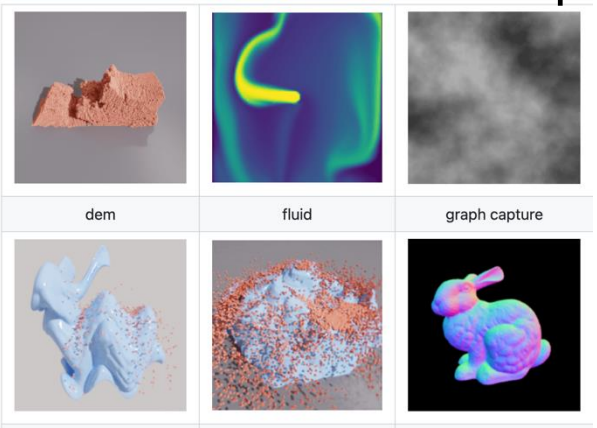
Genie 3



Graph-Based Neural Dynamics (GBND)



Nvidia Warp



Interact with World Simulator

Key Challenges

Scalable Data Generation for Policy Training

Faithful Policy Evaluation

YC-backed companies are using our work for fund raising

Backed by Y Combinator

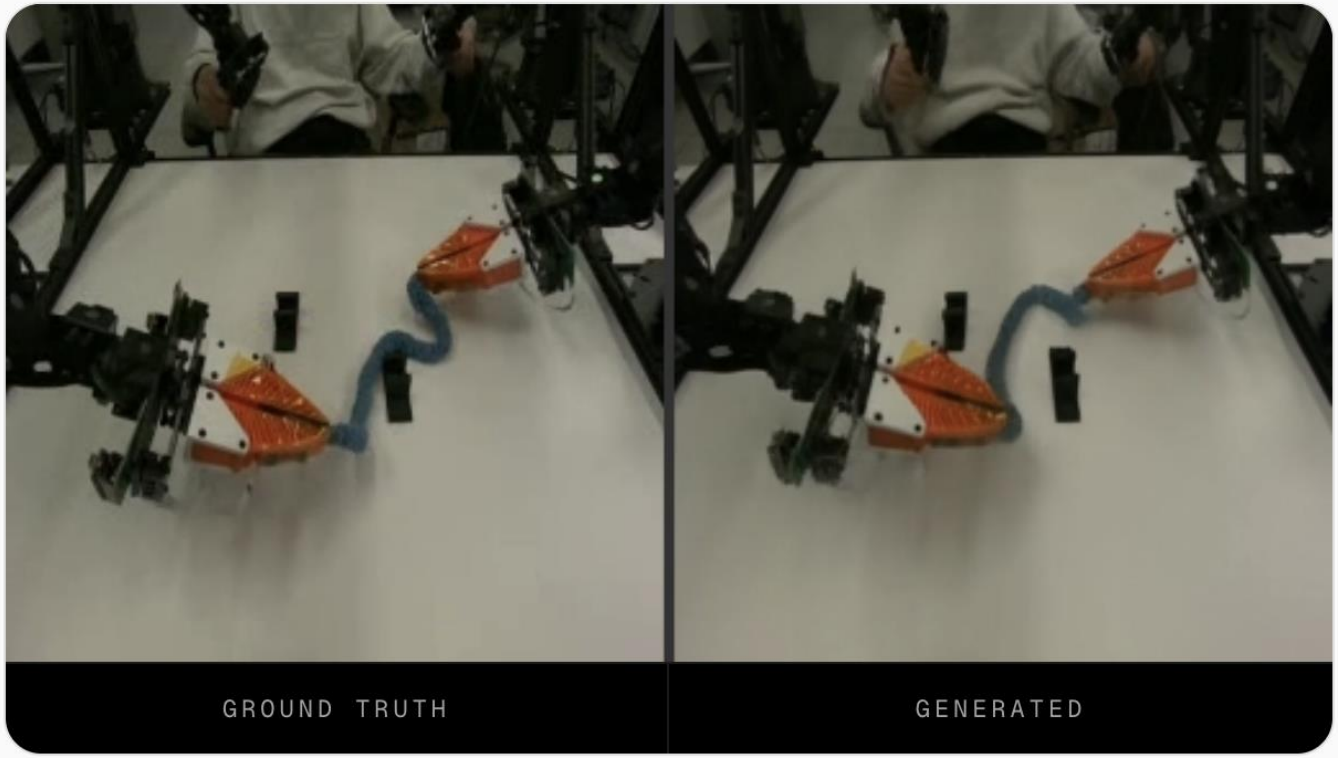
Enabling physical AI at scale

Run your robot evals on high-fidelity environments built from your training data, or on real hardware.

SCHEDULE A MEETING → WHITEPAPER →

What it looks like

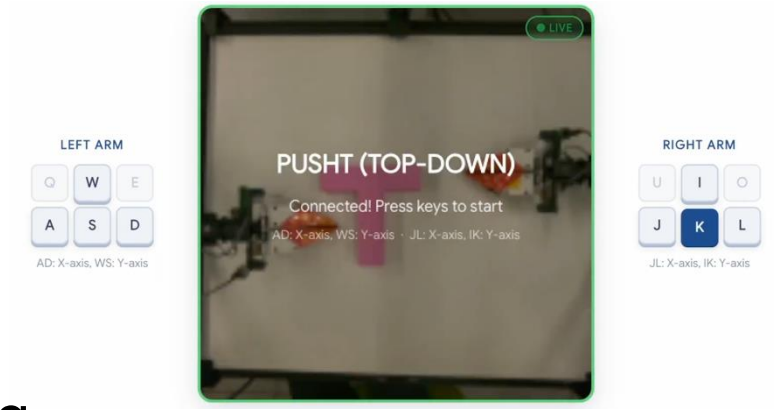
Two clips from the validation set. **Left half** is the real robot recording; **right half** is the model's rollout from the same starting frame and the same action stream — never seen during training.



BIMANUAL ROPE · VALIDATION EPISODE BR_0000

What's next?

- More complicated tasks/objects
- Heterogeneous interactions
- Scale up to in-the-wild videos (e.g., Open X-Embodiment, DROID)
- Towards scalable data/evaluation engines



Interactive WS

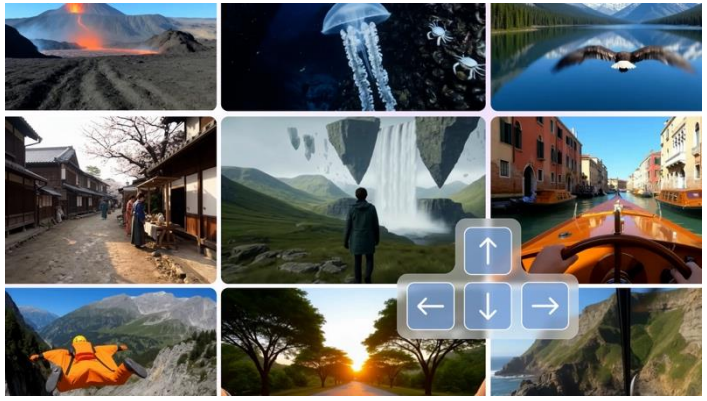


PhysTwin

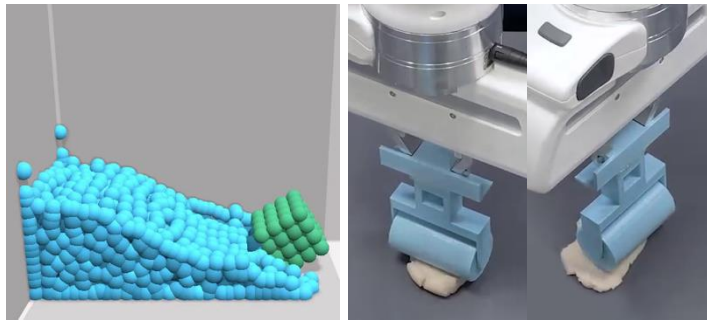
More Learning

More Physics

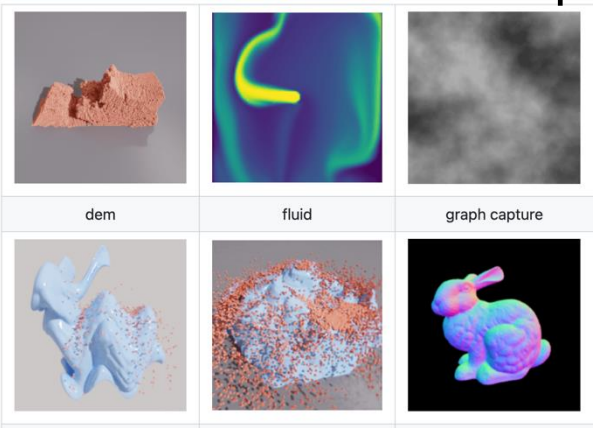
Genie 3



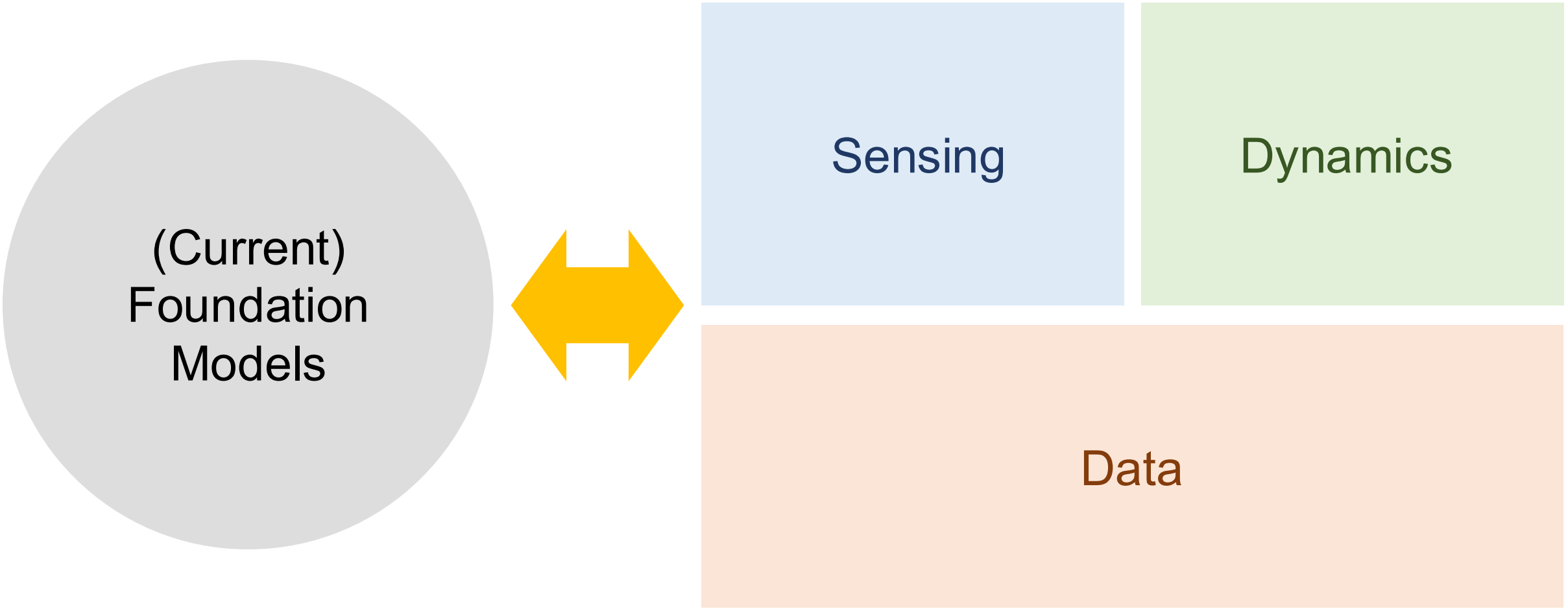
Graph-Based Neural Dynamics (GBND)



Nvidia Warp

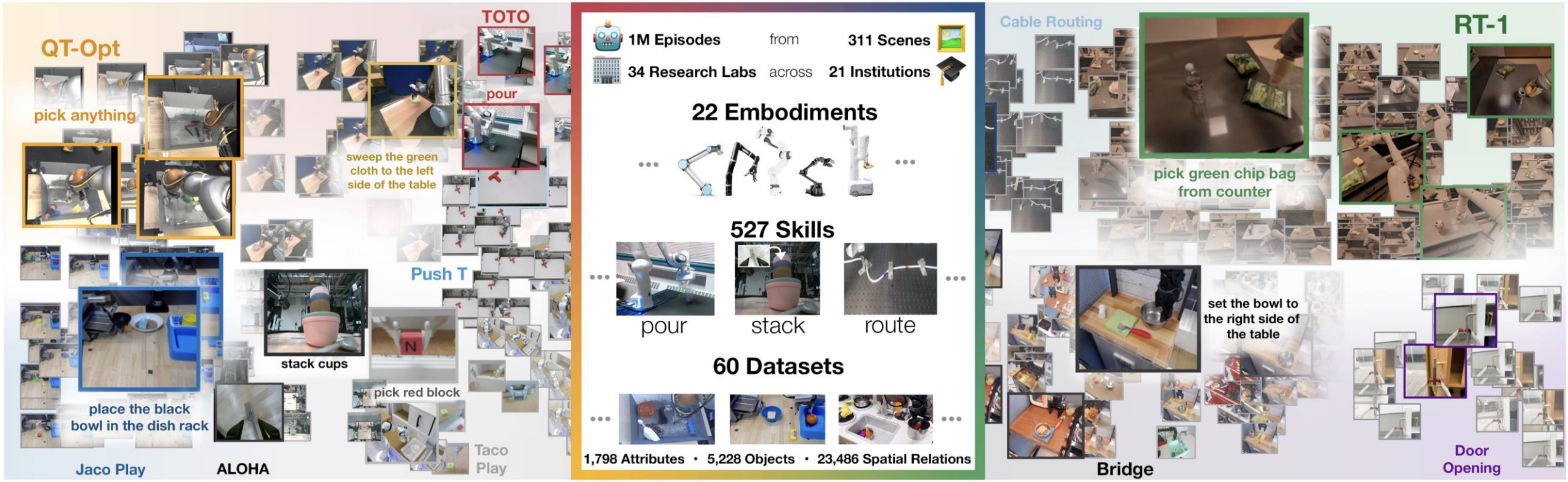


Towards **foundational robotic intelligence**



Towards foundational robotic models: **Real Data**

Open X-Embodiment: Robotic Learning Datasets and RT-X Models



Data collected by over 170 authors across 34 research labs

Best Paper Award, ICRA-24

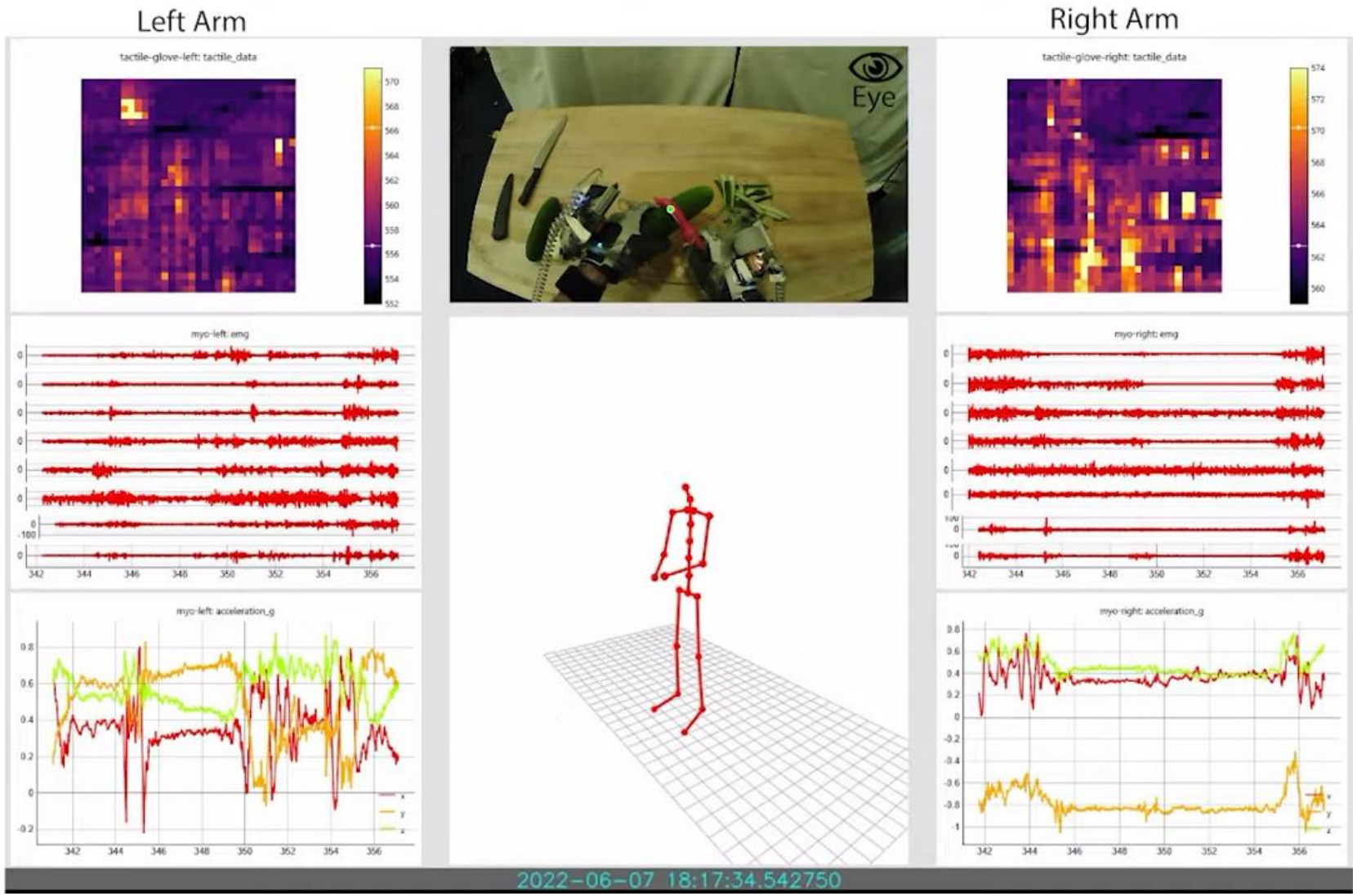
Towards foundational robotic models: **Simulated Data**



Li et al., BEHAVIOR-1K: A Human-Centered, Embodied AI Benchmark with 1,000 Everyday Activities and Realistic Simulation

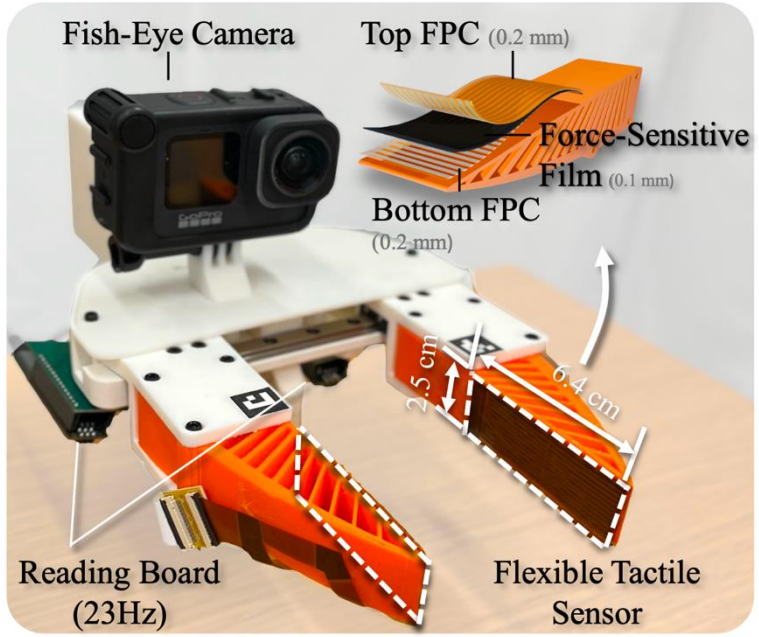
Towards foundational robotic models: **Multimodal Data**

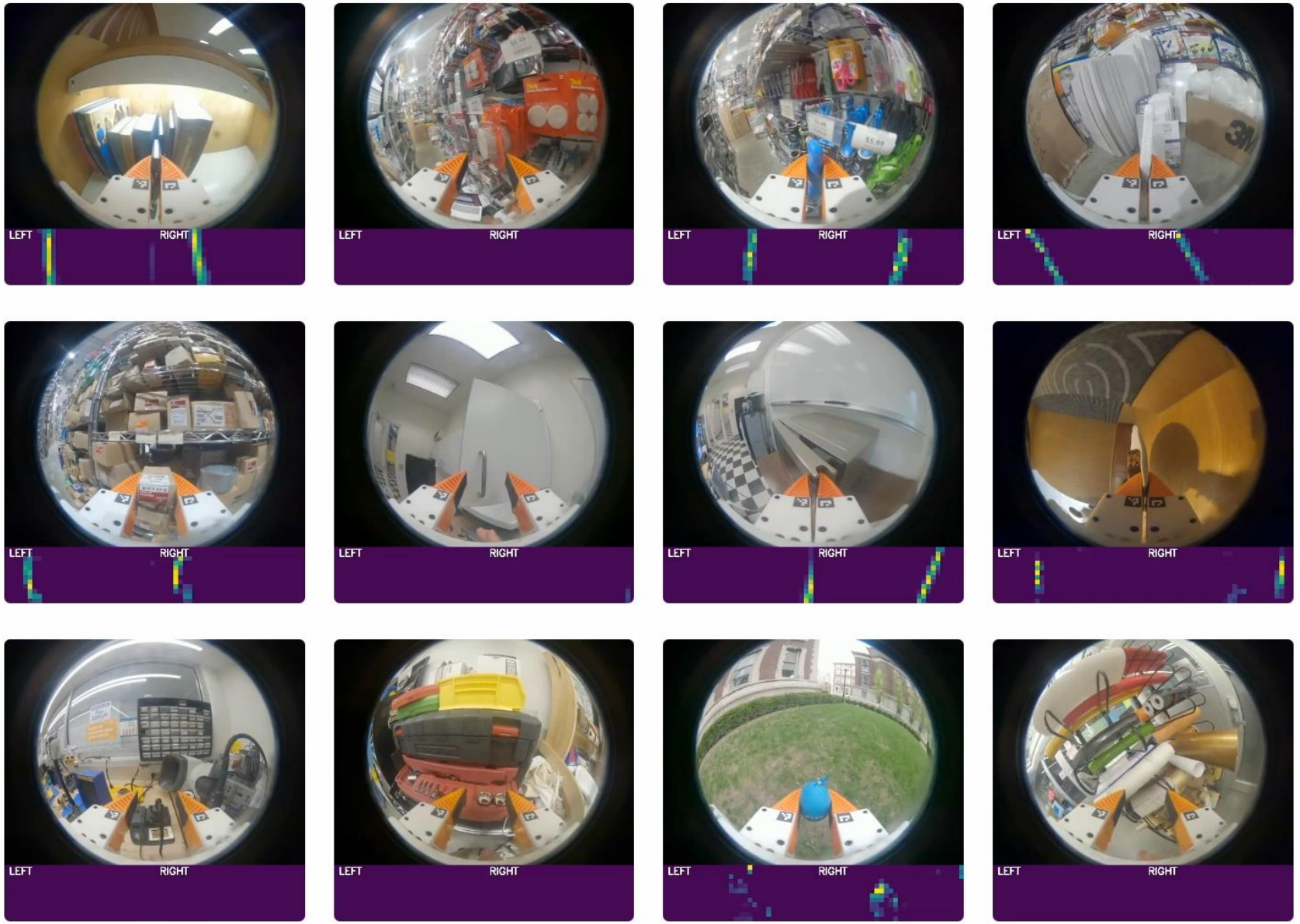




Joseph DelPreto*, Chao Liu*, Yiyue Luo, Michael Foshey, Yunzhu Li, Antonio Torralba, Wojciech Matusik, and Daniela Rus
ActionSense: A Multimodal Dataset and Recording Framework for Human Activities Using Wearable Sensors in a Kitchen Environment, NeurIPS-22

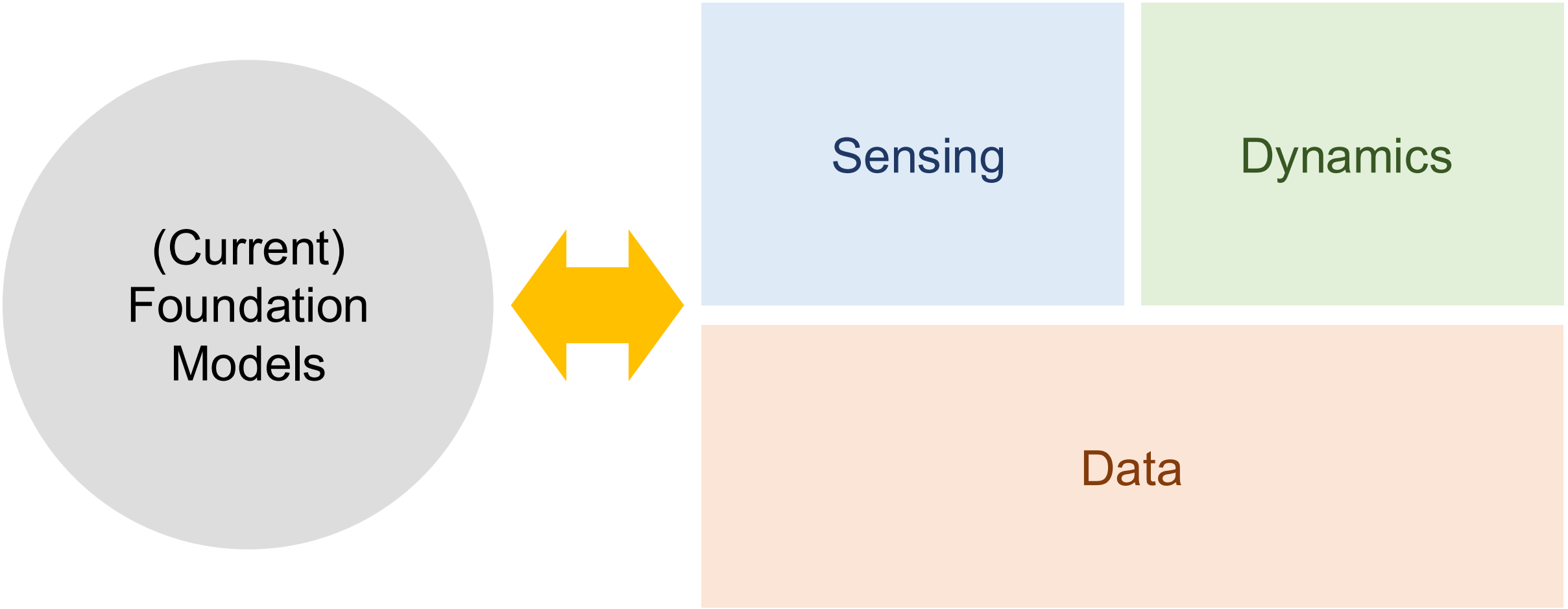
Towards foundational robotic models: **Multimodal Data**





Xinyue Zhu, Binghao Huang, **Yunzhu Li**
Touch in the Wild: Learning Fine-Grained Manipulation with a Portable Visuo-Tactile Gripper, NeurIPS-25

Towards **foundational robotic intelligence**



Towards **foundational robotic intelligence**

Current foundational vision-and-language models:

- The output may **not** always be **perfect**.
- It will always generate something **reasonable**.

What I hope to achieve with foundational robotic models:

- The synthesized action may **not** always be **optimal**.
- The generated trajectory will always be **beautiful** and **reasonable**.

Towards **foundational robotic intelligence**










Images generated using AI



Sponsors













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Scaling Robotic Manipulation via Structured World Models and Tactile Sensing



